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Chapter 7

Hyperbolic Functions

This chapter is a self-study. You will not be given any formal lectures on this examinable chapter. The references for the section on hyperbolic functions in Thomas' textbook is Section 7.3 pages 436–441.

LEARNING OUTCOMES:

On completion of this section you should be able to

1. know the definitions of the hyperbolic functions $\sinh x$, $\cosh x$, $\tanh x$, $\operatorname{sech} x$, $\coth x$ and $\operatorname{cosech} x$ and know the sketches of the graphs of $\sinh x$, $\cosh x$ and $\tanh x$ given on page 436 of Thomas,
2. know the hyperbolic identities and their derivations given on page 437 of Thomas,
3. know the derivatives (and the proofs) of the hyperbolic functions given on page 437 of Thomas,
4. know the Inverse Hyperbolic functions $\sinh^{-1} x$, $\cosh^{-1} x$ and $\tanh^{-1} x$ and their sketches given on pages 438 and 439 of Thomas,
5. know how to express the inverse hyperbolic functions in terms of natural logarithms, i. e., know how to prove and use $\sinh^{-1} x = \ln(x + \sqrt{x^2+1})$, $\cosh^{-1} x = \ln(x + \sqrt{x^2-1})$ and $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ as given on page 442 of Thomas,
6. know the derivatives (and the proofs) of the inverse hyperbolic functions given on page 490 of Stewart and know how to use the inverse hyperbolic functions in integration techniques as given by Example 6 on page 442 of Thomas.
7. know how to solve the **examinable** worked out problems on this section.

Tutorial 7.0.1. Thomas Exercises 11th ed. 7.4, pp. 530–532: 1, 3, 5, 6, 7, 9–12, 14, 15, 17, 21, 23, 25, 27, 31, 37, 40, 41, 45, 55, 61, 62, 67, 69, 71 **or** 12th ed. 7.3, pp. 441–442: 1, 3, 5, 6, 7, 9–12, 14, 15, 17, 21, 23, 25, 27, 31, 37, 40, 41, 45, 55, 61, 62, 67, 69, 71.
2. Prove the following formulas:

(a) $\sinh^{-1} x = \ln(x + \sqrt{x^2+1})$,

(b) $\cosh^{-1} x = \ln(x + \sqrt{x^2-1})$,

(c) $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$.

THIS SECTION IS EXAMINABLE!

Chapter 8

Integration Theory and Applications

8.1 Riemann Sums and Area Approximation

LEARNING OUTCOMES:

On completion of this section you should be able to

1. understand the concept of Riemann Sums and its application to areas,
2. use $\sum$-notation and summation formulæ to calculate the area under a curve using Riemann Sums,
3. know the definition of the definite integral and find the definite integral of a function using Riemann Sums,
4. know how to solve the **examinable** worked out problems from this section.

We recall Sigma Notation from Algebra.

The sum of n terms $a_1, a_2, a_3, \ldots, a_n$ is written as

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \cdots + a_n$$

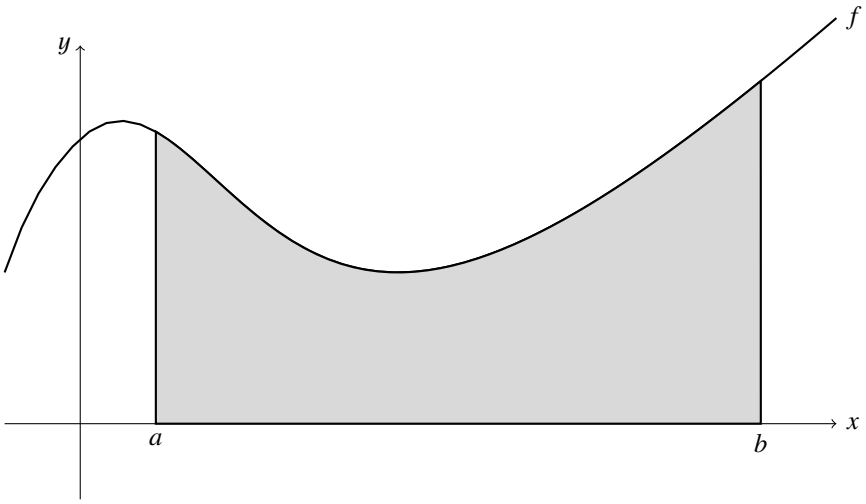
where i is the **index of summation**, a_i is the ith **term** of the sum, the **upper bound of summation** is n and the **lower bound of summation** is 1

and the following formula (proved by Mathematical Induction)

$$\sum_{i=1}^{n} c = nc \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Consider a plane region bounded above by the graph of a nonnegative, continuous function $y = f(x)$. The region is bounded below by the x-axis, and the left and right boundaries are the vertical lines $x = a$ and $x = b$.



Definition of a Riemann Sum

A Riemann Sum of f over the interval $[a, b]$ is any sum of the form

$$\sum_{i=1}^{n} f(x_i^*)\Delta x_i = f(x_1^*)\Delta x_1 + f(x_2^*)\Delta x_2 + f(x_3^*)\Delta x_3 + \cdots + f(x_n^*)\Delta x_n$$

where the points $x_0, x_1, x_2, \ldots, x_n$ with $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ divide the interval $[a, b]$ into n subintervals

$$[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n],$$

the length of each subinterval $[x_{j-1}, x_j]$ is

$$\Delta x_j = x_j - x_{j-1},$$

and x_j^* is any **sample point** in the interval $[x_{j-1}, x_j]$.

We will mostly divide the interval $[a, b]$ into n subintervals of *equal* length. The common length of each subinterval $[x_{j-1}, x_j]$ is then

$$\Delta x = \frac{b-a}{n}$$

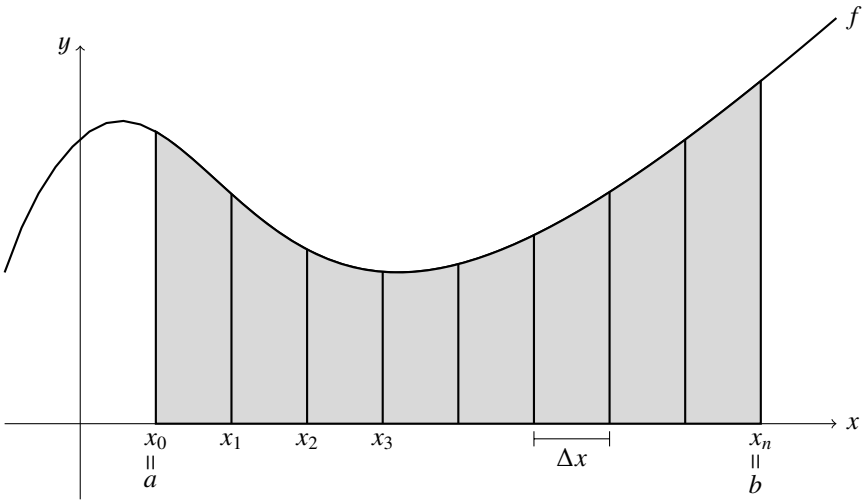
and the points of the subdivision are given by

$$x_j = a + j\Delta x = a + j\left(\frac{b-a}{n}\right).$$

The endpoints of the subintervals are thus

$$\overbrace{a + 0\Delta x}^{a = x_0} < \overbrace{a + 1\Delta x}^{x_1} < \overbrace{a + 2\Delta x}^{x_2} < \cdots < \overbrace{a + n\Delta x}^{x_n = b}$$

We thus divide the region into subregions bounded by the vertical lines $x = x_j$, $j = 0, 1, 2, \ldots, n$.

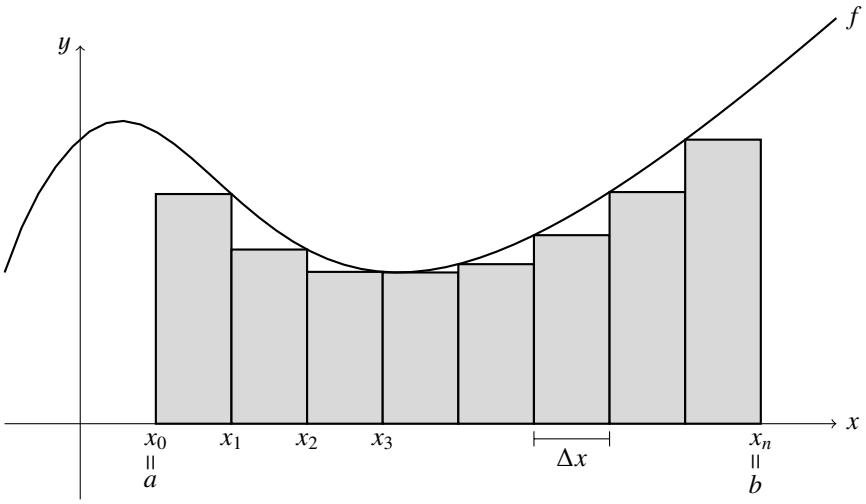


We can now approximate the area of the region using Riemann Sums. Since f is continuous, the Extreme Value Theorem guarantees the existence of a maximum value M_j and a minimum value m_j of $f(x)$ in *each* subinterval $[x_{j-1}, x_j]$. Let

$$f(m_j) \text{ be the minimum value of } f(x) \text{ in } [x_{j-1}, x_j],$$
$$f(M_j) \text{ be the maximum value of } f(x) \text{ in } [x_{j-1}, x_j],$$

Then we define two particular Riemann Sums as follows:

1. **Lower Sums**



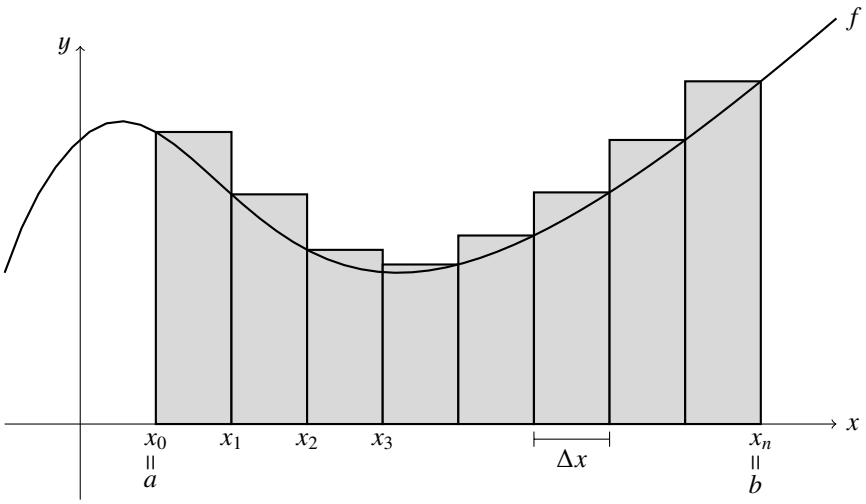
Define an **inscribed rectangle** lying inside the region above the jth subinterval whose height is $f(m_j)$. The area of the jth inscribed rectangle is

$$R_j = f(m_j)\Delta x.$$

Then the sum of the areas of all inscribed rectangles is called the **Lower Sum** and is given by

$$L(n) = \sum_{j=1}^{n} R_j = \sum_{j=1}^{n} f(m_j)\Delta x.$$

2. **Upper Sums**



Define a **circumscribed rectangle** lying above the jth subinterval whose height is $f(M_j)$. The area of the jth inscribed rectangle is

$$R_j = f(M_j)\Delta x.$$

Then the sum of the areas of all circumscribed rectangles is called the **Upper Sum** and is given by

$$U(n) = \sum_{j=1}^{n} R_j = \sum_{j=1}^{n} f(M_j)\Delta x.$$

The Lower and Upper Riemann Sums approximate the area under f

$$\boxed{L(n) \leq \text{ Area under } f \leq U(n)}$$

Worked Example 8.1.1. Approximate the area bounded by $f(x) = x^2$ and the x-axis between $x = 0$ and $x = 2$ using Riemann Sums with $n = 4$.

Solution. Partition $[0, 2]$ into $n = 4$ subintervals of equal length

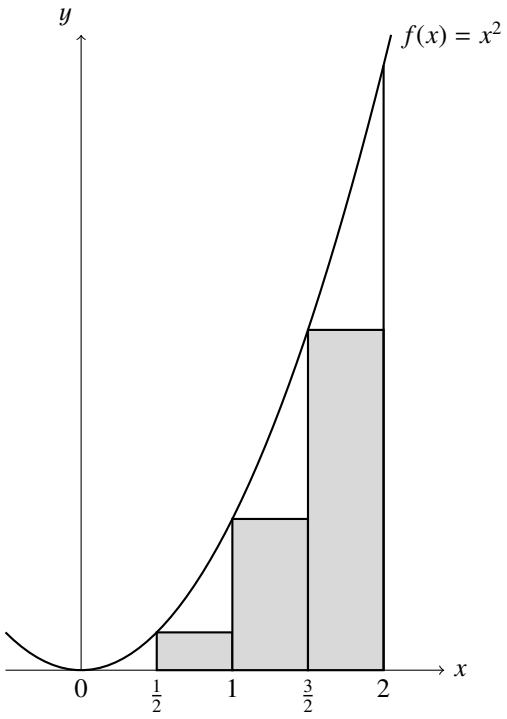
$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = \frac{1}{2}$$

generating subintervals

$$\left[0, \frac{1}{2}\right], \left[\frac{1}{2}, 1\right], \left[1, \frac{3}{2}\right], \left[\frac{1}{2}, 2\right].$$

Since f is increasing in $[0, 2]$, the minimum value of f on each of the above subintervals occurs at the left endpoint. Thus for the Lower Riemann Sums,

$$m_j = a + (j-1)\Delta x = 0 + (j-1)\left(\frac{1}{2}\right) = \frac{j-1}{2}.$$



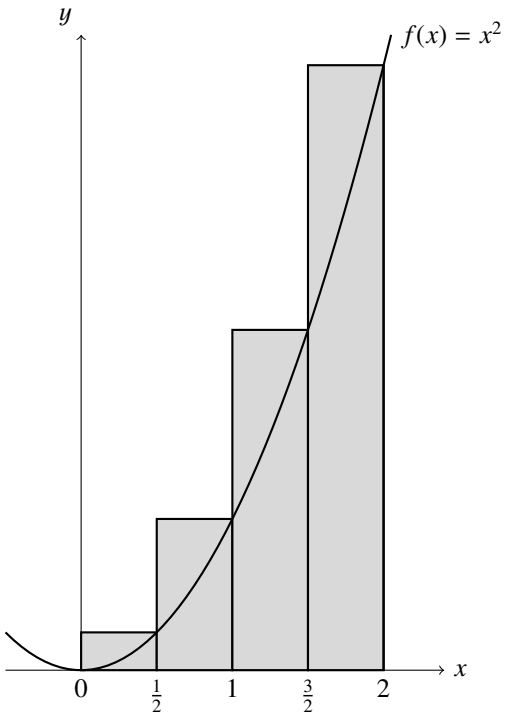
Inscribed rectangle R_j	Minimum value of f $m_j = \dfrac{j-1}{2}$	Height $f(m_j) = m_j^2$	Area $R_j = f(m_j)\Delta x$
R_1	0	$f(0) = 0$	$R_1 = 0$
R_2	$\dfrac{1}{2}$	$f\left(\dfrac{1}{2}\right) = \dfrac{1}{4}$	$R_2 = \left(\dfrac{1}{4}\right)\dfrac{1}{2} = \dfrac{1}{8}$
R_3	1	$f(1) = 1$	$R_3 = (1)\dfrac{1}{2} = \dfrac{1}{2}$
R_4	$\dfrac{3}{2}$	$f\left(\dfrac{3}{2}\right) = \dfrac{9}{4}$	$R_4 = \left(\dfrac{9}{4}\right)\dfrac{1}{2} = \dfrac{9}{8}$

Hence the Lower Sum is

$$L(4) = \sum_{j=1}^{4} f(m_j)\Delta x = \sum_{j=1}^{4} R_j = 0 + \frac{1}{8} + \frac{1}{2} + \frac{9}{8} = \frac{14}{8} = 1.75.$$

Again, since f is increasing in $[0, 2]$, the maximum value of f on each of the above subintervals occurs at the right endpoint. Thus for the Upper Riemann Sums,

$$M_j = a + j\Delta x = 0 + j\left(\frac{1}{2}\right) = \frac{j}{2}.$$



Circumscribed rectangle R_j	Maximum value of f $M_j = \frac{j}{2}$	Height $f(M_j) = M_j^2$	Area $R_j = f(M_j)\Delta x$
R_1	$\frac{1}{2}$	$f\left(\frac{1}{2}\right) = \frac{1}{4}$	$R_1 = \left(\frac{1}{4}\right)\frac{1}{2} = \frac{1}{8}$
R_2	1	$f(1) = 1$	$R_2 = (1)\frac{1}{2} = \frac{1}{2}$
R_3	$\frac{3}{2}$	$f\left(\frac{3}{2}\right) = \frac{9}{4}$	$R_3 = \left(\frac{9}{4}\right)\frac{1}{2} = \frac{9}{8}$
R_4	2	$f(2) = 4$	$R_4 = (4)\frac{1}{2} = 2$

Hence the Upper Sum is

$$U(4) = \sum_{j=1}^{4} f(M_j)\Delta x = \sum_{j=1}^{4} R_j = \frac{1}{8} + \frac{1}{2} + \frac{9}{8} + 2 = \frac{30}{8} = 3.75.$$

Thus

$$\begin{aligned} L(4) &< \text{ Area under } f < U(4) \\ \Rightarrow 1.75 &< \text{ Area under } f < 3.75. \end{aligned}$$

The more rectangles we choose (the larger the value of n) the better the approximation.

Theorem 8.1. *If f is a continuous function on the interval $[a, b]$, then $\lim_{n\to\infty} L(n)$ and $\lim_{n\to\infty} U(n)$ both exist and*

$$\lim_{n\to\infty} L(n) = \lim_{n\to\infty} \sum_{i=1}^{n} f(m_i)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} f(M_i)\Delta x = \lim_{n\to\infty} U(n),$$

where $\Delta x = \dfrac{b-a}{n}$ *and* $f(m_i)$ *is the minimum value of* f *on the* i^{th} *subinterval and* $f(M_i)$ *is the maximum value of* f *on the* i^{th} *subinterval, for each* $i = 1, 2, \ldots, n$.

The area of a region in the plane may now be defined in terms of Riemann Sums.

Let f be a continuous, nonnegative function on the interval $[a, b]$. The area of the region bounded by the graph of f, the x-axis and the vertical lines $x = a$ and $x = b$ is

$$\text{Area} = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x, \quad x_{i-1} \le x_i^* \le x_i,$$

where $\Delta x = \dfrac{b-a}{n}$.

Worked Example 8.1.2. Find the area of the region under the graph of the given function in the given interval using Riemann Sums.

1. $f(x) = 16 - x^2$ in $[0, 3]$.

Solution. We partition $[0, 3]$ into n subintervals of equal length

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{n} = \frac{3}{n}.$$

For convenience we may use Lower or Upper Sums. Since f is decreasing on $[0, 3]$ we see that on each subinterval the minimum value of f occurs at the right endpoint $m_i = x_i^*$ so that

$$x_i^* = a + i\Delta x = 0 + \left(\frac{3}{n}\right) i = \frac{3i}{n}.$$

Then

$$f(x_i^*) = 16 - (x_i^*)^2 = 16 - \frac{9i^2}{n^2}$$

and the area of each ith rectangle is

$$R_i = f(x_i^*)\Delta x = \left(16 - \frac{9i^2}{n^2}\right)\frac{3}{n} = \frac{48}{n} - \frac{27}{n^3} i^2.$$

Thus the area of the region is

$$\begin{aligned}
A &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} R_i \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{48}{n} - \frac{27}{n^3} i^2\right) \\
&= \lim_{n\to\infty} \left[\frac{1}{n} \sum_{i=1}^{n} 48 - \frac{27}{n^3} \sum_{i=1}^{n} i^2\right] \\
&= \lim_{n\to\infty} \left[\frac{1}{n}(48n) - \frac{27}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right)\right] \\
&= \lim_{n\to\infty} \left[48 - \frac{9}{2}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right] \\
&= 48 - \lim_{n\to\infty} \frac{9}{2}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) \\
&= 48 - \frac{9}{2}(1)(2) \\
&= 48 - 9 \\
&= 39.
\end{aligned}$$

To take limits, we may consider n to be a real number and then apply the limit theorems we know from semester 1.

2. $g(x) = 1 + 2x^3$ in $[0, 5]$.

Solution. We partition $[0, 5]$ into n subintervals of equal length

$$\Delta x = \frac{b-a}{n} = \frac{5-0}{n} = \frac{5}{n}.$$

For convenience we take x_i^* to be the right endpoint of the i-th subinterval so that

$$x_i^* = a + i\Delta x = 0 + \left(\frac{5}{n}\right) i = \frac{5i}{n}.$$

Then

$$f(x_i^*) = 1 + 2(x_i^*)^3 = 1 + 2\frac{125i^3}{n^3}$$

and the area of each ith rectangle is

$$R_i = f(x_i^*)\Delta x = \left(1 + \frac{250i^3}{n^3}\right)\frac{5}{n} = \frac{5}{n} + \frac{1250}{n^4} i^3.$$

Thus the area of the region is

$$\begin{aligned}
A &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} R_i \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{5}{n} + \frac{1250}{n^4} i^3\right) \\
&= \lim_{n\to\infty} \left[\frac{1}{n}\sum_{i=1}^{n} 5 + \frac{1250}{n^4}\sum_{i=1}^{n} i^3\right] \\
&= \lim_{n\to\infty} \left[\frac{1}{n}(5n) - \frac{1250}{n^4}\left(\frac{n(n+1)}{2}\right)^2\right] \\
&= \lim_{n\to\infty} \left[5 + \frac{625}{2}\left(1 + \frac{1}{n}\right)^2\right] \\
&= 5 + \lim_{n\to\infty} \frac{625}{2}\left(1 + \frac{1}{n}\right)^2 \\
&= 5 + \frac{625}{2} \\
&= \frac{635}{2}.
\end{aligned}$$

Tutorial 8.1.1. 1. Thomas Exercises: 11th ed. 5.1 p. 360: 1, 3, 5, 7, 10 **or** 12th ed. 5.1 pp. 304–305: 1, 3, 5, 7, 10.

8.2 Area with the Definite Integral

LEARNING OUTCOMES:

On completion of this section you should be able to

1. understand and know the proofs of the Fundamental Theorem of Calculus2 (FTC2) and the Mean Vaue Theorem for integrals,
2. know the properties of the definite integral,
3. know how to evaluate the integrals of symmetric functions,
4. know how to compute the average value of a function,
5. compute the area of a region between two curves using integration,
6. find the area of a region between intersecting curves using integration,
7. interpret a definite integral as the area of a known geometric region,
8. know how to solve the examinable worked out problems from this section.

8.2.1 The Definite Integral

Extending the formula for the area from the previous section to functions which are not necessarily positive we obtain the following definition:

Definition (Definite Integral). Let f be a continuous function defined on the interval $[a, b]$ and divide the interval $[a, b]$ into n subintervals of equal length $\Delta x = \dfrac{b-a}{n}$. We let

$$a = x_0 < x_1 < x_2 < x_3 < \cdots < x_n = b$$

be the endpoints of these subintervals and let $x_1^*, x_2^*, \ldots, x_n^*$ be any sample points in these subintervals so that x_i^* lies in the ith subinterval $[x_{i-1}, x_i]$. Then the definite integral of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

It is clear then from Riemann Sums and the area under a curve that if f is continuous and non-negative on $[a, b]$ (i.e., $f(x) \geq 0$ for all $x \in [a, b]$), then the area bounded by the graph of f, the x-axis and the lines $x = a$ and $x = b$ is

$$\text{Area} = \int_a^b f(x)\,dx.$$

In Chapter 5 of the first semester study guide we used the Fundamental Theorem of Calculus 2 (FTC2) in conjunction with some integration techniques to evaluate a definite integral. We now may interpret the definite integral as the area under the curve of a non-negative continuous function. The following is a formal proof of the Fundamental Theorem of Calculus.

Theorem 8.2 (Fundamental Theorem of Calculus 2, FTC2). *If f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on the interval $[a, b]$, then*

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof. Partition the interval $[a, b]$ into n subintervals with endpoints

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

of length $\Delta x = x_i - x_{i-1}$.
By pairwise subtraction and addition of like terms, we can write as a telescoping series

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_{n-1}) + F(x_{n-1}) - \cdots - F(x_1) + F(x_1) - F(x_0) \\ &= \sum_{i=1}^{n} [F(x_i) - F(x_{i-1})]. \end{aligned}$$

Since F is an antiderivative, F is continuous on $[x_{i-1}, x_i]$ and differentiable on (x_{i-1}, x_i) for each i. By the Mean Value Theorem, applied to F on these subintervals, for each $i = 1, \ldots, n$ there exist $c_i \in (x_{i-1}, x_i)$ such that

$$\begin{aligned} F'(c_i) &= \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \\ &= \frac{F(x_i) - F(x_{i-1})}{\Delta x} \\ \Rightarrow f(c_i) &= \frac{F(x_i) - F(x_{i-1})}{\Delta x} \quad \because F \text{ is an antiderivative of } f,\ F'(c_i) = f(c_i) \\ \Rightarrow f(c_i)\,\Delta x &= F(x_i) - F(x_{i-1}) \\ \Rightarrow \sum_{i=1}^{n} f(c_i)\,\Delta x &= \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = F(b) - F(a) \\ \Rightarrow \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\,\Delta x &= F(b) - F(a) \\ \Rightarrow \int_a^b f(x)\,dx &= F(b) - F(a). \end{aligned}$$

□

The properties of $\sum$ and that of the limit enable us to realise the following properties for the definite integral.

Theorem 8.3 (Properties of the Definite Integral). *We assume that $a \le b$ and that f and g are continuous functions on $[a, b]$.*

1. $\boxed{\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx}$

Proof. Formally, we can take $b = x_0 > x_1 > x_2 > \cdots > a = x_n$. Then

$$-\int_b^a f(x)\,dx = -\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x = -\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\left(\frac{a-b}{n}\right) = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\left(\frac{b-a}{n}\right) = \int_a^b f(x)\,dx.$$

2. $\boxed{\int_a^a f(x)\,dx = 0}$

Proof. Note that $\Delta x = 0$. So the result follows.

3. $\boxed{\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx \quad \text{for any } c \in \mathbb{R}}$

Proof.

$$\int_a^b cf(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} cf(x_i^*)\,\Delta x = \lim_{n\to\infty}\left[c\sum_{i=1}^{n} f(x_i^*)\,\Delta x\right] = c\left[\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x\right] = c\int_a^b f(x)\,dx.$$

4. $\boxed{\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx}$

Proof.

$$\begin{aligned}
\int_a^b [f(x)+g(x)]\,dx &= \lim_{n\to\infty}\sum_{i=1}^{n}[f(x_i^*)+g(x_i^*)]\,\Delta x \\
&= \lim_{n\to\infty}\left[\sum_{i=1}^{n} f(x_i^*)\,\Delta x + \sum_{i=1}^{n} g(x_i^*)\,\Delta x\right] \\
&= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x + \lim_{n\to\infty}\sum_{i=1}^{n} g(x_i^*)\,\Delta x \\
&= \int_a^b f(x)\,dx + \int_a^b g(x)\,dx.
\end{aligned}$$

5. $\boxed{\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx}$

Proof.

$$\int_a^b [f(x) - g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b [-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx.$$

6. $\displaystyle\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$ if f is continuous on an interval containing a, b and c

Proof. We assume that $a < c < b$ and choose a partition of $[a, b]$ that includes c as a point of subdivision, say $c = x_m$. Then

$$\begin{aligned}
\int_a^b f(x)\,dx &= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i \\
&= \lim_{n\to\infty} \left[\sum_{i=1}^{m} f(x_i^*)\,\Delta x_i + \sum_{i=m+1}^{n} f(x_i^*)\,\Delta x_i \right] \\
&= \lim_{n\to\infty} \sum_{i=1}^{m} f(x_i^*)\,\Delta x_i + \lim_{n\to\infty} \sum_{i=m+1}^{n} f(x_i^*)\,\Delta x_i \\
&= \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.
\end{aligned}$$

Note that because of $x_m = c$ for some m, in general, the subintervals cannot be chosen to be of equal length.

7. If $m \le f(x) \le M$ for all $a \le x \le b$, then $m(b-a) \le \displaystyle\int_a^b f(x)\,dx \le M(b-a)$

Proof. Since $f(x_i^*) \le M$ for each choice of x_i^* and $\Delta x \ge 0$, we have

$$\begin{aligned}
& f(x_i^*)\,\Delta x \le M\Delta x \\
\Rightarrow & \sum_{i=1}^{n} f(x_i^*)\,\Delta x \le \sum_{i=1}^{n} M\,\Delta x = M \sum_{i=1}^{n} \frac{b-a}{n} = M\left[n\left(\frac{b-a}{n} \right) \right] \\
\Rightarrow & \sum_{i=1}^{n} f(x_i^*)\,\Delta x \le M(b-a) \\
\Rightarrow & \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x \le M(b-a) \\
\Rightarrow & \int_a^b f(x)\,dx \le M(b-a).
\end{aligned}$$

Similary, using the fact that $m \le f(x_i^*)$ for each choice of x_i^* we get $m(b-a) \le \displaystyle\int_a^b f(x)\,dx$ and the result then follows.

8. If $f(x) \ge 0$ for all $a \le x \le b$, then $\displaystyle\int_a^b f(x)\,dx \ge 0$

Proof. Put $m = 0$ in Property 7.

9. If $f(x) \geq g(x)$ for all $a \leq x \leq b$, then $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$

Proof. Since $f(x) \geq g(x)$ for each $a \leq x \leq b$, $f(x) - g(x) \geq 0$ in $[a, b]$. Thus, by Property 8,

$$\int_a^b [f(x) - g(x)]\,dx \geq 0 \Rightarrow \int_a^b f(x)\,dx - \int_a^b g(x)\,dx \geq 0$$
$$\Rightarrow \int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx.$$

10. $\left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx$

Proof. Since $f(x) \leq |f(x)|$ and $-f(x) \leq |f(x)|$ for each $a \leq x \leq b$,

$$\int_a^b f(x)\,dx \leq \int_a^b |f(x)|\,dx \quad \text{and} \quad -\int_a^b f(x)\,dx \leq \int_a^b |f(x)|\,dx$$
$$\Rightarrow \left|\int_a^b f(x)\,dx\right| \leq \int_a^b |f(x)|\,dx.$$ □

We will now use the FTC2 and the techniques of Chapter 5 (Integral Table, Substitution Rule and Integration by Parts) in the 1st Semester Calculus Guidebook to evaluate definite integrals. For the next theorem we recall even and odd functions from Algebra 1st semester.

$f(x)$ is an **EVEN** function $\Leftrightarrow f(-x) = f(x)$ for all $x \in \text{Dom}(f)$

$f(x)$ is an **ODD** function $\Leftrightarrow f(-x) = -f(x)$ for all $x \in \text{Dom}(f)$

Theorem 8.4 (**Integration of Even and Odd funtions**). *Suppose that f is continous on $[-a, a]$.*
1. *If f is an even function, then*

$$\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx.$$

2. *If f is an odd function, then*

$$\int_{-a}^{a} f(x)\,dx = 0.$$

Proof. Using the properties of the definite integral we have

$$\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_0^a f(x)\,dx$$
$$= -\int_0^{-a} f(x)\,dx + \int_0^a f(x)\,dx.$$

We apply the Definite Integral Substitution Rule. Put $u = -x$. Then $du = -dx$. The limits of the first integral change as follows: when $x = -a$, $u = a$ and when $x = 0$, $u = 0$. Thus

$$\int_{-a}^{a} f(x)\,dx = -\int_{0}^{a} f(-u)(-du) + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(-x)\,dx + \int_{0}^{a} f(x)\,dx.$$

1. Suppose that f is an even function. Then

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(-x)\,dx + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.$$

2. If f is an odd function, then

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} f(-x)\,dx + \int_{0}^{a} f(x)\,dx = -\int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(x)\,dx = 0. \qquad \square$$

Worked Example 8.2.1. Evaluate the following definite integrals.
1.

$$\int_{-\pi}^{\pi} (\sin x - x\cos x)^3\,dx$$

Solution. We note that if $f(x) = (\sin x - x\cos x)^3$, then

$$\begin{aligned} f(-x) &= (\sin(-x) - (-x)\cos(-x))^3 \\ &= (-\sin x + x\cos x)^3 \\ &= [-(\sin x - x\cos x)]^3 \\ &= -(\sin x - x\cos x)^3 \\ &= -f(x). \end{aligned}$$

Thus the integrand is an odd function so that immediately (without having to use integration techniques) we can conclude that $\displaystyle\int_{-\pi}^{\pi} (\sin x - x\cos x)^3\,dx = 0$.

2.

$$\int_{-2}^{2} x^2 \cosh x\,dx$$

Solution. Note that $\cosh x = \dfrac{e^x + e^{-x}}{2}$ so that if $f(x) = x^2\cosh x$, then

$$f(-x) = (-x)^2\cosh(-x) = x^2\left(\frac{e^{-x} + e^{-(-x)}}{2}\right) = x^2\left(\frac{e^x + e^{-x}}{2}\right) = x^2\cosh x = f(x),$$

i.e., $f(x) = x^2\cosh x$ is an even function. Thus

$$\int_{-2}^{2} x^2\cosh x\,dx = 2\int_{0}^{2} x^2\cosh x\,dx.$$

Using IBP, let $u = x^2$ and $dv = \cosh x\,dx$. Then $du = 2x\,dx$ and $v = \int \cosh x\,dx = \sinh x$ so that

$$\begin{aligned}
\int x^2 \cosh x\,dx &= \int u\,dv \\
&= uv - \int v\,du \\
&= x^2 \sinh x - \int 2x \sinh x\,dx \\
&= x^2 \sinh x - [2x\cosh x - \int 2\cosh x\,dx] \qquad (\text{IBP on } \int 2x\sinh x\,dx) \\
&= x^2 \sinh x - [2x\cosh x - 2\sinh x] + C \\
&= x^2 \sinh x - 2x\cosh x + 2\sinh x + C.
\end{aligned}$$

Thus

$$\begin{aligned}
\int_0^2 x^2 \cosh x\,dx &= x^2 \sinh x - 2x\cosh x + 2\sinh x\Big]_0^2 \\
&= 4\sinh 2 - 4\cosh 2 + 2\sinh 2 \\
&= 6\sinh 2 - 4\cosh 2.
\end{aligned}$$

Hence,

$$\int_{-2}^{2} x^2 \sinh x\,dx = 2\int_0^2 x^2 \cosh x\,dx = 12\sinh 2 - 8\cosh 2.$$

We know that the area of the region under a curve is greater or equal than the area of the inscribed rectangles (lower sums) and less or equal than the area of the circumscribed rectangles (upper sums). We now have the Mean Value Theorem for integral calculus which loosely states that somewhere between the inscribed and circumscribed rectangles for $n = 1$, there is a rectangle whose area is precisely equal to the region under the curve.

Theorem 8.5 (**Mean Value Theorem for Integrals**). *If f is continuous on the closed interval $[a, b]$, then there exists a number c in the closed interval $[a, b]$ such that*

$$\int_a^b f(x)\,dx = f(c)(b - a).$$

Proof. By the Extreme Value Theorem, Theorem 6.2, f attains its minimum and maximum on $[a, b]$. Hence there are $y, z \in [a, b]$ such that

$$f(y) \le f(x) \le f(z) \text{ for all } x \in [a, b].$$

Since f is continuous, also the function g given by

$$g(x) = f(x)(b - a)$$

is continuous on $[a, b]$.
In view of Theorem 8.3, 7,

$$g(y) = f(y)(b - a) \le \int_a^b f(x)\,dx \le f(z)(b - a).$$

With $k = \int_a^b f(x)\,dx$, it follows from the Intermediate Value Theorem, Theorem 2.4, that there is c between y and

z such that $g(c) = k$. In particular, $c \in [a, b]$. Hence

$$\int_a^b f(x)\,dx = k = g(c) = f(c)(b-a). \qquad \square$$

The value of $f(c)$ given by the Mean Value Theorem for Integrals is called the **average value** of f on $[a, b]$ denoted by f_{ave} and given by

$$f_{\text{ave}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Worked Example 8.2.2. Find the average value of $f(x) = 3x^2 - 2x$ on $[1, 4]$.

Solution.

$$f_{\text{ave}} = \frac{1}{4-1}\int_1^4 (3x^2 - 2x)\,dx = \frac{1}{3}\left[x^3 - x^2\right]_1^4 = \frac{1}{3}[64 - 16 - (1-1)] = 16.$$

Tutorial 8.2.1. 1. Thomas Exercises 11th ed. 5.3 pp. 379–381: 1, 3, 5, 10–14, 63 **or** 12th ed. 5.3 pp. 321–323: 1, 3, 5, 10–14, 64, 68, 69, 71.
2. Thomas Exercises 11th ed. 5.4 p. 392: 5, 14, 21, 34 **or** 12th ed. 5.4 p. 333: 3, 15, 16, 38.

8.2.2 Interpreting the Definite Integral as a Known Area

The definite integral $\int_a^b f(x)\,dx$ for $f(x) \geq 0$ in $[a, b]$ gives the area under f. If the region has a known geometry, then the definite integral is easily calculated using the known geometric area formula.

Worked Example 8.2.3. Evaluate the following definite integrals by interpreting them in terms of areas.

1. $\int_{-2}^{2} \sqrt{4 - x^2}\,dx.$

Solution. The positive function $\sqrt{4-x^2}$ in $[-2, 2]$ is the upper semicirle at the origin of radius 2. Thus

$$\int_{-2}^{2} \sqrt{4-x^2}\,dx = \text{Area of semicircle } = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi.$$

2. $\int_0^2 |2x - 1|\,dx.$

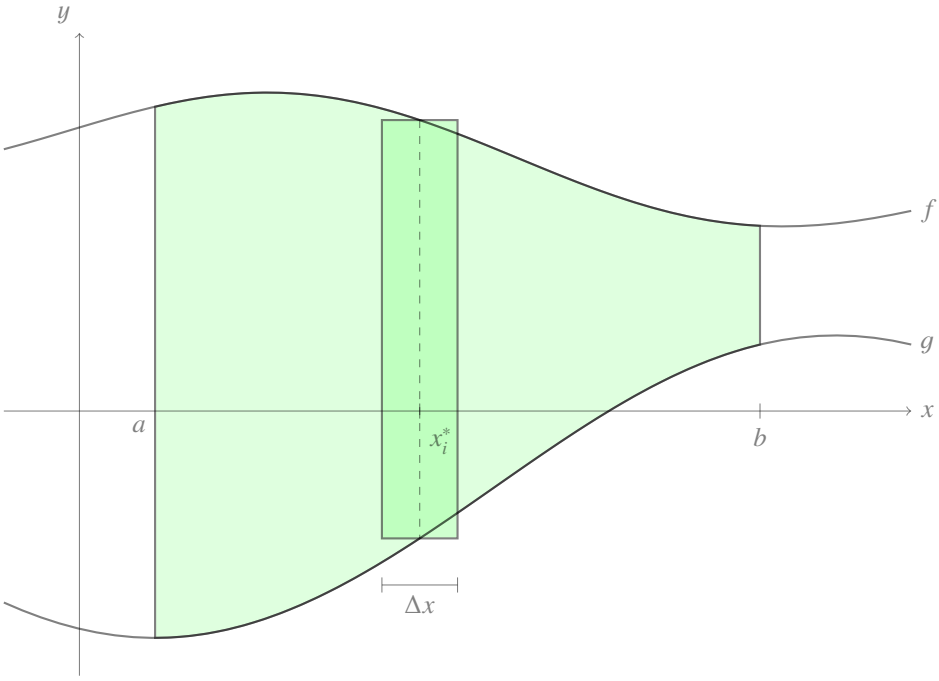
Solution. The definite integral gives an area corresponding to that of the sum of two triangular regions, A_1 with base $b_1 = \frac{1}{2}$ and height $h_1 = 1$ and A_2 with base $b_2 = \frac{3}{2}$ and height $h_2 = 3$. Thus

$$\int_0^2 |2x-1|\,dx = A_1 + A_2 = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 = \frac{1}{2}\left(\frac{1}{2}\right)(1) + \frac{1}{2}\left(\frac{3}{2}\right)(3) = \frac{1}{4} + \frac{9}{4} = \frac{5}{2}.$$

Tutorial 8.2.2. 1. Thomas Exercises 11th ed. 5.3 p. 380: 15, 18, 19, 20, 22 **or** 12th ed. 5.3 p. 323: 15, 18, 19, 20, 22.

8.2.3 Area between two Curves: x-axis

We consider the region between the continuous curves $f(x)$ and $g(x)$ and the vertical lines $x = a$ and $x = b$ with $f(x) \geq g(x)$ for each $x \in [a, b]$.



Using Riemann Sums, we partition $[a, b]$ into n subintervals of equal length Δx. Then for the ith subinterval $[x_{i-1}, x_i]$, the ith vertical rectangular strip has area

$$R_i = [f(x_i^*) - g(x_i^*)]\Delta x$$

with x_i^* any sample point in $[x_{i-1}, x_i]$. Hence the area between the two curves is the limit as $n \to \infty$ of the sum of all these rectangular strips (Riemann Sum) so that

$$\begin{aligned}\text{Area} &= \lim_{n\to\infty} \sum_{i=1}^{n} R_i \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} [f(x_i^*) - g(x_i^*)]\Delta x \\ &= \int_a^b [f(x) - g(x)]\, dx.\end{aligned}$$

It is easy to remember that if f is the curve on the **TOP** and g is the **BOTTOM** curve then the area between f and g is

$$\text{Area} = \int_a^b [\textbf{TOP} - \textbf{BOTTOM}]\, dx.$$

Worked Example 8.2.4. Find the area of the region bounded by $y = \sec^2 x$, $y = \sin x$, $x = 0$ and $x = \dfrac{\pi}{4}$.

Solution.

$$A = \int_0^{\frac{\pi}{4}} (\sec^2 x - \sin x)dx = \tan x + \cos x\Big]_0^{\frac{\pi}{4}} = \tan\frac{\pi}{4} + \cos\frac{\pi}{4} - (\tan 0 + \cos 0) = 1 + \frac{1}{\sqrt{2}} - (0 + 1) = \frac{1}{\sqrt{2}}.$$

If the lines $x = a$ and $y = b$ are not given but the curves f and g intersect in at least two points, i.e., there are $a < b$ such that $f(a) = g(a)$ and $f(b) = g(b)$, then we first have to find these points a and b in order to find the area bounded by the two curves.

Worked Example 8.2.5. 1. Find the area of the region bounded by $y = x^2 + 2x + 1$ and $y = 2x + 5$.

Solution. First we find the points of intersection:

$$\begin{aligned} x^2 + 2x + 1 &= 2x + 5 \\ \Leftrightarrow x^2 - 4 &= 0 \\ \Leftrightarrow x &= \pm 2. \end{aligned}$$

Thus a $= -2$ and $b = 2$. Since $2x+5$ lies above x^2+2x+1 in $[-2, 2]$, $2x+5 =$ **TOP** and $x^2+2x+1 =$ **BOTTOM**. Then

$$\begin{aligned} \text{Area} &= \int_a^b [\textbf{TOP} - \textbf{BOTTOM}]dx \\ &= \int_{-2}^{2} [(2x + 5) - (x^2 + 2x + 1)]dx \\ &= \int_{-2}^{2} [-x^2 + 4]dx \\ &= -\frac{x^3}{3} + 4x\Big]_{-2}^{2} \\ &= \left(-\frac{8}{3} + 8\right) - \left(\frac{8}{3} - 8\right) \\ &= -\frac{16}{3} + 16 \\ &= \frac{32}{3}. \end{aligned}$$

2. Find the area of the region bounded by $y = 5 \ln x$ and $y = x \ln x$.

Solution. Points of intersection:

$$\begin{aligned} 5 \ln x &= x \ln x \\ \Leftrightarrow 5 \ln x - x \ln x &= 0 \\ \Leftrightarrow (5 - x) \ln x &= 0 \\ \Leftrightarrow \ln x = 0 \text{ or } 5 - x &= 0 \\ \Leftrightarrow x = 1 \text{ or } x &= 5. \end{aligned}$$

Since $5 \ln x \geq x \ln x$ for $x \in [1, 5]$, the area of the region bounded by these two functions is

$$A = \int_1^5 [\textbf{TOP- BOTTOM}]\, dx = \int_1^5 [5 \ln x - x \ln x]\, dx = \int_1^5 (5 - x) \ln x \, dx.$$

Using IBP, let

$$\begin{aligned} u = \ln x \text{ and } dv &= (5 - x)dx \\ \Rightarrow du = \frac{1}{x}dx \text{ and } v &= \int (5 - x)dx = 5x - \frac{x^2}{2}. \end{aligned}$$

Then

$$\begin{aligned}
A &= \int_1^5 (5-x)\ln x\,dx \\
&= \int_1^5 u\,dv = uv\Big]_1^5 - \int_1^5 v\,du \\
&= \left(5x - \frac{x^2}{2}\right)\ln x\Big]_1^5 - \int_1^5 \left(5x - \frac{x^2}{2}\right)\frac{1}{x}\,dx \\
&= \left(5x - \frac{x^2}{2}\right)\ln x\Big]_1^5 - \int_1^5 \left(5 - \frac{x}{2}\right)dx \\
&= \left(5x - \frac{x^2}{2}\right)\ln x\Big]_1^5 - \left(5x - \frac{x^2}{4}\right)\Big]_1^5\,dx \\
&= \left(25 - \frac{25}{2}\right)\ln 5 - \left(5 - \frac{1}{2}\right)\ln 1 - \left[\left(25 - \frac{25}{4}\right) - \left(5 - \frac{1}{4}\right)\right] \\
&= \frac{25\ln 5}{2} - 14.
\end{aligned}$$

3. (Bounded region with changing formula) Find the area of the region in the first quadrant bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

Solution. We note here that the region is partitioned at $x = 2$ (the x-intercept of $y = x-2$). Obviously, the left-hand limit is 0. We find the right-hand limit by simultaneous solution:

$$\begin{aligned}
\sqrt{x} = x - 2 &\Leftrightarrow x - \sqrt{x} - 2 = 0 \\
&\Leftrightarrow (\sqrt{x})^2 - \sqrt{x} - 2 = 0 \\
&\Leftrightarrow (\sqrt{x} - 2)(\sqrt{x} + 1) = 0 \\
&\Leftrightarrow \sqrt{x} = 2 \quad (\sqrt{x} + 1 \neq 0) \\
&\Leftrightarrow x = 4.
\end{aligned}$$

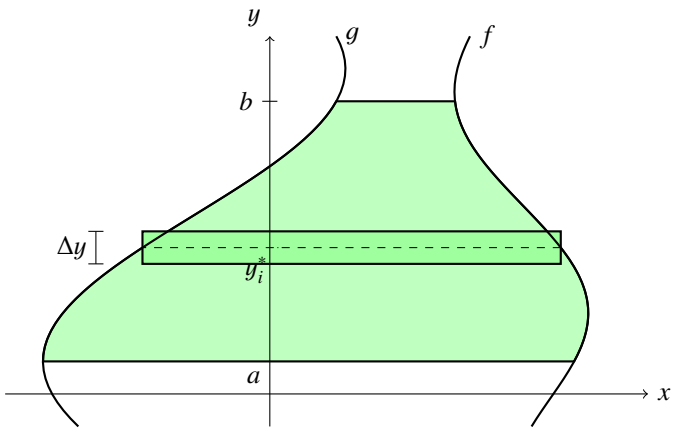
Hence,

$$\begin{aligned}
\text{Area} &= \int_0^2 (\sqrt{x} - 0)\,dx + \int_2^4 [\sqrt{x} - (x-2)]\,dx \\
&= \int_0^2 \sqrt{x}\,dx + \int_2^4 \sqrt{x}\,dx + \int_2^4 (2-x)\,dx \\
&= \int_0^4 \sqrt{x}\,dx + \int_2^4 (2-x)\,dx \\
&= \left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^4 + \left[2x - \frac{x^2}{2}\right]_2^4 \\
&= \frac{2}{3}(8) - 2 \\
&= \frac{10}{3}.
\end{aligned}$$

Tutorial 8.2.3. 1. Thomas Exercises 5.6 11th ed. pp. 411–413 **or** 12th ed. pp. 351–352: 48, 49, 53, 54, 63, 67, 70, 81, 89, 103, 105.

8.2.4 Area between two Curves: y-axis

We consider the region between the continuous curves $x = f(y)$ and $x = g(y)$ and the horizontal lines $y = a$ and $y = b$ with $f(y)$ further to the right than $g(y)$ for each $y \in [a, b]$.



Using Riemann Sums, we partition $[a, b]$ into n subintervals of equal length Δy. Then for the ith subinterval $[y_{i-1}, y_i]$ the ith horizontal rectangular strip has area

$$R_i = [f(y_i^*) - g(y_i^*)]\Delta y$$

with y_i^* any sample point in $[y_{i-1}, y_i]$. Hence the area between the two curves is the limit as $n \to \infty$ of the sum of all these rectangular strips (Riemann Sum) so that

$$\begin{aligned}\text{Area} &= \lim_{n\to\infty} \sum_{i=1}^{n} R_i \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} [f(y_i^*) - g(y_i^*)]\Delta y \\ &= \int_a^b [f(y) - g(y)]\, dy.\end{aligned}$$

It is easy to remember that if f is the curve on the **RIGHT** and g is the **LEFT** curve then the area between f and g is

$$\text{Area} = \int_a^b [\textbf{RIGHT} - \textbf{LEFT}]\, dy.$$

Worked Example 8.2.6. 1. Find the area of the region bounded by $x = 3 - y^2$ and $x = y + 1$.

Solution. Points of intersection:

$$\begin{aligned}3 - y^2 = y + 1 &\Leftrightarrow y^2 + y - 2 = 0 \\ &\Leftrightarrow (y + 2)(y - 1) = 0 \\ &\Leftrightarrow y = -2 \text{ or } y = 1.\end{aligned}$$

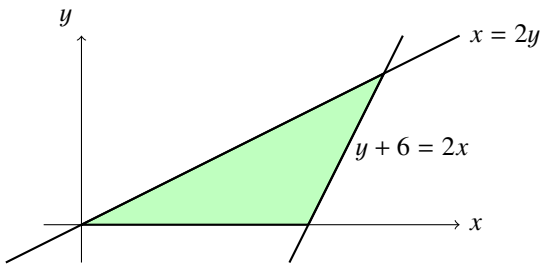
We see that for $y \in [-2, 1]$, $3 - y^2$ is further to the right than $y + 1$ (e.g. take $y = 0$, use the above calculations or

the shape of the graphs). Hence

$$\begin{aligned}
\text{Area} &= \int_{-2}^{1} [\textbf{RIGHT} - \textbf{LEFT}]\, dy \\
&= \int_{-2}^{1} [3 - y^2 - (y+1)]\, dy \\
&= \int_{-2}^{1} [2 - y^2 - y]\, dy \\
&= 2y - \frac{y^3}{3} - \frac{y^2}{2}\Big]_{-2}^{1} \\
&= 2 - \frac{1}{3} - \frac{1}{2} - \left(-4 + \frac{8}{3} - \frac{4}{2}\right) \\
&= 8 - 3 - \frac{1}{2} \\
&= \frac{9}{2}.
\end{aligned}$$

2. Set up the integral with respect to y for the area of the region bounded by $x = 2y$, $y + 6 = 2x$ and the x-axis. Then set up the integral, for the same region, but with respect to x. Do not evaluate the area.

Solution. A sketch of the region is helpful.



(a) With respect to y: Point of intersection:

$$\begin{aligned}
2y = \frac{1}{2}(y+6) &\Leftrightarrow 4y = y + 6 \\
&\Leftrightarrow y = 2.
\end{aligned}$$

Hence the y-axis interval of integration is [0,2]. Since $\frac{1}{2}(y+6) = \textbf{RIGHT}$ and $2y = \textbf{LEFT}$, the integral with respect to y representing the area of the region is

$$\int_0^2 [\textbf{RIGHT} - \textbf{LEFT}]\, dy = \int_0^2 \left[\frac{1}{2}(y+6) - 2y\right] dy = \int_0^2 \left[-\frac{3}{2}y + 3\right] dy.$$

(b) With respect to x: The point of intersection of the two lines is at $x = 4$. Since $y + 6 = 2x$ has x-intercept 3, the integral with respect to x representing the area of the region is

$$\int_0^3 \frac{x}{2}\, dx + \int_3^4 \left[\frac{x}{2} - (2x-6)\right] dx = \int_0^4 \frac{x}{2}\, dx - \int_3^4 (2x-6)\, dx.$$

Tutorial 8.2.4. 1. Thomas Exercises 5.6 11th ed. pp. 411–412 **or** 12th ed. pp. 351–352: 55, 57, 73, 79, 83.

8.2.5 Area with Crossing over Functions

In order to find the area bounded by intersecting curves $y = f(x)$ and $y = g(x)$ in an interval $[a, b]$ we have to distinguish the regions in $[a, b]$ where $f(x) \geq g(x)$ and $g(x) \geq f(x)$. Thus we first find the points of intersection of f and g (equate $f(x) = g(x)$ and solve for $x \in [a, b]$) which will produce finitely many points $x_1, x_2, \ldots, x_n$, say, generating the subintervals $[a, x_1], [x_1, x_2], [x_2, x_3], \ldots, [x_n, b]$. We then have regions corresponding to these subintervals with area A_i in which we can identify the TOP and BOTTOM graphs. Then the area of the required region is

$$A = A_1 + A_2 + \cdots + A_n\,.$$

Since

$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{if } f(x) \geq g(x) \\ g(x) - f(x) & \text{if } g(x) \geq f(x) \end{cases}$$

the area between f and g in $[a, b]$ is

$$A = \int_a^b |f(x) - g(x)|\,dx.$$

We do not explicitly use the above formula, but rather split the region $[a, b]$ into subintervals where we can identify which graph is on TOP and which is on the BOTTOM.

Worked Example 8.2.7.

1. Find the area of the region bounded by the graphs $y = x^2$ and $y = 2 - x^2$ for $0 \leq x \leq 2$.

Solution. Points of intersection: In $[0, 2]$,

$$\begin{aligned} x^2 = 2 - x^2 &\Leftrightarrow 2x^2 - 2 = 0 \\ &\Leftrightarrow x^2 = 1 \\ &\Leftrightarrow x = \pm 1 \end{aligned}$$

Since -1 is outside the interval of interest, this gives the subintervals $[0, 1]$ and $[1, 2]$. The area is then

$$\begin{aligned} A &= \int_0^1 [(2 - x^2) - x^2]\,dx + \int_1^2 [x^2 - (2 - x^2)]\,dx \\ &= \int_0^1 [2 - 2x^2]\,dx + \int_1^2 [2x^2 - 2]\,dx \\ &= \left[2x - \frac{2x^3}{3}\right]_0^1 + \left[\frac{2x^3}{3} - 2x\right]_1^2 \\ &= \left(2 - \frac{2}{3}\right) - (0 - 0) + \left(\frac{16}{3} - 4\right) - \left(\frac{2}{3} - 2\right) \\ &= \frac{4}{3} + \frac{4}{3} + \frac{4}{3} \\ &= 4. \end{aligned}$$

2. Find the area of the region bounded by the graphs $y = \sin x$ and $y = \cos x$ in $[0, 2\pi]$.

Solution. Points of intersection: For $x \in [0, 2\pi]$,

$$\begin{aligned} \sin x = \cos x &\Leftrightarrow \tan x = 1 \\ &\Leftrightarrow x = \arctan 1 + k\pi \qquad (k \in \mathbb{Z}) \\ &\Leftrightarrow x = \frac{\pi}{4} + k\pi \qquad (k \in \mathbb{Z}) \\ &\Leftrightarrow x = \frac{\pi}{4}, \frac{5\pi}{4} \end{aligned}$$

This gives the subintervals $\left[0, \frac{\pi}{4}\right]$, $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ and $\left[\frac{5\pi}{4}, 2\pi\right]$. Taking specific values in these subintervals or a sketch of the graphs will tell us which graph is TOP and which is BOTTOM. In $\left[0, \frac{\pi}{4}\right]$, $\cos x \geq \sin x$. In $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$, $\sin x \geq \cos x$ and in $\left[\frac{5\pi}{4}, 2\pi\right]$, $\cos x \geq \sin x$. Hence

$$\begin{aligned} A &= \int_0^{2\pi} |\cos x - \sin x|\, dx \\ &= \int_0^{\frac{\pi}{4}} (\cos x - \sin x)\, dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\, dx + \int_{\frac{5\pi}{4}}^{2\pi} (\cos x - \sin x)\, dx \\ &= \left[\sin x + \cos x\right]_0^{\frac{\pi}{4}} + \left[-\cos x - \sin x\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}} + \left[\sin x + \cos x\right]_{\frac{5\pi}{4}}^{2\pi} \\ &= 4\sqrt{2}. \end{aligned}$$

Tutorial 8.2.5. 1. Find the area enclosed by the curves $y = x^3$, $y = 3x^2 - 2x$, $x = -1$ and $x = 4$.
2. Thomas Exercises 5.6 11th ed. pp. 411–413 **or** 12th ed. pp. 351–352: 50, 59, 60, 61, 62.
3. Thomas Chapter 5 Practice 11th ed. pp. 416–418 **or** 12th ed. pp. 355–357: 18, 113, 114, 119, 120.

8.3 Solids of Revolution

LEARNING OUTCOMES:

On completion of this section you should be able to

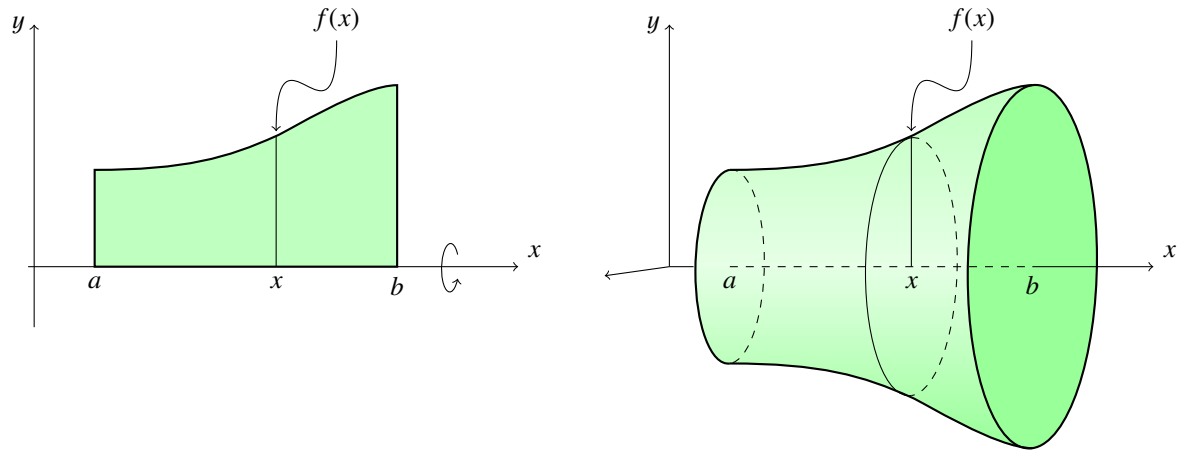
1. understand and be able to use the Disk Method in computing the volume of a solid of revolution about the coordinate axes and about lines of rotation.
2. understand and be able to use the Washer Method in computing the volume of a solid of revolution about the coordinate axes and about lines of rotation.
3. know how to solve the examinable worked out problems from this section.

If a region in the plane is revolved about a line, the resulting solid is called a **solid of revolution**, and the line is called the **axis of revolution**.

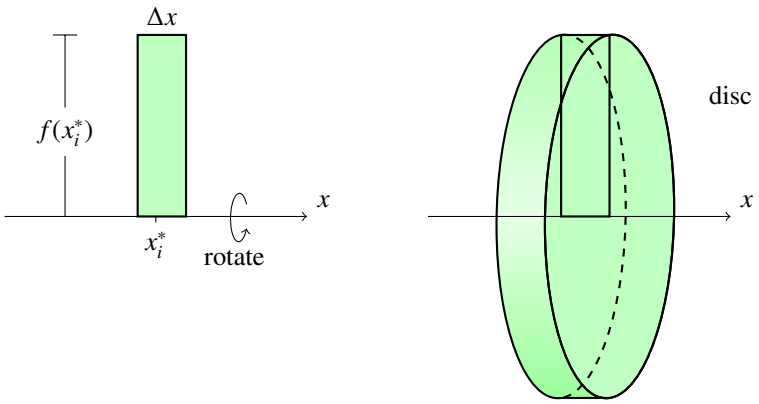
8.3.1 The Disk Method

Rotation about the x-axis

Consider the solid of revolution formed by revolving the plane region below about the x-axis.



To find the volume of this solid of revolution with the **Disk Method** we approximate the region bounded by the curve $y = f(x)$, the x axis and the lines $x = a$ and $x = b$ by rectangles. We consider a representative rectangle in the plane region. Spinning this rectangle once around the x-axis produces a representative disc depicted below.



Rotating the line with length $f(x_i^*)$ about the x-axis results in a circle with radius $f(x_i^*)$. The area of this circle is $A(x_i^*) = \pi[f(x_i^*)]^2$. Therefore, the volume of the representative disc is

$$V_i = A(x_i^*)\Delta x = \pi[f(x_i^*)]^2\Delta x.$$

We may then approximate the volume V of the solid by n such discs of equal width Δx and radius $f(x_i^*)$ so that

$$V \approx \sum_{i=1}^{n} V_i = \sum_{i=1}^{n} \pi[f(x_i^*)]^2\Delta x.$$

By letting $n \to \infty$ we then have the volume of the solid of revolution (by the disk method) as

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} \pi[f(x_i^*)]^2\Delta x = \pi \lim_{n\to\infty} \sum_{i=1}^{n} [f(x_i^*)]^2\Delta x = \pi \int_a^b [f(x)]^2\, dx.$$

Hence

$$V = \pi \int_a^b [f(x)]^2 \, dx.$$

We may replace $f(x)$ by $R(x)$ where (for convenience) $R(x)$ is the **radius** of the disk. Thus

$$V = \pi \int_a^b [R(x)]^2 \, dx.$$

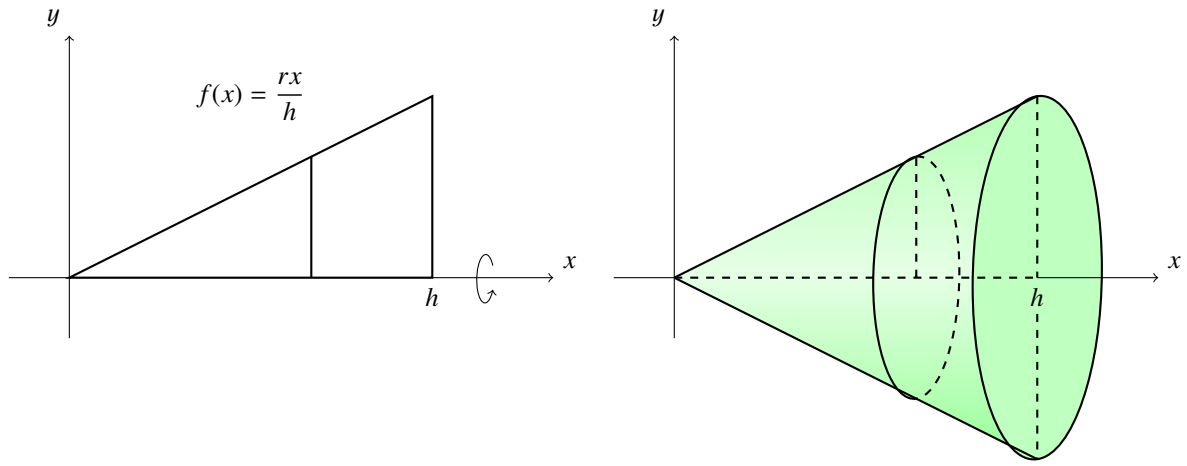
The circle generated by rotating the line with length $f(x)$ about the x-axis is also obtained by taking the **cross-section** of the solid with the plane x = constant, which we call the cross-section of the solid at x. Since the area of this cross-section is $\pi[f(x)]^2$, the above formula for the volume V of the solid of revolution can informally be written as

$$V = \int_a^b [\text{area of the cross-section of the solid at } x] \, dx.$$

Worked Example 8.3.1.

1. (Volume of a cone) The region bounded by the graph of $f(x) = \dfrac{rx}{h}$, $0 \le x \le h$ is revolved about the x-axis. Find the volume of the resulting solid.

The following sketch is for illustration only; it is not needed to do the calculations.

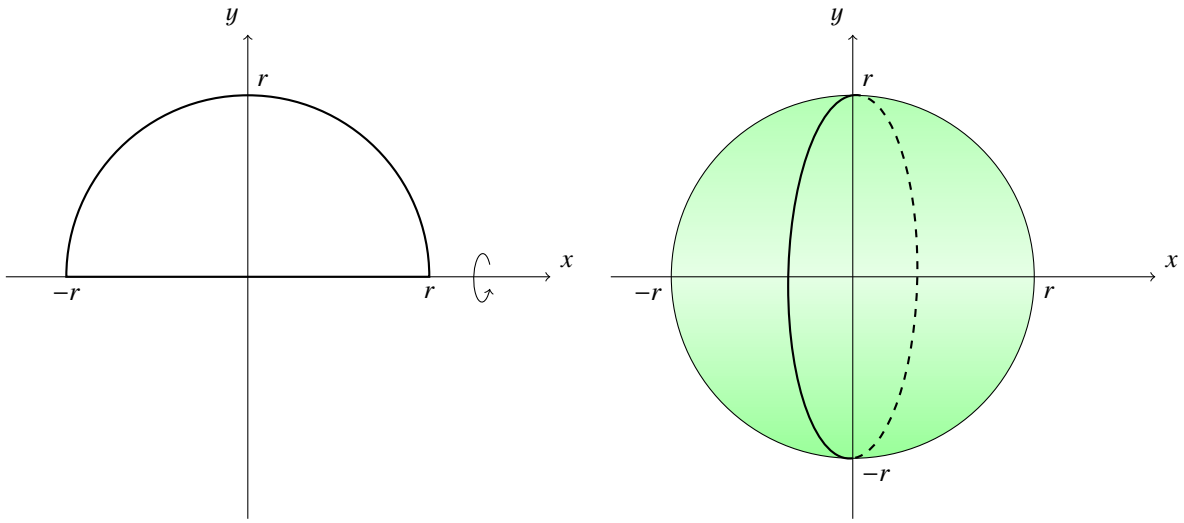


Solution.

$$V_{\text{cone}} = \pi \int_0^h \left[\frac{rx}{h}\right]^2 dx = \pi \int_0^h \frac{r^2x^2}{h^2} \, dx = \frac{\pi r^2}{h^2} \int_0^h x^2 \, dx = \frac{\pi r^2}{h^2} \left[\frac{x^3}{3}\right]_0^h = \frac{\pi r^2}{h^2} \left[\frac{h^3}{3}\right] = \frac{1}{3}\pi r^2 h.$$

2. (Volume of a sphere) The region bounded by the graph of $f(x) = \sqrt{r^2 - x^2}$, $-r \le x \le r$, is revolved about the x-axis. Find the volume of the resulting solid.

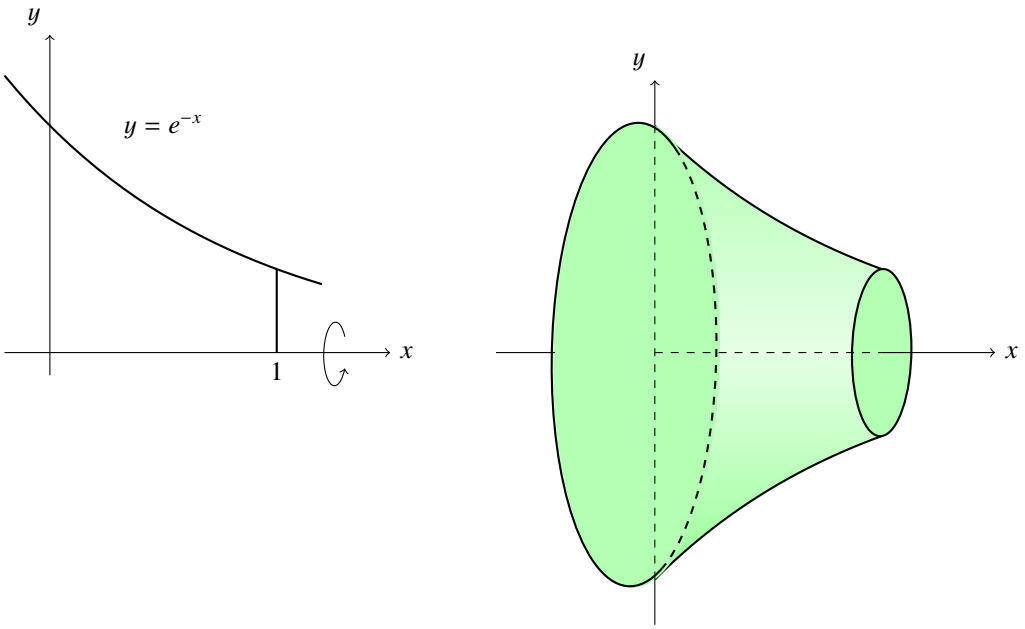
Solution. The following sketch is for illustration only; it is not needed to do the calculations.



$$
\begin{aligned}
V_{\text{sphere}} &= \pi \int_{-r}^{r} \left[\sqrt{r^2 - x^2} \right]^2 dx \\
&= \pi \int_{-r}^{r} (r^2 - x^2)\, dx \\
&= 2\pi \int_{0}^{r} (r^2 - x^2)\, dx \qquad (\because\ r^2 - x^2 \text{ is an even function}) \\
&= 2\pi \left[r^2 x - \frac{x^3}{3} \right]_0^r \\
&= 2\pi \left(r^3 - \frac{r^3}{3} \right) \\
&= 2\pi \left(\frac{2r^3}{3} \right) \\
&= \frac{4}{3}\pi r^3.
\end{aligned}
$$

3. A vuvuzela is constructed by revolving the region bounded by the curve $y = e^{-x}$ between $x = 0$ and $x = 1$ about the x-axis. Find the volume of the vuvuzela.

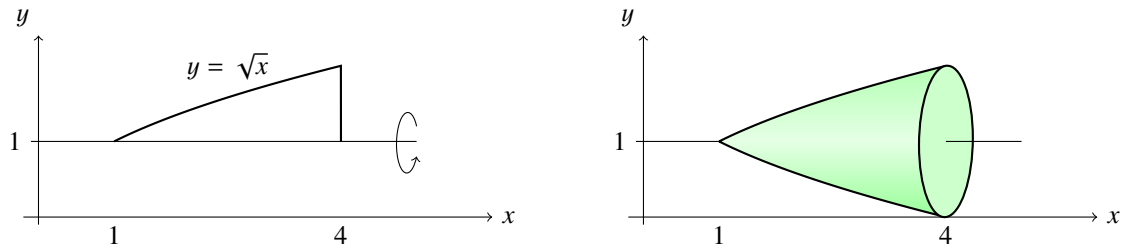
Solution. The following sketch is for illustration only; it is not needed to do the calculations.



$$V = \pi \int_0^1 [e^{-x}]^2 \, dx = \pi \int_0^1 e^{-2x} \, dx = \pi \left[-\frac{e^{-2x}}{2} \right]_0^1 = -\frac{\pi}{2}(e^{-2} - e^0) = \frac{\pi}{2}(1 - e^{-2}) \approx 1.36 \, .$$

4. (Rotation about a line parallel to the x-axis) Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, $y = 1$ and $x = 4$ about the line $y = 1$.

Solution.



$$\begin{aligned}
V &= \pi \int_1^4 [R(x)]^2 \, dx \\
&= \pi \int_1^4 \left[\sqrt{x} - 1 \right]^2 \, dx \\
&= \pi \int_1^4 (x - 2\sqrt{x} + 1) \, dx \\
&= \pi \int_1^4 \left[\frac{x^2}{2} - \frac{4}{3} x^{\frac{3}{2}} + x \right]_1^4 \\
&= \pi \int_1^4 \left[8 - \frac{32}{3} + 4 - \left(\frac{1}{2} - \frac{4}{3} + 1 \right) \right] \\
&= \pi \left[\frac{21}{2} - \frac{28}{3} \right] \\
&= \frac{7}{6} \pi .
\end{aligned}$$

Rotation about the y-axis

To find the volume of the solid generated by revolving the region between the y-axis and a curve $x = f(y)$, $a \leq y \leq b$, about the y-axis we use

$$V = \pi \int_a^b [f(y)]^2 \, dy$$

or taking $R(y)$ as the radius of the disk at y we may write

$$V = \pi \int_a^b [R(y)]^2 \, dy.$$

Again, we may memorize this as

$$V = \int_a^b [\text{area of the cross-section of the solid at } y] \, dy.$$

Worked Example 8.3.2. Find the volume of the solid generated by revolving the region bounded by $x = 6y - y^2$ and the y-axis about the y-axis.

Solution. Since $x = y(6 - y)$, the curve intersects the y-axis, $x = 0$, at $y = 0$ and $y = 6$. Therefore, the volume of

the solid is

$$\begin{aligned} V &= \pi \int_0^6 [R(y)]^2\, dy \\ &= \pi \int_0^6 [6y - y^2]^2\, dy \\ &= \pi \int_0^6 [36y^2 - 12y^3 + y^4]\, dy \\ &= \pi \left[12y^3 - 3y^4 + \frac{y^5}{5} \right]_0^6 \\ &= \pi y^3 \left[12 - 3y + \frac{y^2}{5} \right]_0^6 \\ &= 216\pi \left[12 - 18 + \frac{36}{5} \right] \\ &= \frac{1296\pi}{5}\,. \end{aligned}$$

Tutorial 8.3.1. 1. Thomas Exercises 6.1 11th ed. p. 436 **or** 12th ed. p. 372: 15, 17, 19, 22, 24, 29.

2. The region bounded by the graphs of $y = x^3$, $x = 8$ and the x-axis is rotated about the line $x = 8$. Find the volume of the resulting solid.

3. Find the volume of the solid generated by revolving the regions bounded by the given curves about the the y-axis using the disk method.

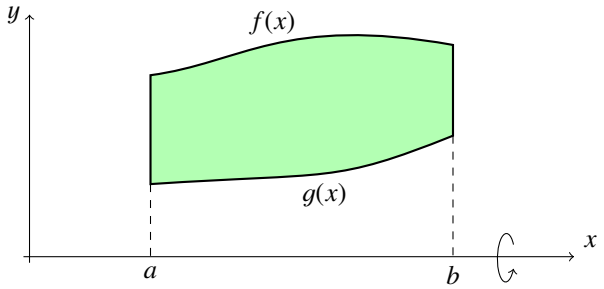
(a) $x^2 + 4y^2 = 4$ (in the first quadrant).

(b) $y = 2\sqrt{x}$, $x = 0$ and $y = 2$.

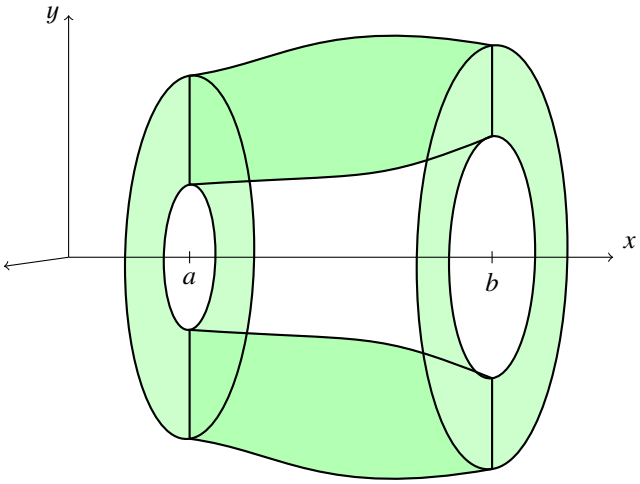
8.3.2 The Washer Method

Rotation about the x-axis

Consider a region bounded by the graphs of $x = a$, $x = b$ and two continuous functions f and g with $f(x) \geq g(x) \geq 0$ for all $x \in [a, b]$.



Revolving this region about the x-axis produces a solid of revolution with a hole through the solid.



The volume V of this solid may be obtained by subtracting the volume of the solid generated by the smaller region from the volume of the solid generated by the larger region (subtraction of two disks) to give

$$V = \pi \int_a^b [f(x)]^2\,dx - \pi \int_a^b [g(x)]^2\,dx = \pi \int_a^b \left\{[f(x)]^2 - [g(x)]^2\right\}\,dx.$$

Note that the cross-section at x is a washer with inner radius $g(x)$ and outer radius $f(x)$. Denoting the inner radius by $r(x)$ and the outer radius by $R(x)$, we may also write

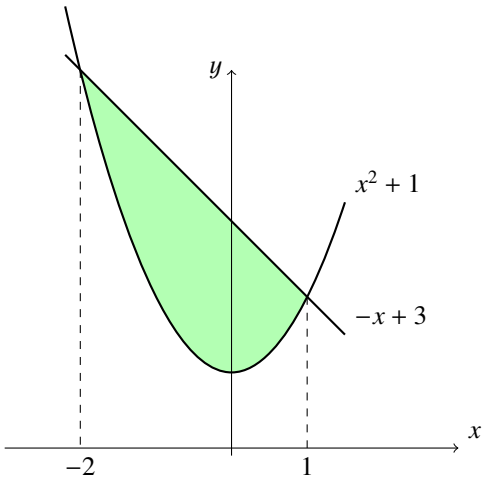
$$V = \pi \int_a^b \left\{[R(x)]^2 - [r(x)]^2\right\}\,dx.$$

Noting that the area of the cross-section washer at x is $\pi[f(x)]^2 - \pi[g(x)]^2$, we may again write

$$V = \int_a^b [\text{area of the cross-section of the solid at } x]\,dx.$$

Worked Example 8.3.3. 1. The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the x-axis to generate a solid. Find the volume of the solid.

Solution.



The x-values of the points of intersection are given by

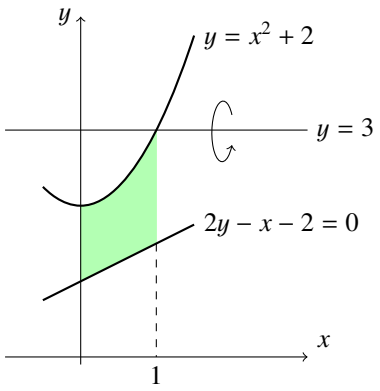
$$x^2 + 1 = -x + 3 \Leftrightarrow x^2 + x - 2 = 0 \Leftrightarrow (x-1)(x+2) = 0 \Leftrightarrow x = -2 \text{ or } x = 1.$$

Thus the interval of integration is $[-2, 1]$. Then

$$\begin{aligned}
V &= \pi \int_{-2}^{1} \left\{ [R(x)]^2 - [r(x)]^2 \right\} dx = \pi \int_{-2}^{1} \left\{ (-x+3)^2 - (x^2+1)^2 \right\} dx \\
&= \pi \int_{-2}^{1} (x^2 - 6x + 9 - x^4 - 2x^2 - 1)\, dx \\
&= \pi \int_{-2}^{1} (8 - 6x - x^2 - x^4)\, dx \\
&= \pi \left[8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5} \right]_{-2}^{1} \\
&= \pi \left[8 - 3 - \frac{1}{3} - \frac{1}{5} + 16 + 12 - \frac{8}{3} - \frac{32}{5} \right] \\
&= \pi \left[33 - \frac{9}{3} - \frac{33}{5} \right] \\
&= \frac{117\pi}{5}.
\end{aligned}$$

2. (Rotation about a line parallel to the x-axis) The region bounded by the graphs $y = x^2 + 2$, $2y - x - 2 = 0$, $x = 0$ and $x = 1$ is revolved about the line $y = 3$. Find the volume of the solid generated.

Solution.



From the sketch or from

$$\frac{x}{2} + 1 < 2 \le x^2 + 2 \le 3 \text{ for } 0 \le x \le 1$$

we obtain that

$$R(x) = 3 - \left(1 + \frac{x}{2}\right) = 2 - \frac{x}{2} \text{ and } r(x) = 3 - (x^2 + 2) = 1 - x^2.$$

Then

$$\begin{aligned}
V &= \pi \int_0^1 \left\{ [R(x)]^2 - [r(x)]^2 \right\} dx = \pi \int_0^1 \left\{ \left(2 - \frac{x}{2}\right)^2 - (1 - x^2)^2 \right\} dx \\
&= \pi \int_0^1 \left(4 - 2x + \frac{x^2}{4} - 1 + 2x^2 - x^4 \right) dx \\
&= \pi \int_0^1 \left(3 - 2x + \frac{9x^2}{4} - x^4 \right) dx \\
&= \pi \left[3x - x^2 + \frac{3x^3}{4} - \frac{x^5}{5} \right]_0^1 \\
&= \pi \left[3 - 1 + \frac{3}{4} - \frac{1}{5} \right] \\
&= \frac{51\pi}{20}.
\end{aligned}$$

Rotation about the y-axis

By interchanging the roles of x and y we can apply the above techniques to solids generated by revolving regions about the y-axis using the washer method. Therefore the volume of the solid is

$$V = \pi \int_a^b \left\{ [R(y)]^2 - [r(y)]^2 \right\} dy.$$

We may again memorize this as

$$V = \int_a^b [\text{area of the cross-section of the solid at } y]\, dy.$$

Worked Example 8.3.4. The region in the first quadrant bounded by $y = \dfrac{x^3}{8}$ and $y = 2x$ is revolved about the y-axis. Find the volume of the resulting solid.

Solution. The x-values of the points of intersection are given by

$$\frac{x^3}{8} = 2x \Leftrightarrow x^3 - 16x = 0 \Leftrightarrow x(x^2 - 16) = 0 \Leftrightarrow x = 0 \text{ or } x = \pm 4.$$

Since the region is in the first quadrant, $x = 0$ and $x = 4$ are the relevant values, with the corresponding y-values

$y = 0$ and $y = 8$, so that the y-interval of integration is $[0, 8]$. Then

$$\begin{aligned}
V &= \pi \int_0^8 \left\{ [R(y)]^2 - [r(y)]^2 \right\} dy \\
&= \pi \int_0^8 \left\{ \left[2y^{\frac{1}{3}} \right]^2 - \left[\frac{y}{2} \right]^2 \right\} dy \\
&= \pi \int_0^8 \left[4y^{\frac{2}{3}} - \frac{y^2}{4} \right] dy \\
&= \pi \left[\frac{12}{5} y^{\frac{5}{3}} - \frac{y^3}{12} \right]_0^8 \\
&= \pi \left[\frac{12}{5} 8^{\frac{5}{3}} - \frac{8^3}{12} \right]_0^8 \\
&= 128\pi \left[\frac{3}{5} - \frac{1}{3} \right]_0^8 \\
&= \frac{512\pi}{15}.
\end{aligned}$$

2. (Rotation about a line parallel to the y-axis) The region bounded by the parabola $y = x^2$ and the line $y = 2x$ in the first quadrant is revolved about the line $x = 2$. Find the volume of the solid generated.

Solution. The x-values of the points of intersection are given by

$$x^2 = 2x \Leftrightarrow x^2 - 2x = 0 \Leftrightarrow x(x-2) = 0 \Leftrightarrow x = 0 \text{ or } x = 2.$$

The y-values of the points of intersection are $y = 0$ and $y = 4$, so that the y-interval of integration is $[0, 4]$. Then

$$\begin{aligned}
V &= \pi \int_0^4 \left\{ [R(y)]^2 - [r(y)]^2 \right\} dy \\
&= \pi \int_0^4 \left\{ \left[2 - \frac{y}{2} \right]^2 - [2 - \sqrt{y}]^2 \right\} dy \\
&= \pi \int_0^4 \left[4 - 2y + \frac{y^2}{4} - 4 + 4\sqrt{y} - y \right] dy \\
&= \pi \left[\frac{y^2}{4} - 3y + 4\sqrt{y} \right]_0^4 \\
&= \pi \left[\frac{y^3}{12} - \frac{3y^2}{2} + \frac{8y^{\frac{3}{2}}}{3} \right]_0^4 \\
&= \pi \left[\frac{64}{12} - \frac{48}{2} + \frac{64}{3} \right] \\
&= \frac{8\pi}{3}.
\end{aligned}$$

Tutorial 8.3.2. 1. Thomas Exercises 6.1 11th ed. pp. 436–437 **or** 12th ed. pp. 372-373: 37, 38, 42, 48, 49, 51.
2. A manufacturer drills a hole through the center of a metal sphere of radius 15 cm. The hole has a radius of 9 cm. What is the volume of the resulting metal ring?
3. Find the volume of the solid formed by revolving the region bounded by the graphs of $y = x^2 + 1$, $y = 0$, $x = 0$ and $x = 1$ about the y-axis.

8.4 Volumes by Slicing

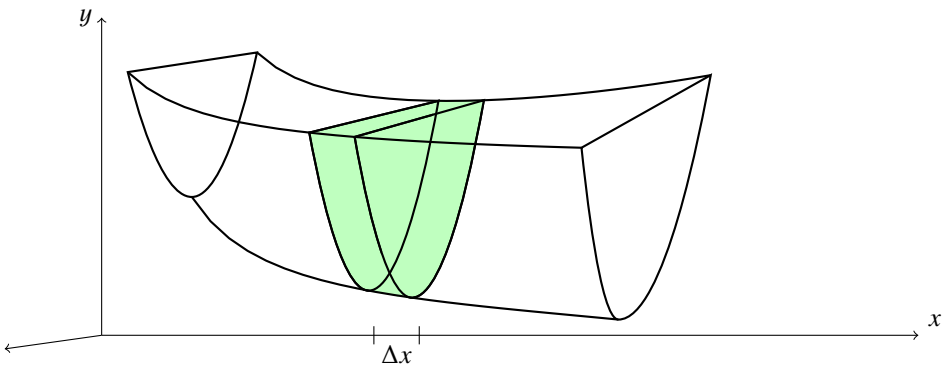
LEARNING OUTCOMES:

On completion of this section you should be able to

1. understand and be able to use the Method of Slicing in computing the volume of a given solid with known cross-section.
2. know how to solve the examinable worked out problems from this section.

If a plane intersects a solid, then the region common to the plane and the solid is called a cross-section of the solid. With the disk method, we found the volume of a solid having a circular cross-section. This method can be generalised to solids of any shape with known cross-sectional area. Common cross-sections are squares, triangles, semicircles and trapezoids.

Therefore consider solids with the property that for each x in a closed interval $[a, b]$ on the x-axis, the plane perpendicular to the x-axis at the point with coordinate x intersects the solid in a cross-section, called cross-section at x, whose area is given by $A(x)$.



Like before, to find the volume of the solid we partition $[a, b]$ into n subintervals with partition points $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ of equal length $\Delta x = x_i - x_{i-1}$ generating slices. The volume of each ith slice is

$$V_i \approx A(x_i^*)\Delta x, \quad x_{i-1} \le x_i^* \le x_i.$$

Consequently the volume of the solid is approximated by the Riemann sum

$$V \approx \sum_{i=1}^{n} A(x_i^*)\Delta x.$$

Hence as $n \to \infty$ we get the precise volume of the solid by slicing

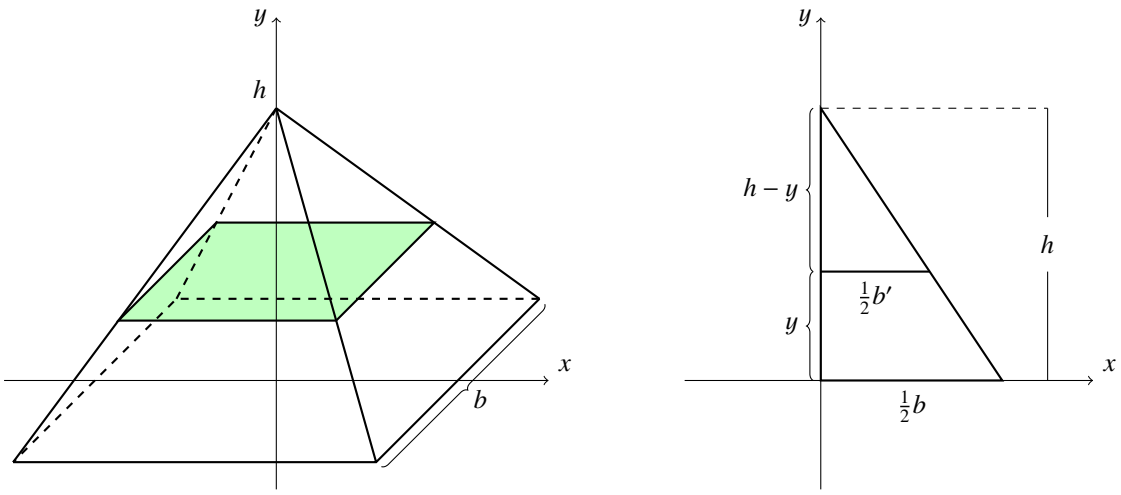
$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\Delta x = \int_a^b A(x)\,dx.$$

We may again memorize this as

$$V = \int_a^b [\text{area of the cross-section of the solid at } x]\,dx.$$

Worked Example 8.4.1. 1. Find the volume of a pyramid that has altitude h and square base of side b.

Solution.



We slice the pyramid with a plane parallel to the base at y to form a square cross-section whose sides are of length b'. Using similar triangles we have

$$\frac{b'}{b} = \frac{h-y}{h}.$$

The slice cross-sectional area is then

$$A(y) = (b')^2 = \frac{b^2(h-y)^2}{h^2}.$$

Hence the volume of the pyramid by slicing is

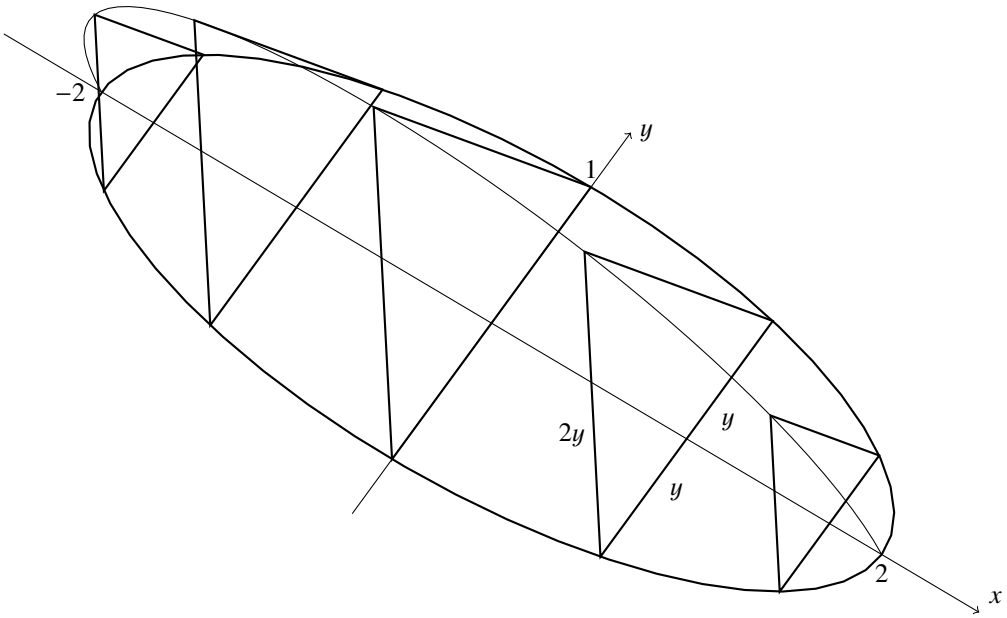
$$\begin{aligned} V &= \int_0^h A(y)\,dy \\ &= \int_0^h \frac{b^2(h-y)^2}{h^2}\,dy \\ &= \frac{b^2}{h^2}\int_0^h (h-y)^2\,dy \\ &= \frac{b^2}{h^2}\left[-\frac{(h-y)^3}{3}\right]_0^h \\ &= \frac{b^2}{h^2}\left[0-\left(-\frac{h^3}{3}\right)\right] \\ &= \frac{1}{3}b^2h. \end{aligned}$$

2. A solid has, as its base, the region bounded by the ellipse

$$\frac{x^2}{4} + y^2 = 1.$$

Find the volume of the solid given that cross-sections perpendicular to the x-axis are equilateral triangles.

Solution.



Let b be the base of the equilateral triangular slice. Since the equation of the ellipse can be written as

$$y^2 = \frac{1}{4}(4 - x^2)$$

we have

$$b = 2|y| = \sqrt{4 - x^2}, \quad h = \frac{\sqrt{3}}{2}b = \frac{\sqrt{3}}{2}\sqrt{4 - x^2}.$$

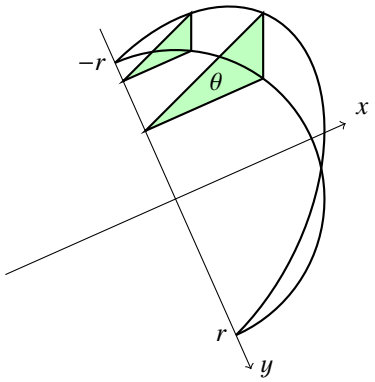
The cross-sectional area of the equilateral triangular slice is then

$$A(x) = \frac{1}{2}bh = \frac{\sqrt{3}}{4}(4 - x^2).$$

Hence, the volume of the solid is

$$\begin{aligned}
V &= \int_{-2}^{2} \frac{\sqrt{3}}{4}(4 - x^2)\,dx \\
&= 2\int_{0}^{2} \frac{\sqrt{3}}{4}(4 - x^2)\,dx \qquad \because A(x) \text{ is even} \\
&= \frac{\sqrt{3}}{2}\int_{0}^{2} (4 - x^2)\,dx \\
&= \frac{\sqrt{3}}{2}\left[4x - \frac{x^3}{3}\right]_0^2 \\
&= \frac{\sqrt{3}}{2}\left[8 - \frac{8}{3}\right] \\
&= \frac{8}{\sqrt{3}}.
\end{aligned}$$

3. A curved wedge is cut from a right circular cylinder of radius r by two planes. One plane is perpendicular to the axis of the cylinder. The second plane makes an acute angle θ with the first and intersects it at the centre of the cylinder. The line of intersection is the y-axis. The line perpendicular to this line and the axis of the cylinder is the x-axis. Find the volume of the wedge where
(a) cross-sections are perpendicular to the y-axis



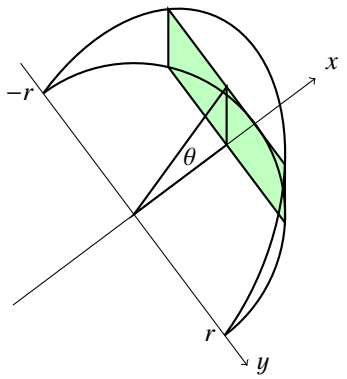
Solution. The cross-section is a triangle of area $A = \frac{1}{2}xh$. Using trigonometry, $h = x\tan\theta$ and by Pythagoras, $x^2 = r^2 - y^2$. Hence

$$A(y) = \frac{1}{2}x(x\tan\theta) = \frac{1}{2}(r^2 - y^2)\tan\theta.$$

Thus the volume of the wedge is

$$\begin{aligned} V &= \int_{-r}^{r} A(y)\,dy \\ &= \frac{1}{2}\tan\theta \int_{-r}^{r} (r^2 - y^2)\,dy \\ &= \tan\theta \int_{0}^{r} (r^2 - y^2)\,dy \\ &= \tan\theta \left[r^2 y - \frac{y^3}{3} \right]_0^r \\ &= \frac{2}{3}r^3 \tan\theta. \end{aligned}$$

(b) cross-sections are perpendicular to the x-axis.



Solution. The cross-sections are rectangles with height $h = x\tan\theta$ and width $2\sqrt{r^2 - x^2}$. Thus the cross-section of a typical rectangular slice is

$$A(x) = (x\tan\theta)(2\sqrt{r^2 - x^2}).$$

Hence the volume is

$$V = \int_0^r A(x)\,dx = 2\tan\theta \int_0^r x\sqrt{r^2 - x^2}\,dx.$$

Let $u = r^2 - x^2$. Then $du = -2x\,dx$. Thus

$$\begin{aligned}\int x\sqrt{r^2 - x^2}\,dx &= -\frac{1}{2}\int \sqrt{u}\,du \\ &= -\frac{1}{2}\left[\frac{2}{3}u^{\frac{3}{2}}\right] + C \\ &= -\frac{1}{3}(r^2 - x^2)^{\frac{3}{2}} + C,\end{aligned}$$

and therefore

$$\begin{aligned}\int_0^r x\sqrt{r^2 - x^2}\,dx &= -\frac{1}{3}(r^2 - x^2)^{\frac{3}{2}}\Big]_0^r \\ &= \frac{1}{3}r^3.\end{aligned}$$

Thus

$$V = 2\tan\theta \int_0^r x\sqrt{r^2 - x^2}\,dx = \frac{2}{3}r^3\tan\theta.$$

Tutorial 8.4.1. 1. Thomas Exercises 6.1 11th ed. p. 435: 4, 5, 10, 13, 14 **or** 12th ed. pp. 371–372: 2, 3, 7, 10, 11, 14.
2. The base of a solid is the region between the parabolas $x = y^2$ and $x = 3 - 2y^2$. Find the volume of the solid given that cross-sections perpendicular to the x-axis are squares.
3. Explain why a stack of coins has the same total volume whether or not the coins are stacked directly on top of one another.

8.5 The Definite Integral as a Function

The definite integral of any continuous function f from $t = a$ to $t = x$ defines a number

$$g(x) = \int_a^x f(t)\,dt$$

that we can treat as a function of x. The next theorem brings out the remarkable fact that the function g defined above is an antiderivative of f which provides the link between the definite integral and antiderivatives.

Theorem 8.6 (Fundamental Theorem of Calculus 1, FTC1)**.** *If f is continuous on $[a, b]$, then the function defined by*

$$g(x) = \int_a^x f(t)\,dt, \quad a < x < b$$

is continuous on $[a, b]$ and differentiable on (a, b) and $g'(x) = f(x)$, i.e.

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

Proof. By the definition of the derivative

$$\begin{aligned} g'(x) &= \lim_{h\to 0} \frac{g(x+h)-g(x)}{h} \\ &= \lim_{h\to 0} \frac{\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left[\int_a^x f(t)\,dt + \int_x^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right] \\ &= \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt. \end{aligned}$$

Now apply the Mean Value Theorem for Integrals to the integral $\int_x^{x+h} f(t)\,dt$. Assuming that $h > 0$ there exists $c \in [x, x+h]$ such that $\int_x^{x+h} f(t)\,dt = f(c)h$. Moreover, since $x \le c \le x+h$, as $h \to 0$, $c \to x$. Thus

$$g'_+(x) = \lim_{h\to 0^+} \frac{1}{h}\int_x^{x+h} f(t)\,dt = \lim_{h\to 0^+} \frac{1}{h}\,(f(c)h) = \lim_{h\to 0^+} f(c) = f(x)$$

where the last identity follows from the continuity of f.
A similar argument can be made for $h < 0$. □

Worked Example 8.5.1. Complete the following first derivatives
1.

$$\frac{d}{dx}\int_\pi^x \sin t\,dt = \sin x.$$

2.

$$\frac{d}{dx}\int_\pi^x \sin t^2\,dt = \sin x^2.$$

Worked Example 8.5.2. Let f be continuous on $[a,b]$ and assume that $u, v : [c,d] \to [a,b]$ are differentiable. Show that

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = f(v(x))v'(x) - f(u(x))u'(x).$$

Proof. Let

$$g(x) = \int_a^x f(t)\,dt.$$

Then g is an antiderivative of f by FTC 1 and therefore, by FTC 2,

$$\int_{u(x)}^{v(x)} f(t)\,dt = g(v(x)) - g(u(x)).$$

Hence, by the chain rule,

$$\begin{aligned}\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt &= \frac{d}{dx}\left[g(v(x)) - g(u(x))\right]\\ &= \frac{d}{dx}g(v(x)) - \frac{d}{dx}g(u(x))\\ &= g'(v(x))v'(x) - g'(u(x))u'(x)\\ &= f(v(x))v'(x) - f(u(x))u'(x).\end{aligned}$$

Worked Example 8.5.3. Evaluate the following derivatives.
1.

$$\frac{d}{dx}\int_0^{x^3} \frac{dt}{1+t}\,.$$

Solution.

$$\frac{d}{dx}\int_0^{x^3} \frac{dt}{1+t} = \frac{1}{1+x^3}(3x^2) = \frac{3x^2}{1+x^3}.$$

2.

$$\frac{d}{dx}\int_x^{2x} \frac{dt}{1+t^2}\,.$$

Solution.

$$\begin{aligned}\frac{d}{dx}\int_x^{2x} \frac{dt}{1+t^2} &= \frac{1}{1+(2x)^2}(2) - \frac{1}{1+x^2}\\ &= \frac{2}{1+4x^2} - \frac{1}{1+x^2}.\end{aligned}$$

Tutorial 8.5.1. 1. Thomas Exercises 5.4 11th ed. pp. 392–393: 35, 38, 48, 61–64 **or** 12th ed. p. 334: 39, 42, 48, 49, 65–68.
2. Find the derivatives of the following functions:
(a) $f(x) = \displaystyle\int_{\tan x}^{\sec x} \frac{1}{1+t^2}\,dt$, (b) $g(x) = \displaystyle\int_{x^2}^{x^3} \sin t^4\,dt$.

Integration Review

$\int e^x \, dx = e^x + C$
$\int a^x \, dx = \frac{a^x}{\ln a} + C$
$\int \ln x \, dx = x \ln x - x + C$
$\int \sin x \, dx = -\cos x + C$
$\int \cos x \, dx = \sin x + C$
$\int \tan x \, dx = \ln \lvert \sec x \rvert + C$
$\int \operatorname{cosec} x \, dx = \ln \lvert \operatorname{cosec} x - \cot x \rvert + C$
$\int \sec x \, dx = \ln \lvert \sec x + \tan x \rvert + C$
$\int \cot x \, dx = \ln \lvert \sin x \rvert + C$
$\int \sec^2 x \, dx = \tan x + C$
$\int \sec x \tan x \, dx = \sec x + C$
$\int \operatorname{cosec}^2 x \, dx = -\cot x + C$

Chapter 9

Advanced Integration Techniques

9.1 Inverse Trigonometric, Exponential and Logarithmic Integrals

LEARNING OUTCOMES:

On completion of this section you should be able to

1. identify an integral as that of an inverse trigonometric type,
2. use the Method of Completing the Square in reducing a quadratic in the integrand and solve such integrals,
3. identify that an integrand is of the logarithmic type where the numerator is the derivative of the denominator and solve such integrals,
4. know how to solve the examinable worked out problems from this section.

This section is a follow up to the derivatives and antiderivatives of inverse trigonometric functions. We recall

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right)+C \quad \text{and} \quad \int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+C.$$

Worked Example 9.1.1. Evaluate

$$\int_0^1 \frac{x+3}{\sqrt{4-x^2}}\,dx.$$

Solution.

$$\begin{aligned}
\int \frac{x+3}{\sqrt{4-x^2}}\,dx &= \int \frac{x}{\sqrt{4-x^2}}\,dx + \int \frac{3}{\sqrt{4-x^2}}\,dx \qquad \text{let } u = \sqrt{4-x^2};\ \text{then } du = \frac{-2x}{2\sqrt{4-x^2}} \\
&= -\int du + 3\arcsin\left(\frac{x}{2}\right) \\
&= -u + 3\arcsin\left(\frac{x}{2}\right)+C \\
&= -\sqrt{4-x^2} + 3\arcsin\left(\frac{x}{2}\right)+C.
\end{aligned}$$

Hence,

$$\begin{aligned}
\int_0^1 \frac{x+3}{\sqrt{4-x^2}}\,dx &= \left[-\sqrt{4-x^2} + 3\arcsin\left(\frac{x}{2}\right)\right]_0^1 \\
&= -\sqrt{3} + 3\arcsin\left(\frac{1}{2}\right) - \left(-\sqrt{4} + 3\arcsin 0\right) \\
&= -\sqrt{3} + 3\frac{\pi}{6} + 2 - 3(0) \\
&= 2 - \sqrt{3} + \frac{\pi}{2}.
\end{aligned}$$

9.1.1 Completing the Square

Given an integrand with a polynomial in the numerator and an irreducible quadratic or the square root of a reducible quadratic with a negative quadratic term in the denominator, we complete the square of the quadratic. The outcome of the integral is an inverse trigonometric function.

Worked Example 9.1.2. Evaluate the following by completing the square in the denominator.
1.

$$\int \frac{dx}{2x^2 - 8x + 10}.$$

Solution.

$$2x^2 - 8x + 10 = 2(x^2 - 4x + 5) = 2[(x-2)^2 - 4 + 5] = 2[(x-2)^2 + 1].$$

Then

$$\int \frac{dx}{2x^2 - 8x + 10} = \frac{1}{2}\int \frac{dx}{1 + (x-2)^2} = \frac{1}{2}\arctan(x-2) + C.$$

2.

$$\int_{\frac{3}{2}}^{\frac{9}{4}} \frac{dx}{\sqrt{3x - x^2}}.$$

Solution.

$$3x - x^2 = -(x^2 - 3x) = -\left[\left(x - \frac{3}{2}\right)^2 - \frac{9}{4}\right] = \frac{9}{4} - \left(x - \frac{3}{2}\right)^2.$$

Then

$$\begin{aligned}\int \frac{dx}{\sqrt{3x - x^2}} &= \int \frac{dx}{\sqrt{\frac{9}{4} - \left(x - \frac{3}{2}\right)^2}} \\ &= \arcsin\left(\frac{x - \frac{3}{2}}{\frac{3}{2}}\right) + C \\ &= \arcsin\left(\frac{2x-3}{3}\right) + C.\end{aligned}$$

Hence,

$$\begin{aligned}\int_{\frac{3}{2}}^{\frac{9}{4}} \frac{dx}{\sqrt{3x - x^2}} &= \arcsin\left(\frac{2x-3}{3}\right)\Big]_{\frac{3}{2}}^{\frac{9}{4}} \\ &= \arcsin\left(\frac{2\left(\frac{9}{4}\right) - 3}{3}\right) - \arcsin\left(\frac{2\left(\frac{3}{2}\right) - 3}{3}\right) \\ &= \arcsin\left(\frac{1}{2}\right) - \arcsin 0 \\ &= \frac{\pi}{6} - 0 \\ &= \frac{\pi}{6}.\end{aligned}$$

9.1.2 Exponential and Logarithmic Integrals

Recall that in the first semester we proved that $\frac{d}{dx}\ln|x| = \frac{1}{x}$ and consequently, $\int \frac{1}{x}\,dx = \ln|x| + C$ (see First Semester Calculus Guidebook Theorem 4.20 page 45 and Theorem 5.2 page 50). We also recall

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C.$$

Worked Example 9.1.3. Evaluate the following integrals.
1.

$$\int \tan x\,dx.$$

Solution.

$$\begin{aligned}\int \tan x\,dx &= \int \frac{\sin x}{\cos x}\,dx \\ &= -\int \frac{-\sin x}{\cos x}\,dx \qquad \text{integrand is of the type } \frac{f'(x)}{f(x)} \\ &= -\ln|\cos x| + C.\end{aligned}$$

Thus

$$\boxed{\int \tan x\,dx = -\ln|\cos x| + C}$$

Also

$$-\ln|\cos x| = \ln\left|\frac{1}{\cos x}\right| = \ln|\sec x|,$$

so that we can also write

$$\boxed{\int \tan x\,dx = \ln|\sec x| + C}$$

2. (a)

$$\int \sec x\,dx.$$

Solution.

$$\begin{aligned}\int \sec x\,dx &= \int \sec x\left(\frac{\sec x + \tan x}{\sec x + \tan x}\right)dx \\ &= \int \frac{\sec x\tan x + \sec^2 x}{\sec x + \tan x}\,dx \qquad \text{integrand is of the type } \frac{f'(x)}{f(x)} \\ &= \ln|\sec x + \tan x| + C.\end{aligned}$$

Hence,

$$\boxed{\int \sec x\,dx = \ln|\sec x + \tan x| + C}$$

(b)

$$\int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\,dx.$$

Solution.

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\,dx &= \int_0^{\frac{\pi}{4}} |\sec x|\,dx \\ &= \int_0^{\frac{\pi}{4}} \sec x\,dx \qquad \because \sec x > 0 \text{ for } x \in \left[0, \frac{\pi}{4}\right] \\ &= \ln|\sec x + \tan x|\Big]_0^{\frac{\pi}{4}} \\ &= \ln\left|\sec\frac{\pi}{4} + \tan\frac{\pi}{4}\right| - \ln|\sec 0 + \tan 0| \\ &= \ln|\sqrt{2}+1| - \ln 1 \\ &= \ln(\sqrt{2}+1).\end{aligned}$$

3.
$$\int \cot x\,dx.$$

Solution.
$$\begin{aligned}\int \cot x\,dx &= \int \frac{\cos x}{\sin x}\,dx \qquad \text{integrand is of the type } \frac{f'(x)}{f(x)}\\ &= \ln|\sin x| + C.\end{aligned}$$

Hence,
$$\boxed{\int \cot x\,dx = \ln|\sin x| + C}$$

4.
$$\int \operatorname{cosec} x\,dx.$$

Solution, Similar to that of $\int \sec x\,dx$. Complete as tutorial.

We will also encounter substitutions (and integrals) involving $\ln x$ as the following examples illustrate.

Worked Example 9.1.4. Evaluate the following integrals.
1.
$$\int_1^e \frac{(1+\ln x)^2}{x}\,dx.$$

Solution. Let $u = 1 + \ln x$. Then $du = \dfrac{dx}{x}$. When $x = 1$, $u = 1 + \ln 1 = 1$ and when $x = e$, $u = 1 + \ln e = 2$. Thus
$$\begin{aligned}\int_1^e \frac{(1+\ln x)^2}{x}\,dx &= \int_1^2 u^2\,du\\ &= \frac{u^3}{3}\Big]_1^2\\ &= \frac{8}{3} - \frac{1}{3}\\ &= \frac{7}{3}.\end{aligned}$$

2.
$$\int \frac{2x}{(x+1)^2}\,dx.$$

Solution. Let $u = x + 1$. Then $x = u - 1$ and $du = dx$. Hence
$$\begin{aligned}\int \frac{2x}{(x+1)^2}\,dx &= \int \frac{2(u-1)}{u^2}\,du\\ &= \int \left[\frac{2}{u} - 2u^{-2}\right] du\\ &= 2\ln|u| + \frac{2}{u} + C\\ &= 2\ln|x+1| + \frac{2}{x+1} + C.\end{aligned}$$

3.
$$\int \frac{2x-1}{x^2-6x+13}\,dx.$$

Solution.

$$\begin{aligned}\int \frac{2x-1}{x^2-6x+13}\,dx &= \int \frac{(2x-6)+5}{x^2-6x+13}\,dx \\ &= \int \frac{2x-6}{x^2-6x+13}\,dx + \int \frac{5}{x^2-6x+13}\,dx \\ &= \ln|x^2-6x+13| + \int \frac{5}{(x-3)^2+4}\,dx \\ &= \ln(x^2-6x+13) + \frac{5}{2}\arctan\left(\frac{x-3}{2}\right) + C.\end{aligned}$$

Recall that $\dfrac{d}{dx}a^x = a^x \ln a$. Then the integrals of the exponential functions are

$$\boxed{\int a^x\,dx = \frac{a^x}{\ln a} + C}$$

and in particular

$$\boxed{\int e^x\,dx = e^x + C}$$

Worked Example 9.1.5. Evaluate the following integrals.
1.

$$\int 5x2^{-x^2}\,dx.$$

Solution. Let $u = -x^2$. Then $du = -2x\,dx$ so that $-\dfrac{du}{2} = x\,dx$ and thus

$$\begin{aligned}\int 5x2^{-x^2}\,dx &= 5\int \left(-\frac{2^u}{2}\right)du \\ &= -\frac{5}{2}\int 2^u\,du \\ &= -\frac{5}{2}\left(\frac{2^u}{\ln 2}\right) + C \\ &= -\frac{(5)2^u}{2\ln 2} + C \\ &= -\frac{5}{2\ln 2}2^{-x^2} + C.\end{aligned}$$

2. Find the volume of the solid obtained by revolving the region bounded by $y = e^x$ and $y = \sqrt{x}$ on $[0,1]$ about the x-axis.
Solution. (Washer Method)

$$\begin{aligned}V &= \pi\int_0^1 \left[(e^x)^2 - \left(\sqrt{x}\right)^2\right]dx \\ &= \pi\int_0^1 \left[e^{2x} - x\right]dx \\ &= \pi\left[\frac{e^{2x}}{2} - \frac{x^2}{2}\right]_0^1 \\ &= \pi\left[\frac{e^2}{2} - \frac{1}{2} - \left(\frac{1}{2} - 0\right)\right] \\ &= \pi\left(\frac{e^2}{2} - 1\right).\end{aligned}$$

Tutorial 9.1.1. 1. Evaluate the following integrals by completing the square, if necessary.
(a)

$$\int_0^2 \frac{dx}{x^2 - 2x + 2}$$

(b)

$$\int \frac{2x}{x^2 + 6x + 13}\, dx$$

(c)

$$\int \frac{x + 2}{\sqrt{-x^2 - 4x}}\, dx$$

(d)

$$\int_2^3 \frac{2x - 3}{\sqrt{4x - x^2}}\, dx$$

2. Thomas Exercises 7.1 11th ed. pp. 506–507: 7, 8, 15, 25, 28, 34, 42 **or** 12th ed. p. 426: 7, 8, 12, 15, 18, 25, 28, 34, 42.

9.2 Integration by Parts Revisited

LEARNING OUTCOMES:

On completion of this section you should be able to

1. use the technique of Integration by Parts in solving integrals involving hyperbolic functions,
2. prove the various reduction formulas using IBP and use them in evaluating a given integral,
3. know how to solve the examinable worked out problems from this section.

We revisit IBP but focus on some additional problems on integration by parts. In particular, we consider the proofs of various reduction formulas that are useful in simplifying certain integrals with integrand having just powers of known functions. We recall the IBP formula

$$\boxed{\int u\, dv = uv - \int v\, du}$$

and

$$\int_a^b u\, dv = uv\Big]_a^b - \int_a^b v\, du.$$

Worked Example 9.2.1. Use IBP to evaluate the following integrals.
1.

$$\int_0^1 x \sinh x\, dx.$$

Solution. Let

$$\begin{array}{lll} u = x & \text{and} & dv = \sinh x\, dx \\ \Rightarrow du = dx & \text{and} & v = \displaystyle\int \sinh x\, dx = \cosh x. \end{array}$$

Then

$$\begin{aligned}\int_0^1 x\sinh x\,dx &= \int_0^1 u\,dv\\ &= uv\Big]_0^1 - \int_0^1 v\,du\\ &= x\cosh x\Big]_0^1 - \int_0^1 \cosh x\,dx\\ &= [(1)\cosh(1) - 0\cosh(0)] - \sinh x\Big]_0^1\\ &= \cosh(1) - [\sinh(1) - \sinh(0)]\\ &= \cosh(1) - \sinh(1)\\ &= e^{-1}\end{aligned}$$

2.

$$I = \int (\ln x)^2\,dx.$$

Solution. Let

$$\begin{aligned} u &= (\ln x)^2 && \text{and} && dv = dx\\ \Rightarrow du &= 2\frac{\ln x}{x}\,dx && \text{and} && v = \int dx = x.\end{aligned}$$

Then

$$\begin{aligned}I &= \int (\ln x)^2\,dx\\ &= \int uv\,dx\\ &= uv - \int v\,du\\ &= x(\ln x)^2 - \int x\left(2\frac{\ln x}{x}\right)dx\\ &= x(\ln x)^2 - 2\int \ln x\,dx\\ &= x(\ln x)^2 - 2(x\ln x - x) + C\\ &= x(\ln x)^2 - 2x\ln x + 2x + C\end{aligned}$$

3. Use the reduction formula

$$\int (\ln x)^n\,dx = x(\ln x)^n - n\int (\ln x)^{n-1}\,dx$$

to evaluate $\int (\ln x)^3\,dx$ together with the solution of example 2 above.

Solution.

$$\begin{aligned}\int (\ln x)^3\,dx &= x(\ln x)^3 - 3\int (\ln x)^2\,dx\\ &= x(\ln x)^3 - 3(x(\ln x)^2 - 2x\ln x + 2x) + C\\ &= x(\ln x)^3 - 3x(\ln x)^2 + 6x\ln x - 6x + C.\end{aligned}$$

4. Prove the reduction formula

$$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2} x\tan x + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx$$

and then use this formula to evaluate $\int \sec^3 x\,dx$.

Proof. Write

$$\int \sec^n x\,dx = \int \sec^{n-2} x\sec^2 x\,dx$$

and let

$$\begin{aligned} u &= \sec^{n-2} x\,dx \quad \text{and} \quad dv = \sec^2 x\,dx \\ \Rightarrow du &= (n-2)\sec^{n-3} x\sec x\tan x\,dx \quad \text{and} \quad v = \int \sec^2 x\,dx = \tan x \\ \Rightarrow du &= (n-2)\sec^{n-2} x\tan x\,dx \quad \text{and} \quad v = \tan x. \end{aligned}$$

Then

$$\begin{aligned} \int \sec^n x\,dx &= \int \sec^{n-2} x\sec^2 x\,dx \\ &= \int u\,dv \\ &= uv - \int v\,du \\ &= \sec^{n-2} x\tan x - \int \tan x\,(n-2)\sec^{n-2} x\tan x\,dx \\ &= \sec^{n-2} x\tan x - (n-2)\int \sec^{n-2} x\tan^2 x\,dx \\ &= \sec^{n-2} x\tan x - (n-2)\int \sec^{n-2} x(\sec^2 x - 1)\,dx \\ &= \sec^{n-2} x\tan x - (n-2)\int \sec^{n-2} x\sec^2 x\,dx + (n-2)\int \sec^{n-2} x\,dx \\ &= \sec^{n-2} x\tan x - (n-2)\int \sec^n x\,dx + (n-2)\int \sec^{n-2} x\,dx. \end{aligned}$$

Therefore

$$\begin{aligned} \int \sec^n x\,dx + (n-2)\int \sec^n x\,dx &= \sec^{n-2} x\tan x + (n-2)\int \sec^{n-2} x\,dx \\ \Rightarrow (n-1)\int \sec^n x\,dx &= \sec^{n-2} x\tan x + (n-2)\int \sec^{n-2} x\,dx \\ \Rightarrow \int \sec^n x\,dx &= \frac{1}{n-1}\sec^{n-2} x\tan x + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx. \end{aligned}$$

Thus

$$\int \sec^3 x\,dx = \frac{1}{2}\sec x\tan x + \frac{1}{2}\int \sec x\,dx = \frac{1}{2}\sec x\tan x + \frac{1}{2}\ln|\sec x + \tan x| + C.$$

All the proofs of the reduction formulas are examinable! You must not learn the proof by rote but rather know the technique in proving the reduction formula as this is easier. You do not need to remember these formulae.

Tutorial 9.2.1. 1. Thomas Exercises 7.4 11th ed. p. 531 **or** 7.3 12th ed. p. 442: 53, 54.
2. Thomas Exercises 11th ed. 8.2 p. 553 **or** 12th ed. p. 459–460: 25–27.
3. Prove the following reduction formulas
(a)

$$\int \cos^n x\,dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

and then evaluate $\int \cos^4 x\,dx$.
(b)

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(b\sin bx + a\cos bx) + C$$

and then evaluate $I = \int e^x \cos x\,dx$.
(c)

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$$

and then evaluate $I = \int \sin(\ln x)\,dx$.
(d)

$$\int (\ln x)^n\,dx = x(lnx)^n - n\int (\ln x)^{n-1}\,dx$$

9.3 Trigonometric Substitutions

LEARNING OUTCOMES:

On completion of this section you should be able to

1. identify that a trigonometric substitution is applicable in reducing the integrand,
2. use the appropriate trigonometric substitution in reducing a definite or indefinite integral,
3. evaluate a trigonometric integral,
4. use a sketch of a representative right angle triangle with angle θ to transform from functions of θ back to functions of x in resolving the integral,
5. incorporate other integration techniques, in particular completing the square of an irreducible quadratic in the integrand and then applying a trigonometric substitution to resolve a given integral,
6. know how to solve the examinable worked out problems from this section.

If the integrand contains one of the expressions $a^2 - x^2$, $a^2 + x^2$, $x^2 - a^2$, $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$ with $a > 0$, we can eliminate the expression by using the appropriate trigonometric substitution listed in the following table.

Expression	**Trigonometric substitution**
$a^2 - x^2$ or $\sqrt{a^2 - x^2}$, $\quad \lvert x\rvert < a$	Let $x = a\sin\theta$, $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$
$a^2 + x^2$ or $\sqrt{a^2 + x^2}$	Let $x = a\tan\theta$, $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$
$x^2 - a^2$ or $\sqrt{x^2 - a^2}$, $\quad \lvert x\rvert > a$	Let $x = a\sec\theta$, $0 \le \theta < \dfrac{\pi}{2}$ or $\pi \le \theta < \dfrac{3\pi}{2}$

Worked Example 9.3.1.
1.

$$\int \frac{dx}{x^2\sqrt{9-x^2}}.$$

Solution. Let $x = 3\sin\theta$, $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$. Then

$$9 - x^2 = 9 - 9\sin^2\theta = 9(1 - \sin^2\theta) = 9\cos^2\theta.$$

Hence

$$\sqrt{9 - x^2} = 3\cos\theta \text{ and } dx = 3\cos\theta\, d\theta$$

since $\cos\theta > 0$ if $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$. Then

$$\begin{aligned}\int \frac{dx}{x^2\sqrt{9-x^2}} &= \int \frac{3\cos\theta}{(9\sin^2\theta)(3\cos\theta)}\, d\theta \\ &= \frac{1}{9}\int \operatorname{cosec}^2\theta\, d\theta \\ &= -\frac{1}{9}\cot\theta + C.\end{aligned}$$

Observe that $x = 3\sin\theta$ gives

$$\cot\theta = \frac{3\cos\theta}{3\sin\theta} = \frac{\sqrt{9-(3\sin\theta)^2}}{3\sin\theta} = \frac{\sqrt{9-x^2}}{x}.$$

Then

$$\int \frac{dx}{x^2\sqrt{9-x^2}} = -\frac{1}{9}\frac{\sqrt{9-x^2}}{x} + C = -\frac{\sqrt{9-x^2}}{9x} + C.$$

2.

$$\int \frac{dx}{2e^{-x} - e^{x}}.$$

Solution.

$$\int \frac{dx}{2e^{-x} - e^{x}} = \int \frac{dx}{e^{-x}(2 - e^{2x})} = \int \frac{e^{x}}{2 - e^{2x}}\, dx.$$

Note that here the domain is $x \neq \dfrac{1}{2}\ln 2$. If $x > \dfrac{1}{2}\ln 2$ then $e^x > \sqrt{2}$. In this case, use the substitution $e^x = \sqrt{2}\sec\theta$. Then

$$e^x\, dx = \sqrt{2}\sec\theta\tan\theta\, d\theta$$

and

$$2 - e^{2x} = 2 - (\sqrt{2}\sec\theta)^2 = 2 - 2\sec^2\theta = -2(\sec^2\theta - 1) = -2\tan^2\theta.$$

Then

$$\begin{aligned}
\int \frac{dx}{2e^{-x} - e^{x}} &= \int \frac{e^{x}}{2 - e^{2x}}\, dx \\
&= \int \frac{\sqrt{2}\sec\theta\tan\theta}{-2\tan^2\theta}\, d\theta \\
&= -\frac{1}{\sqrt{2}} \int \frac{\sec\theta}{\tan\theta}\, d\theta \\
&= -\frac{1}{\sqrt{2}} \int \operatorname{cosec}\theta\, d\theta \\
&= -\frac{1}{\sqrt{2}} \ln|\operatorname{cosec}\theta - \cot\theta| + C \\
&= -\frac{1}{\sqrt{2}} \ln\left|\frac{\sec\theta - 1}{\tan\theta}\right| + C \\
&= -\frac{1}{\sqrt{2}} \ln\left|\frac{\sqrt{2}\sec\theta - \sqrt{2}}{\sqrt{2\tan^2\theta}}\right| + C \\
&= -\frac{1}{\sqrt{2}} \ln\left|\frac{e^{x} - \sqrt{2}}{\sqrt{e^{2x} - 2}}\right| + C \\
&= -\frac{1}{2\sqrt{2}} \ln\left|\frac{(e^{x} - \sqrt{2})^2}{e^{2x} - 2}\right| + C \\
&= \frac{1}{2\sqrt{2}} \ln\left|\frac{e^{x} + \sqrt{2}}{e^{x} - \sqrt{2}}\right| + C.
\end{aligned}$$

If $x < \frac{1}{2}\ln 2$, let $e^{x} = \sqrt{2}\sin\theta$. Then

$$e^{x}\, dx = \sqrt{2}\cos\theta\, d\theta$$

and

$$\begin{aligned}
\int \frac{dx}{2e^{-x} - e^{x}} &= \int \frac{\sqrt{2}\cos\theta}{2(1 - \sin^2\theta)}\, d\theta \\
&= \frac{1}{\sqrt{2}} \int \sec\theta\, d\theta \\
&= \frac{1}{\sqrt{2}} \ln|\sec\theta + \tan\theta| + C \\
&= \frac{1}{\sqrt{2}} \ln\left|\frac{1}{\sqrt{1 - \frac{e^{2x}}{2}}} + \frac{\frac{1}{\sqrt{2}}e^{x}}{\sqrt{1 - \frac{e^{2x}}{2}}}\right| + C \\
&= \frac{1}{\sqrt{2}} \ln\left|\frac{\sqrt{2} + e^{x}}{\sqrt{2 - e^{2x}}}\right| + C \\
&= \frac{1}{2\sqrt{2}} \ln\left|\frac{e^{x} + \sqrt{2}}{e^{x} - \sqrt{2}}\right| + C
\end{aligned}$$

Although we got the solution, it is quite cumbersome, in pa rticular since we have to consider two different cases to get a solution which is represented by the same formula for both cases.

3.
$$\int \frac{dx}{(x^2+1)^2}.$$

Solution. Let $x = \tan\theta$. Then $dx = \sec^2\theta\, d\theta$ and $(x^2+1)^2 = \sec^4\theta$. Then

$$\begin{aligned}
\int \frac{dx}{(x^2+1)^2} &= \int \frac{\sec^2\theta}{\sec^4\theta}\, d\theta \\
&= \int \cos^2\theta\, d\theta \\
&= \frac{1}{2}\int (1+\cos 2\theta)\, d\theta \\
&= \frac{1}{2}\left(\theta + \frac{\sin 2\theta}{2}\right) + C \\
&= \frac{1}{2}\left(\theta + \frac{2\sin\theta\cos\theta}{2}\right) + C \\
&= \frac{1}{2}\left(\theta + \frac{\tan\theta}{\sec^2\theta}\right) + C \\
&= \frac{1}{2}\left(\theta + \frac{\tan\theta}{1+\tan^2\theta}\right) + C \\
&= \frac{1}{2}\arctan x + \frac{x}{2(1+x^2)} + C.
\end{aligned}$$

4.
$$\int \sqrt{x^2-2x}\, dx.$$

Solution. We complete the square $x^2 - 2x = (x-1)^2 - 1$ and let $x - 1 = \sec\theta$. Then $\sqrt{x^2-2x} = \sqrt{(x-1)^2-1} = \tan\theta$ [note that $0 < \theta < \frac{\pi}{2}$ or $\pi < \theta < \frac{3\pi}{2}$] and $dx = \sec\theta\tan\theta\, d\theta$. Then

$$\begin{aligned}
\int \sqrt{x^2-2x}\, dx &= \int \tan\theta\sec\theta\tan\theta\, d\theta \\
&= \int \tan^2\theta\sec\theta\, d\theta \\
&= \int (\sec^2\theta - 1)\sec\theta\, d\theta \\
&= \int \sec^3\theta\, d\theta - \int \sec\theta\, d\theta \\
&\overset{9.2.1.4}{=} \frac{1}{2}\sec\theta\tan\theta + \left(\frac{1}{2}-1\right)\int \sec\theta\, d\theta \\
&= \frac{1}{2}\sec\theta\tan\theta - \frac{1}{2}\ln|\sec\theta+\tan\theta| + C \\
&= \frac{1}{2}(x-1)\sqrt{x^2-2x} - \frac{1}{2}\ln\left|x-1+\sqrt{x^2-2x}\right| + C.
\end{aligned}$$

Tutorial 9.3.1. 1. Thomas Exercises 11th ed. 8.5 p.575: 2, 4, 6, 11 12, 13, 15, 16, 24, 27, 29, 32 **or** 12th ed. 8.3 pp. 470–471: 2, 4, 6, 11, 12, 13, 17, 18, 28, 33, 35, 38, 46, 57.

9.4 Trigonometric Integrals

LEARNING OUTCOMES:

On completion of this section you should be able to

1. identify the type of trigonometric integral that is given and use the appropriate technique to solve the given trigonometric integral,
2. combine previous techniques, in particular the Substitution Rule and IBP, together with the reduction formulas to solve the given trigonometric integral,
3. know how to solve the examinable worked out problems from this section.

9.4.1 Integrals Involving $\sin kx$ and $\cos kx$ with $k \neq 0$

The solution follows by direct substitution. It may be easier to remember the following formulae.

$$\int \sin kx\, dx = -\frac{1}{k}\cos kx + C$$

and

$$\int \cos kx\, dx = \frac{1}{k}\sin kx + C.$$

Worked Example 9.4.1.

1.

$$\int \sin 4x\, dx.$$

Solution. $\int \sin 4x\, dx = -\frac{1}{4}\cos 4x + C.$

2.

$$\int \cos 7x\, dx.$$

Solution. $\int \cos 7x\, dx = \frac{1}{7}\sin 7x + C.$

9.4.2 Powers of $\sin x$ and Powers of $\cos x$

You must revise all the trigonometric identities, in particular

$$\sin^2 x = \frac{1-\cos 2x}{2} \qquad \text{and} \qquad \cos^2 x = \frac{1+\cos 2x}{2}.$$

Let n be any positive integer.
1. To evaluate $\int \cos^n x\, dx$ we proceed as follows
(a) n **is even**: Use the identity $\cos^2 x = \frac{1+\cos 2x}{2}$.
Repeat this procedure until you have odd powers.
(b) n **is odd**: We write

$$\int \cos^n x\, dx = \int \cos^{n-1} x \cos x\, dx$$

and, since $n-1$ is even, we proceed with using the identity $\cos^2 x = 1 - \sin^2 x$ followed by the substitution $u = \sin x$.

Worked Example 9.4.2. Evaluate $\int \cos^6 x\,dx$.
Solution.

$$\begin{aligned}
\int \cos^6 x\,dx &= \int (\cos^2 x)^3\,dx \\
&= \int \left(\frac{1+\cos 2x}{2}\right)^3\,dx \\
&= \frac{1}{8}\int (1+3\cos 2x+3\cos^2 2x+\cos^3 2x)\,dx \qquad \text{Binomial Theorem} \\
&= \frac{1}{8}\int \left(1+3\cos 2x+3\frac{1+\cos 4x}{2}+\cos^3 2x\right)dx \\
&= \frac{5}{16}\int dx+\frac{3}{8}\int \cos 2x\,dx+\frac{3}{16}\int \cos 4x\,dx+\frac{1}{8}\int (1-\sin^2 2x)\cos 2x\,dx \\
&= \frac{5}{16}\int dx+\frac{1}{2}\int \cos 2x\,dx+\frac{3}{16}\int \cos 4x\,dx-\frac{1}{8}\int \sin^2 2x\cos 2x\,dx
\end{aligned}$$

Now let $u=\sin 2x$. Then $du=2\cos 2x\,dx$, and

$$\begin{aligned}
\int \sin^2 2x\cos 2x\,dx &= \frac{1}{2}\int u^2\,du \\
&= \frac{u^3}{6}+C \\
&= \frac{1}{6}\sin^3 2x+C.
\end{aligned}$$

Hence

$$\begin{aligned}
\int \cos^6 x\,dx &= \frac{5}{16}\int dx+\frac{1}{2}\int \cos 2x\,dx+\frac{3}{16}\int \cos 4x\,dx-\frac{1}{8}\int \sin^2 2x\cos 2x\,dx \\
&= \frac{5}{16}x+\frac{1}{4}\sin 2x+\frac{3}{64}\sin 4x-\frac{1}{48}\sin^3 2x+C.
\end{aligned}$$

2. To evaluate $\int \sin^n x\,dx$ we proceed as follows
(a) n **is even**: Use the identity $\sin^2 x=\dfrac{1-\cos 2x}{2}$, which reduces the integral to the case 1 or to direct integration.

Worked Example 9.4.3. Evaluate $\int \sin^4 x\,dx$.
Solution.

$$\begin{aligned}
\int \sin^4 x\,dx &= \int (\sin^2 x)^2\,dx \\
&= \int \left(\frac{1-\cos 2x}{2}\right)^2\,dx \\
&= \frac{1}{4}\int (1-2\cos 2x+\cos^2 2x)\,dx \\
&= \frac{1}{4}\int \left[1-2\cos 2x+\frac{1}{2}(1+\cos 4x)\right]dx \\
&= \frac{3}{8}\int dx-\frac{1}{2}\int \cos 2x\,dx+\frac{1}{8}\int \cos 4x\,dx \\
&= \frac{3}{8}x-\frac{1}{4}\sin 2x+\frac{1}{32}\sin 4x+C.
\end{aligned}$$

(b) n **is odd**: We write

$$\int \sin^n x\,dx=\int \sin^{n-1} x\sin x\,dx.$$

Then $n-1$ is even and we proceed with using the identity $\sin^2 x=1-\cos^2 x$. Then make the substitution $u=\cos x$.

Worked Example 9.4.4. Evaluate $\int \sin^5 x\,dx$.

Solution.

$$\begin{aligned}\int \sin^5 x\,dx &= \int \sin^4 x \sin x\,dx \\ &= \int (\sin^2 x)^2 \sin x\,dx \\ &= \int (1-\cos^2 x)^2 \sin x\,dx.\end{aligned}$$

Now let $u = \cos x$. Then $du = -\sin x\,dx$ and thus

$$\begin{aligned}\int \sin^5 x\,dx &= \int (1-\cos^2 x)^2 \sin x\,dx \\ &= -\int (1-u^2)^2\,du \\ &= -\int (1-2u^2+u^4)\,du \\ &= -\left(u - \frac{2}{3}u^3 + \frac{1}{5}u^5\right) + C \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C.\end{aligned}$$

9.4.3 Products of Powers of $\sin x$ and Powers of $\cos x$

There are two specific cases.

1. $\boxed{\sin^m x \cos^n x\,dx \text{ for } m \text{ odd or } n \text{ odd}}$

Here one of the two powers is odd. Suppose that n is odd. If $n = 1$ then we have easily the solution by direct substitution $u = \sin x$:

$$\int \sin^m x \cos x\,dx = \frac{1}{m+1}\sin^{m+1} x + C. \qquad (*)$$

If $n > 1$ then we write $\cos^n x = \cos^{n-1} x \cos x$. Since $n-1$ is even we may resort to subsection 9.4.2(1b). The integral then takes the form

$$\int (\text{sum of powers of } \sin x)\cos x\,dx$$

which is then broken up into integrals of the form (*).

If m is odd, then resort to subsection 9.4.2(2b) which realises integrals of the form

$$\int (\text{sum of powers of } \cos x)\sin x\,dx$$

which can be broken up into integrals of the type

$$\int \sin x \cos^m x\,dx = -\frac{1}{m+1}\cos^{m+1} x + C$$

Worked Example 9.4.5. 1. Identify the odd power and proceed as described above to evaluate the following.

(a) $\int \sin^2 x \cos^5 x\,dx$.

Solution.

$$\begin{aligned}\int \sin^2 x\cos^5 x\,dx &= \int \sin^2 x\cos^4 x\cos x\,dx\\ &= \int \sin^2 x(\cos^2 x)^2\cos x\,dx\\ &= \int \sin^2 x(1-\sin^2 x)^2\cos x\,dx\\ &= \int (\sin^2 x - 2\sin^4 x + \sin^6 x)\cos x\,dx\\ &= \frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C.\end{aligned}$$

(b) $\displaystyle\int \frac{\sin^3 x}{\sqrt{\cos x}}\,dx.$

Solution. Here write the integral as $\int \sin^3 x(\cos x)^{-\frac{1}{2}}\,dx$ and use the odd power discussion to reduce $\sin^3 x$. This technique is valid even though we only have one integer power.

$$\begin{aligned}\int \frac{\sin^3 x}{\sqrt{\cos x}}\,dx &= \int \sin^3 x(\cos x)^{-\frac{1}{2}}\,dx\\ &= \int \sin^2 x\sin x(\cos x)^{-\frac{1}{2}}\,dx\\ &= \int (1-\cos^2 x)\sin x(\cos x)^{-\frac{1}{2}}\,dx.\end{aligned}$$

Now let $u = \cos x$. Then $du = -\sin x\,dx$ so that

$$\begin{aligned}\int \frac{\sin^3 x}{\sqrt{\cos x}}\,dx &= \int (1-\cos^2 x)\sin x(\cos x)^{-\frac{1}{2}}\,dx\\ &= -\int (1-u^2)u^{-\frac{1}{2}}\,du\\ &= -\int \left(u^{-\frac{1}{2}} - u^{\frac{3}{2}}\right)du\\ &= -\left(2u^{\frac{1}{2}} - \frac{2}{5}u^{\frac{5}{2}}\right) + C\\ &= -2\sqrt{\cos x} + \frac{2}{5}\sqrt{\cos^5 x} + C.\end{aligned}$$

2. $\boxed{\sin^m x\cos^n x\,dx \text{ for } m \text{ even and } n \text{ even}}$

We use the identities

$$\sin x\cos x = \frac{1}{2}\sin 2x \quad \text{or} \quad \sin^2 x = \frac{1-\cos 2x}{2} \quad \text{or} \quad \cos^2 x = \frac{1+\cos 2x}{2}.$$

Worked Example 9.4.6. Evaluate $\displaystyle\int \sin^4 x\cos^4 x\,dx.$

Solution.

$$\begin{aligned}
\int \sin^4 x \cos^4 x\, dx &= \int (\sin x \cos x)^4\, dx \\
&= \int \left[\frac{1}{2}\sin 2x\right]^4\, dx \\
&= \frac{1}{16}\int \sin^4 2x\, dx \\
&= \frac{1}{16}\int (\sin^2 2x)^2\, dx \\
&= \frac{1}{16}\int \left(\frac{1-\cos 4x}{2}\right)^2\, dx \\
&= \frac{1}{64}\int (1 - 2\cos 4x + \cos^2 4x)\, dx \\
&= \frac{1}{64}\int \left[1 - 2\cos 4x + \frac{1}{2}(1+\cos 8x)\right]\, dx \\
&= \frac{1}{128}\int [3 - 4\cos 4x + \cos 8x]\, dx \\
&= \frac{3}{128}x - \frac{1}{128}\sin 4x + \frac{1}{1024}\sin 8x + C.
\end{aligned}$$

9.4.4 Products of sin and cos with Different Arguments

1. To evaluate $\boxed{\int \sin mx \sin nx\, dx}$ use the identity

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x].$$

Worked Example 9.4.7. Evaluate

$$\int_0^{\frac{\pi}{4}} \sin 2x \sin 3x \, dx.$$

Solution.

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} \sin 2x \sin 3x \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos(2-3)x - \cos(2+3)x] \, dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos(-x) - \cos 5x] \, dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos x - \cos 5x] \, dx \\
&= \frac{1}{2} \left[\sin x - \frac{1}{5} \sin 5x \right]_0^{\frac{\pi}{4}} \\
&= \frac{1}{2} \left[\sin \frac{\pi}{4} - \frac{1}{5} \sin \frac{5\pi}{4} - \sin 0 + \frac{1}{5} \sin 0 \right] \\
&= \frac{1}{2} \left[\frac{1}{\sqrt{2}} - \frac{1}{5} \left(-\frac{1}{\sqrt{2}} \right) - 0 + 0 \right] \\
&= \frac{3}{5\sqrt{2}}.
\end{aligned}$$

2. To evaluate $\boxed{\int \sin mx \cos nx \, dx}$ use the identity

$$\sin mx \cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x].$$

Worked Example 9.4.8. Evaluate

$$\int \sin 5x \cos 4x \, dx.$$

Solution.

$$\begin{aligned}
\int \sin 5x \cos 4x \, dx &= \frac{1}{2} \int [\sin(5+4)x + \sin(5-4)]x \, dx \\
&= \frac{1}{2} \int [\sin 9x + \sin x] \, dx \\
&= \frac{1}{2} \left[-\frac{1}{9} \cos 9x - \cos x \right] + C \\
&= -\frac{1}{18} \cos 9x - \frac{1}{2} \cos x + C.
\end{aligned}$$

3. To evaluate $\boxed{\int \cos mx \cos nx \, dx}$ use the identity

$$\cos mx \cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x].$$

Worked Example 9.4.9. Evaluate $\int \cos 5x \cos 2x \, dx$.

Solution.

$$\begin{aligned}\cos 5x \cos 2x\,dx &= \frac{1}{2}\int [\cos(5+2)x + \cos(5-2)x]\,dx \\ &= \frac{1}{2}\int [\cos 7x + \cos 3x]\,dx \\ &= \frac{1}{2}\left[\frac{1}{7}\sin 7x + \frac{1}{3}\cos 3x\right] + C \\ &= \frac{1}{14}\sin 7x + \frac{1}{6}\sin 3x + C\end{aligned}$$

In the above, note the use of subsection 9.4.1.

9.4.5 Powers of $\tan x$ and $\cot x$

Note that students must know that

$$\int \tan x\,dx = \ln|\sec x| + C \quad \text{and} \quad \int \cot x\,dx = \ln|\sin x| + C.$$

1. $\boxed{\tan^n x\,dx}$

We factor out $\tan^2 x$, use the identity $\sec^2 x = 1 + \tan^2 x$ and write

$$\tan^n x\,dx = \int \tan^{n-2} x \tan^2 x\,dx = \int \tan^{n-2} x\,(\sec^2 x - 1)\,dx.$$

This then is separated to give an integral of the following type, which is easily resolved by direct substitution $u = \tan x$:

$$\int \tan^m x \sec^2 x\,dx = \frac{\tan^{m+1} x}{m+1} + C.$$

Worked Example 9.4.10. Evaluate $\int \tan^4 x\,dx$.

Solution.

$$\begin{aligned}\int \tan^4 x\,dx &= \int \tan^2 x \tan^2 x\,dx \\ &= \int \tan^2 x(\sec^2 x - 1)\,dx \\ &= \int \tan^2 x \sec^2 x\,dx - \int \tan^2 x\,dx \\ &= \int \tan^2 x \sec^2 x\,dx - \int (\sec^2 x - 1)\,dx \\ &= \int \tan^2 x \sec^2 x\,dx - (\tan x - x).\end{aligned}$$

For $\int \tan^2 x \sec^2 x\,dx$ let $u = \tan x$. Then $du = \sec^2 x\,dx$ so that

$$\begin{aligned}\int \tan^4 x\,dx &= \int \tan^2 x \sec^2 x\,dx - (\tan x - x) \\ &= \int u^2\,du - \tan x + x \\ &= \frac{1}{3}u^3 - \tan x + x + C \\ &= \frac{1}{3}\tan^3 x - \tan x + x + C.\end{aligned}$$

2. $\boxed{\cot^n x\,dx}$
Similar to above but we use $\operatorname{cosec}^2 x = 1 + \cot^2 x$.

Worked Example 9.4.11. Evaluate $\int \cot^3 x\,dx$.
Solution:

$$\begin{aligned}\int \cot^3 x\,dx &= \int \cot x \cot^2 x\,dx \\ &= \int \cot x(\operatorname{cosec}^2 x - 1)\,dx \\ &= \int \cot x \operatorname{cosec}^2 x\,dx - \int \cot x\,dx \\ &= \int \cot x \operatorname{cosec}^2 x\,dx - \ln|\sin x|.\end{aligned}$$

For $\int \cot x \operatorname{cosec}^2 x\,dx$ let $u = \cot x$. Then $du = -\operatorname{cosec}^2 x\,dx$ so that

$$\begin{aligned}\int \cot^3 x\,dx &= \int \cot x \operatorname{cosec}^2 x\,dx - \ln|\sin x| \\ &= -\int u\,du - \ln|\sin x| \\ &= -\frac{1}{2}u^2 - \ln|\sin x| + C \\ &= -\frac{1}{2}\cot^2 x - \ln|\sin x| + C.\end{aligned}$$

Tutorial 9.4.1. 1. Thomas Exercises 11th ed. 8.4 pp. 569–570: 1, 5, 7, 8, 16, 17, 21, 30, 33, 34 **or** 12th ed. pp. 466–467: 1, 4, 7, 18–22, 27, 28, 33, 46, 51, 52, 56, 57, 60, 62, 63, 64, 67.

9.5 Partial Fraction Decomposition

This section provides a technique that can be used to evaluate rational integrals of the form $\int \frac{p(x)}{q(x)}\,dx$ with $p(x)$ and $q(x)$ polynomials, $q(x) \neq 0$.

LEARNING OUTCOMES:

On completion of this section you should be able to

1. understand the concept of partial fraction decomposition,
2. identify an improper rational function [deg(numerator) ≥ deg(denominator)] and perform Long Division reducing it to a proper rational function,
3. meet the prerequisite requirement of factorizing the denominator of a rational function,
4. identify the type of factors in the denominator of a rational function as either non-repeating linear, linear repeated, irreducible non-repeating quadratic or irreducible repeated quadratic,
5. decompose any rational function into partial fractions using the appropriate outcome of denominator factorization,
6. use the technique of partial fraction decomposition in evaluating an integral of a rational function,
7. know how to solve the examinable worked out problems from this section.

1. Long division first

$$\int \frac{p(x)}{q(x)}\,dx \text{ and } \deg(p(x)) \geq \deg(q(x)) \Rightarrow \textbf{Do Long Division}$$

Worked Example 9.5.1. Evaluate
(a)

$$\int \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1}\,dx$$

Solution. We note that degree(numerator) ≥ degree(denominator). So we MUST perform long division.

$$\begin{array}{r|l} & x - 6 \\ \hline x^2 - 1 & x^3 - 6x^2 + 5x - 3 \\ & x^3 - x \\ & \hline & -6x^2 + 6x - 3 \\ & -6x^2 + 6 \\ & \hline & 6x - 9 \end{array}$$

Hence

$$\begin{aligned} \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1} &= x - 6 + \frac{6x - 9}{x^2 - 1} \\ &= x - 6 + \frac{6x}{x^2 - 1} - \frac{9}{x^2 - 1} \\ \Rightarrow \int \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1}\,dx &= \int (x - 6)\,dx + \int \frac{6x}{x^2 - 1}\,dx - \int \frac{9}{x^2 - 1}\,dx \\ &= \int (x - 6)\,dx + 3\int \frac{2x}{x^2 - 1}\,dx - 9\int \frac{1}{x^2 - 1}\,dx \\ &= \frac{1}{2}x^2 - 6x + 3\ln|x^2 - 1| - 9\int \frac{1}{x^2 - 1}\,dx. \end{aligned}$$

For $\int \frac{1}{x^2 - 1}\,dx$ let $x = \sec\theta$, $x^2 > 1$. Then $x^2 - 1 = \tan^2\theta$ and $dx = \sec\theta\tan\theta\,d\theta$ so that

$$\begin{aligned} \int \frac{1}{x^2 - 1}\,dx &= \int \frac{\sec\theta\tan\theta}{\tan^2\theta}\,d\theta \\ &= \int \frac{\sec\theta}{\tan\theta}\,d\theta \\ &= \int \operatorname{cosec}\theta\,d\theta \\ &= \ln|\operatorname{cosec}\theta - \cot\theta| + C \\ &= \ln\left|\frac{x}{\sqrt{x^2 - 1}} - \frac{1}{\sqrt{x^2 - 1}}\right| + C \\ &= \ln\left|\frac{x - 1}{\sqrt{x^2 - 1}}\right| + C \\ &= \frac{1}{2}\ln\left|\frac{x - 1}{x + 1}\right| + C. \end{aligned}$$

Hence,

$$\int \frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1}\,dx = \frac{1}{2}x^2 - 6x + 3\ln|x^2 - 1| - \frac{9}{2}\ln\left|\frac{x-1}{x+1}\right| + C$$
$$= \frac{1}{2}x^2 - 6x - \frac{3}{2}\ln|x-1| + \frac{15}{2}\ln|x+1| + C.$$

(b)

$$\int \frac{\sqrt{x}}{\sqrt{x}+1}\,dx.$$

Solution. Let $u^2 = x$. Then $2u\,du = dx$ and $\sqrt{x} = u$. Thus

$$\int \frac{\sqrt{x}}{\sqrt{x}+1}\,dx = \int \frac{u}{u+1}(2u)\,du = \int \frac{2u^2}{u+1}\,du.$$

Since degree(numerator) ≥ degree(denominator), we carry out long division.

$$\begin{array}{r|l} & 2u-2 \\ \hline u+1 & 2u^2 \\ & 2u^2+2u \\ & \hline -2u \\ & -2u-2 \\ & \hline 2 \end{array}$$

Then

$$\frac{2u^2}{u+1} = 2u - 2 + \frac{2}{u+1}$$

and thus

$$\int \frac{\sqrt{x}}{\sqrt{x}+1}\,dx = \int \frac{2u^2}{u+1}\,du$$
$$= \int \left(2u - 2 + \frac{2}{u+1}\right)du$$
$$= u^2 - 2u + 2\ln|u+1| + C$$
$$= x - 2\sqrt{x} + 2\ln|\sqrt{x}+1| + C.$$

2. $\boxed{\text{degree}(p(x)) < \text{degree}(q(x))}$

Case I $\boxed{\text{Denominator is a product of distinct linear factors}}$

If

$$\frac{p(x)}{q(x)} = \frac{p(x)}{\prod_{i=1}^{n}(a_i x + b_i)} = \frac{p(x)}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)}$$

where $a_i \neq 0$ for all $i = 1, 2, \ldots, n$ and all numbers $\dfrac{b_i}{a_i}$ are distinct, then each linear factor a_ix+b_i in the denominator gives rise to a term of the form $\dfrac{A_i}{a_ix+b}$ with constant A_i, so that the rational function is decomposed as

$$\frac{p(x)}{q(x)} = \frac{A_1}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \cdots + \frac{A_n}{a_nx+b_n}.$$

This is called the partial fractions decomposition of the rational function $\dfrac{p(x)}{q(x)}$.

Worked Example 9.5.2. Evaluate the following integrals by first decomposing the integrand into partial fractions.
(a)

$$\int \frac{3x+4}{x^2+x-6}\,dx.$$

Solution. Note that degree(numerator) < degree(denominator), so there is no need to do long division. Since

$$\frac{3x+4}{x^2+x-6} = \frac{3x+4}{(x+3)(x-2)}$$

the denominator is a product of *distinct* linear factors so that we decompose the rational function as follows

$$\frac{3x+4}{x^2+x-6} = \frac{A}{x+3} + \frac{B}{x-2}$$

and then solve for the constants A and B in one of two methods:
Method 1 – Substitution:

$$\frac{3x+4}{x^2+x-6} = \frac{A}{x+3} + \frac{B}{x-2} \Rightarrow 3x+4 = A(x-2) + B(x+3).$$

To solve for A, let $x = -3$. Then

$$\begin{aligned} 3x+4 = A(x-2) + B(x+3) &\Rightarrow 3(-3)+4 = A(-3-2) + B(0) \\ &\Rightarrow -5 = -5A \\ &\Rightarrow A = 1. \end{aligned}$$

To solve for B, let $x = 2$. Then

$$\begin{aligned} 3x+4 = A(x-2) + B(x+3) &\Rightarrow 3(2)+4 = A(0) + B(2+3) \\ &\Rightarrow 10 = 5B \\ &\Rightarrow B = 2. \end{aligned}$$

Hence

$$\frac{3x+4}{x^2+x-6} = \frac{1}{x+3} + \frac{2}{x-2}.$$

Thus

$$\begin{aligned} \int \frac{3x+4}{x^2+x-6}\,dx &= \int \left(\frac{1}{x+3} + \frac{2}{x-2}\right) dx \\ &= \int \frac{1}{x+3}\,dx + \int \frac{2}{x-2}\,dx \\ &= \ln|x+3| + 2\ln|x-2| + C. \end{aligned}$$

Method 2 – Equate coefficients:

$$\begin{aligned} \frac{3x+4}{x^2+x-6} = \frac{A}{x+3} + \frac{B}{x-2} &\Rightarrow 3x+4 = A(x-2) + B(x+3) \\ &\Rightarrow 3x+4 = (A+B)x + (-2A+3B). \end{aligned}$$

Equating coefficients gives

$$A + B = 3 \quad \text{and} \quad -2A + 3B = 4.$$

We then solve simultaneously to find A and B. Since $A + B = 3$, $A = 3 - B$. Then

$$\begin{aligned} -2A + 3B = 4 &\Rightarrow -2(3-B) + 3B = 4 \\ &\Rightarrow -6 + 2B + 3B = 4 \\ &\Rightarrow 5B = 10 \\ &\Rightarrow B = 2. \end{aligned}$$

Since $A = 3 - B$, $A = 3 - 2 = 1$ and thus

$$\frac{3x+4}{x^2+x-6} = \frac{1}{x+3} + \frac{2}{x-2}.$$

Now proceed as above.
(b)

$$\int \frac{4x^2+13x-9}{x^3+2x^2-3x}\,dx.$$

Solution. Note that deg(numerator) < deg(denominator), so that there is no need for long division. Then we may continue with factorizing the denominator and decompose accordingly to get

$$\frac{4x^2+13x-9}{x^3+2x^2-3x} = \frac{4x^2+13x-9}{x(x^2+2x-3)} = \frac{4x^2+13x-9}{x(x+3)(x-1)} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-1}.$$

Then clearing off the denominator gives the basic equation

$$4x^2+13x-9 = A(x+3)(x-1) + Bx(x-1) + Cx(x+3).$$

By substitution,

$$\begin{aligned} x = 0 &\Rightarrow -9 = -3A \Rightarrow A = 3 \\ x = 1 &\Rightarrow 8 = 4C \Rightarrow C = 2 \\ x = -3 &\Rightarrow -12 = 12B \Rightarrow B = -1. \end{aligned}$$

Hence the partial fraction decomposition is

$$\frac{4x^2+13x-9}{x^3+2x^2-3x} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-1} = \frac{3}{x} - \frac{1}{x+3} + \frac{2}{x-1}.$$

Therefore

$$\begin{aligned} \int \frac{4x^2+13x-9}{x^3+2x^2-3x}\,dx &= \int \frac{3}{x}\,dx - \int \frac{1}{x+3}\,dx + \int \frac{2}{x-1}\,dx \\ &= 3\ln|x| - \ln|x+3| + 2\ln|x-1| + C. \end{aligned}$$

Note: Using the Method of Equating Coefficients leads to a 3×3 system to solve.

Case II Denominator is a product of repeated linear factors

For each factor of $q(x)$ of the form $(ax+b)^r$ where $r \geq 1$, the partial fraction decomposition contains a sum of r partial fractions of the form

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \frac{A_3}{(ax+b)^3} + \cdots + \frac{A_r}{(ax+b)^r}.$$

Worked Example 9.5.3. Decompose the integrand into partial fractions and then evaluate the integral.
(a)

$$\int \frac{2x^2+3}{x(x-1)^2}\,dx$$

Solution. Since deg(numerator) < deg(denominator), there is no need for long division. We decompose the integrand as follows since it has a repeated linear factor

$$\frac{2x^2+3}{x(x-1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

This leads to

$$2x^2+3 = A(x-1)^2 + Bx(x-1) + Cx.$$

Solving for the three constants by an appropriate substitution gives

$$\begin{aligned} x = 0 &\Rightarrow A = 3 \\ x = 1 &\Rightarrow C = 5 \\ x = 2 &\Rightarrow A + 2B + 2C = 11 \Rightarrow 3 + 2B + 10 = 11 \Rightarrow B = -1 \end{aligned}$$

Hence

$$\frac{2x^2+3}{x(x-1)^2} = \frac{3}{x} - \frac{1}{x-1} + \frac{5}{(x-1)^2}.$$

Therefore

$$\begin{aligned} \int \frac{2x^2+3}{x(x-1)^2}\,dx &= \int \frac{3}{x}\,dx - \int \frac{1}{x-1}\,dx + \int \frac{5}{(x-1)^2}\,dx \\ &= 3\ln|x| - \ln|x-1| - \frac{5}{x-1} + C. \end{aligned}$$

(b)

$$\int \frac{12}{x^4 - x^3 - 2x^2}\,dx$$

Solution.

$$\frac{12}{x^4-x^3-2x^2} = \frac{12}{x^2(x^2-x-2)} = \frac{12}{x^2(x-2)(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} + \frac{D}{x+1}.$$

Then

$$12 = Ax(x-2)(x+1) + B(x-2)(x+1) + Cx^2(x+1) + Dx^2(x-2).$$

Solving for the constants, we obtain

$$\begin{aligned} x = 0 &\Rightarrow -2B = 12 \Rightarrow B = -6 \\ x = -1 &\Rightarrow -3D = 12 \Rightarrow D = -4 \\ x = 2 &\Rightarrow 12C = 12 => C = 1 \\ x = 1 &\Rightarrow -2A - 2B + 2C - D = 12 \\ &\Rightarrow -2A - 2(-6) + 2(1) - (-4) = 12 \Rightarrow -2A = -6 \Rightarrow A = 3. \end{aligned}$$

Thus

$$\frac{12}{x^4-x^3-2x^2} = \frac{3}{x} - \frac{6}{x^2} + \frac{1}{x-2} - \frac{4}{x+1}.$$

Hence,

$$\begin{aligned} \int \frac{12}{x^4-x^3-2x^2}\,dx &= \int \frac{3}{x}\,dx - \int \frac{6}{x^2}\,dx + \int \frac{1}{x-2}\,dx - \int \frac{4}{x+1}\,dx \\ &= 3\ln|x| + \frac{6}{x} + \ln|x-2| - 4\ln|x+1| + C. \end{aligned}$$

(c)

$$\int \frac{3x^3 - 18x^2 + 29x - 4}{(x+1)(x-2)^3}\,dx$$

Solution. We decompose the integrand as

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x+1)(x-2)^3} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2} + \frac{D}{(x-2)^3}\,.$$

Solving for the constants gives

$$\begin{aligned}
x = -1 \Rightarrow A &= \frac{3(-1)^3 - 18(-1)^2 + 29(-1) - 4}{(-1-2)^3} = 2\\
x = 2 \Rightarrow D &= \frac{3(2)^3 - 18(2)^2 + 29(2) - 4}{(2+1)} = 2\\
\cdot x,\ x \to \infty &\Rightarrow 3 = A + B\\
&\Rightarrow B = 3 - A = 1\\
x = 0 &\Rightarrow \frac{-4}{1(-2)^3} = A + \frac{B}{-2} + \frac{C}{(-2)^2} + \frac{D}{(-2)^3}\\
&\Rightarrow C = 2 - 4A + 2B + \frac{D}{2} = 2 - 8 + 2 + 1 = -3.
\end{aligned}$$

Then

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x+1)(x-2)^3} = \frac{2}{x+1} + \frac{1}{x-2} - \frac{3}{(x-2)^2} + \frac{2}{(x-2)^3}\,.$$

Hence

$$\begin{aligned}
\int \frac{3x^3 - 18x^2 + 29x - 4}{(x+1)(x-2)^3}\,dx &= \int \frac{2}{x+1}\,dx + \int \frac{1}{x-2}\,dx - \int \frac{3}{(x-2)^2}\,dx + \int \frac{2}{(x-2)^3}\,dx\\
&= 2\ln|x+1| + \ln|x-2| + \frac{3}{x-2} - \frac{1}{(x-2)^2} + C\\
&= 2\ln|x+1| + \ln|x-2| + \frac{3x-7}{(x-2)^2} + C.
\end{aligned}$$

4. **Case III** Denominator contains non-repeating irreducible quadratics

Every nonconstant polynomial with real coefficients can be written as a product of linear and irreducible quadratic polynomials with real coefficients. Here a quadratic polynomial with real coefficients $ax^2 + bx + c$, $a \neq 0$, is called irreducible if it has no real root. This happens if and only if its discriminant is negative, i. e., $\Delta = b^2 - 4ac < 0$. Such a factor is called non-repeating if the polynomial has no other (irreducible) quadratic factor of the form $d(ax^2 + bx + c)$. Each non-repeating quadratic factor $ax^2 + bx + c$ in the denominator gives rise to a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

in the partial fractions decomposition of the rational function $\dfrac{p(x)}{q(x)}$. Completing the square in the denominator, the integral can be evaluated as

$$\begin{aligned}
\int \frac{2Dx + E}{(x-d)^2 + e^2}\,dx &= D\int \frac{2(x-d)}{(x-d)^2 + e^2}\,dx + (E + 2Dd)\int \frac{1}{(x-d)^2 + e^2}\,dx\\
&= D\ln|(x-d)^2 + e^2| + \frac{E + 2Dd}{e}\arctan\left(\frac{x-d}{e}\right) + C.
\end{aligned}$$

Worked Example 9.5.4. Decompose the integrand into partial fractions and then evaluate the integral.
(a)

$$\int \frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)}\, dx$$

Solution. We factorize the denominator and decompose accordingly, taking into account Cases I and II.

$$\frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)} = \frac{2x^3 - 4x - 8}{x(x - 1)(x^2 + 4)} = \frac{A}{x} + \frac{B}{x - 1} + \frac{Cx + D}{x^2 + 4},$$

which gives

$$2x^3 - 4x - 8 = A(x - 1)(x^2 + 4) + Bx(x^2 + 4) + (Cx + D)[x(x - 1)].$$

Then

$$x = 0 \Rightarrow -4A = -8 \Rightarrow A = 2$$

$$x = 1 \Rightarrow 2 - 4 - 8 = 5B \Rightarrow B = -2$$

$$\begin{aligned} \cdot x,\ x \to \infty &\Rightarrow 2 = A + B + C \\ &\Rightarrow C = 2 \end{aligned}$$

$$\begin{aligned} x = -1 &\Rightarrow -2 + 4 - 8 = -10A - 5B + 2(D - C) \\ &\Rightarrow -6 = -10(2) - 5(-2) + 2D - 4 \\ &\Rightarrow 8 = 2D \\ &\Rightarrow D = 4 \end{aligned}$$

Thus

$$\frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)} = \frac{2}{x} - \frac{2}{x - 1} + \frac{2x + 4}{x^2 + 4}.$$

Then

$$\begin{aligned} \int \frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)}\, dx &= \int \frac{2}{x}\, dx - \int \frac{2}{x - 1}\, dx + \int \frac{2x}{x^2 + 4}\, dx + \int \frac{4}{x^2 + 4}\, dx \\ &= 2\ln|x| - 2\ln|x - 1| + \ln(x^2 + 4) + 2\arctan\left(\frac{x}{2}\right) + C. \end{aligned}$$

(b)

$$\int \frac{\sqrt[3]{x + 1}}{x}\, dx.$$

Solution. Let $u^3 = x + 1$. Then $3u^2\, du = dx$ and thus

$$\int \frac{\sqrt[3]{x + 1}}{x}\, dx = \int \frac{u}{u^3 - 1}(3u^2)\, du = \int \frac{3u^3}{u^3 - 1}\, du.$$

In the last integral involving u's deg(numerator) = deg(denominator) so that we have to perform long division first. Obviously,

$$\frac{3u^3}{u^3 - 1} = \frac{3(u^3 - 1) + 3}{u^3 - 1} = 3 + \frac{3}{u^3 - 1}.$$

Now

$$\frac{3}{u^3 - 1} = \frac{3}{(u - 1)(u^2 + u + 1)} = \frac{A}{u - 1} + \frac{Bu + C}{u^2 + u + 1}.$$

Then

$$3 = A(u^2 + u + 1) + (Bu + C)(u - 1).$$

Solving for the constants:

$$\begin{aligned} u = 1 &\Rightarrow 3 = 3A \Rightarrow A = 1 \\ u = 0 &\Rightarrow 3 = A - C \Rightarrow 3 = 1 - C \Rightarrow C = -2 \\ u = -1 &\Rightarrow 3 = A + 2B - 2C \Rightarrow 3 = 1 + 2B + 4 \Rightarrow B = -1. \end{aligned}$$

Consequently,

$$\frac{3u^3}{u^3-1} = 3 + \frac{1}{u-1} - \frac{u+2}{u^2+u+1}.$$

Hence,

$$\begin{aligned} \int \frac{\sqrt[3]{x+1}}{x}\,dx &= \int \frac{3u^3}{u^3-1}\,du \\ &= \int 3\,du + \int \frac{1}{u-1}du - \int \frac{u+2}{u^2+u+1}\,du \\ &= 3u + \ln|u-1| - \frac{1}{2}\int \frac{2u+1}{u^2+u+1}\,du - \frac{3}{2}\int \frac{1}{u^2+u+1}\,du \\ &= 3u + \ln|u-1| - \frac{1}{2}\ln(u^2+u+1) - \frac{3}{2}\int \frac{1}{\left(u+\frac{1}{2}\right)^2+\frac{3}{4}}\,du \\ &= 3u + \ln|u-1| - \frac{1}{2}(\ln|u^3-1| - \ln|u-1|) - \frac{3}{2}\,\frac{2}{\sqrt{3}}\arctan\left(\frac{u+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C \\ &= 3u + \frac{3}{2}\ln|u-1| - \frac{1}{2}\ln|u^3-1| - \sqrt{3}\arctan\left(\frac{2u+1}{\sqrt{3}}\right) + C \\ &= 3\sqrt[3]{x+1} + \frac{3}{2}\ln|\sqrt[3]{x+1}-1| - \frac{1}{2}\ln|x| - \sqrt{3}\arctan\left(\frac{2\sqrt[3]{x+1}+1}{\sqrt{3}}\right) + C. \end{aligned}$$

5. Case IV Denominator contains repeated irreducible quadratics

Each irreducible quadratic factor $(ax^2+bx+c)^r$, $\Delta < 0$, gives rise to an expression of the form

$$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \frac{A_3x+B_3}{(ax^2+bx+c)^3} + \cdots + \frac{A_rx+B_r}{(ax^2+bx+c)^r}\,.$$

Worked Example 9.5.5. Decompose the integrand into partial fractions and then evaluate

$$\int \frac{8x^3+13x}{(x^2+2)^2}\,dx.$$

Solution.

$$\frac{8x^3+13x}{(x^2+2)^2} = \frac{Ax+B}{x^2+2} + \frac{Cx+D}{(x^2+2)^2}\,.$$

This then gives

$$8x^3+13x = (Ax+B)(x^2+2) + Cx + D.$$

Then

$$\begin{aligned} 8x^3+13x &= (Ax+B)(x^2+2) + Cx + D \\ &= Ax^3 + 2Ax + Bx^2 + 2B + Cx + D \\ &= Ax^3 + Bx^2 + (2A+C)x + (2B+D). \end{aligned}$$

Equating coefficients gives

$$\begin{aligned} A &= 8 \\ B &= 0 \\ 2A + C = 13 &\Rightarrow C = 13 - 2A = 13 - 2(8) = -3 \\ 2B + D = 0 &\Rightarrow D = 0. \end{aligned}$$

Thus

$$\frac{8x^3+13x}{(x^2+2)^2} = \frac{8x}{x^2+2} - \frac{3x}{(x^2+2)^2}\,.$$

Hence,

$$\begin{aligned}\int \frac{8x^3+13x}{(x^2+2)^2}\,dx &= \int \frac{8x}{x^2+2}\,dx - \int \frac{3x}{(x^2+2)^2}\,dx \\ &= 4\int \frac{2x}{x^2+2}\,dx - \frac{3}{2}\int \frac{2x}{(x^2+2)^2}\,dx \\ &= 4\ln(x^2+2) + \frac{3}{2}\frac{1}{x^2+2} + C.\end{aligned}$$

Worked Example 9.5.6. Give the partial fraction decomposition of the following:

(a) $\dfrac{3x^4+x^3+20x^2+3x+31}{(x^2+4)^2(x+1)}$,

Solution.

$$\frac{3x^4+x^3+20x^2+3x+31}{(x^2+4)^2(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4} + \frac{Dx+E}{(x^2+4)^2}.$$

(b) $\dfrac{x^6-x^3+1}{x^4+9x^2}$.

Solution. One may be tempted to write

$$\frac{x^6-x^3+1}{x^4+9x^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+9}\,.$$

However, this is incorrect since deg(numerator) > deg(denominator) so that we have to do long division first! Consequently,

$$\begin{aligned}\frac{x^6-x^3+1}{x^4+9x^2} &= x^2 - 9 + \frac{-x^3+81x^2+1}{x^4+9x^2} \\ &= x^2 - 9 + \frac{-x^3+81x^2+1}{x^2(x^2+9)} = x^2 - 9 + \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+9}.\end{aligned}$$

Students have the following table of decompositions in their Study Guide with the middle and third columns blank. Complete in class (on a transparency).

Worked Example 9.5.7. In the following examples complete the partial fraction decomposition in terms of variables $(A, B, C, D, \dots)$ in column 2, then solve for these variables and write the final decomposition in column 3. Thereafter find the integral with integrand column 3.

	$\frac{P(x)}{Q(x)}$	Partial fraction decomposition	Solution
1.	$\frac{2x-1}{(x+1)(3x+2)}$	$\frac{A}{x+1}+\frac{B}{3x+2}$	$\frac{3}{x+1}-\frac{7}{3x+2}$
2.	$\frac{x(x+1)}{(x-1)(x-2)}$	$1+\frac{A}{x-1}+\frac{B}{x-2}$	$1-\frac{2}{x-1}+\frac{6}{x-2}$
3.	$\frac{3}{x(3x-1)^2}$	$\frac{A}{x}+\frac{B}{3x-1}+\frac{C}{(3x-1)^2}$	$\frac{3}{x}-\frac{9}{3x-1}+\frac{9}{(3x-1)^2}$
4.	$\frac{1}{x^3(x+2)}$	$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{D}{x+2}$	$\frac{1}{8x}-\frac{1}{4x^2}+\frac{1}{2x^3}-\frac{1}{8(x+2)}$
5.	$\frac{2}{(x-1)(x^2+1)}$	$\frac{A}{x-1}+\frac{Bx+C}{x^2+1}$	$\frac{1}{x-1}-\frac{x+1}{x^2+1}$
6.	$\frac{x^2-x-1}{x^3-8}$	$\frac{A}{x-2}+\frac{Bx+C}{x^2+2x+4}$	$\frac{1}{12(x-2)}+\frac{11x+8}{12(x^2+2x+4)}$
7.	$\frac{x^3+1}{(x^2+1)^2}$	$\frac{Ax+B}{x^2+1}+\frac{Cx+D}{(x^2+1)^2}$	$\frac{x}{x^2+1}+\frac{-x+1}{(x^2+1)^2}$

Tutorial 9.5.1. 2. Thomas Exercises 11th ed. 8.3 pp. 563-564: 1, 2, 3, 4, 6, 7, 8, 11, 12, 16, 17, 20, 21, 22, 24, 26, 32, 35, 38 **or** 12th ed. 8.4 pp. 479–480: 1, 2, 3, 4, 6, 7, 8, 11, 12, 16, 17, 20, 21, 22, 24, 26, 36, 39, 42, 46.

Chapter 10

Improper Integrals

LEARNING OUTCOMES:

On completion of this section you should be able to

1. understand the meaning of an improper integral,
2. identify when an integral is improper,
3. evaluate an improper integral,
4. use L'Hôpital's Rule in evaluating an improper integral,
5. know when an improper integral converges or diverges,
6. know how to solve the examinable worked out problems from this section.

Recall that the definition of the definite integral on $[a, b]$ in terms of Riemann Sums

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

requires the interval $[a, b]$ to be finite. Furthermore, the FTC requires that f be continuous. In this chapter integrals are considered which fail to meet these two requirements.

The integral $\int_a^b f(x)\,dx$ is called improper if

1. one or both of the limits of integration are infinite or
2. f has a finite number of infinite discontinuities in the interval $[a, b]$.

Note that f has an infinite discontinuity at c if

$$\lim_{x\to c^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x\to c^+} f(x) = \pm\infty.$$

10.1 Type I: Infinite Interval of Integration

Definition of Improper Integrals with Infinite Integration Limits: Type I

1. If f is continuous on $[a, \infty)$, then $\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$.

2. If f is continuous on $(-\infty, b]$, then $\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx$.

3. If f is continuous on $(-\infty, \infty)$, then $\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^t f(x)\,dx + \int_t^\infty f(x)\,dx$ for any $t \in \mathbb{R}$.

In the first two cases, the improper integral **converges** if the limit exists (and is finite). In this case $\lim_{t\to\infty} \int_a^t f(x)\,dx = L$ or $\lim_{t\to\infty} \int_t^b f(x)\,dx = M$. Then:

$\int_a^\infty f(x)\,dx$ *converges* to L, and we write $\int_a^\infty f(x)\,dx = L$;

$\int_{-\infty}^b f(x)\,dx$ *converges* to M, and we write $\int_{-\infty}^b f(x)\,dx = M$.

If the limits do not exist, then the improper integrals **diverge**.

In the third case, the improper integral $\int_{-\infty}^\infty f(x)\,dx$ diverges if either $\int_{-\infty}^t f(x)\,dx$ or $\int_t^\infty f(x)\,dx$ diverges.

If both $\int_{-\infty}^t f(x)\,dx$ and $\int_t^\infty f(x)\,dx$ converge, then $\int_{-\infty}^\infty f(x)\,dx$ is convergent.

Worked Example 10.1.1. (Type I)

Determine whether the following integrals are convergent or divergent.

1.

$$\int_2^\infty \frac{1}{(x-1)^2}\,dx.$$

Solution.

$$\begin{aligned}\int_2^\infty \frac{1}{(x-1)^2}\,dx &= \lim_{t\to\infty}\int_2^t \frac{1}{(x-1)^2}\,dx\\ &= \lim_{t\to\infty}\left[-\frac{1}{x-1}\right]_2^t\\ &= \lim_{t\to\infty}\left[-\frac{1}{t-1}-\left(-\frac{1}{2-1}\right)\right]\\ &= \lim_{t\to\infty}\left[-\frac{1}{t-1}+1\right]\\ &= -\lim_{t\to\infty}\frac{1}{t-1}+\lim_{t\to\infty}1\\ &= 0+1\\ &= 1.\end{aligned}$$

Hence $\displaystyle\int_2^\infty \frac{1}{(x-1)^2}\,dx$ converges and $\displaystyle\int_2^\infty \frac{1}{(x-1)^2}\,dx = 1$.

2.

$$\int_2^\infty \frac{1}{x-1}\,dx.$$

Solution.

$$\begin{aligned}\int_2^\infty \frac{1}{x-1}\,dx &= \lim_{t\to\infty}\int_2^t \frac{1}{x-1}\,dx\\ &= \lim_{t\to\infty}\ln|x-1|\Big]_2^t\\ &= \lim_{t\to\infty}\left[\ln|t-1|-\ln|2-1|\right]\\ &= \lim_{t\to\infty}\ln|t-1|\\ &= \infty.\end{aligned}$$

Thus $\displaystyle\int_2^\infty \frac{1}{x-1}\,dx$ diverges.

Remark: If the region under the graph of $f(x) = \dfrac{1}{x-1}$ from 2 to t is revolved about the x-axis, then the volume of the solid generated (Gabriel's Horn) is (by the Disk Method)

$$V = \pi\int_2^t \frac{1}{(x-1)^2}\,dx.$$

The improper integral

$$\pi\int_2^\infty \frac{1}{(x-1)^2}\,dx$$

may be regarded as the volume of the unbounded solid obtained by revolving, about the x-axis, the graph of $y = \dfrac{1}{x-1}$ for $x \geq 2$. By Worked Example 1 above

$$\pi\int_2^\infty \frac{1}{(x-1)^2}\,dx = \pi(1) = \pi.$$

This gives us a rather curious fact that although the area of the region is infinite, the volume of the solid of revolution it generates is finite!

3.

$$\int_{-\infty}^{0} xe^x\,dx.$$

Solution: Consider the indefinite integral $\int xe^x\,dx$. Let

$$\begin{aligned} u = x \quad &\text{and} \quad dv = e^x\,dx \\ \Rightarrow du = dx \quad &\text{and} \quad v = \int e^x\,dx = e^x. \end{aligned}$$

Thus

$$\int xe^x\,dx = \int u\,dv \overset{IBP}{=} uv - \int v\,du = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Hence

$$\begin{aligned} \int_{-\infty}^{0} xe^x\,dx &= \lim_{t\to\infty} \int_{t}^{0} xe^x\,dx \\ &= \lim_{t\to-\infty} \Big[e^x(x-1) \Big]_t^0 \\ &= \lim_{t\to-\infty} \Big[e^0(0-1) - e^t(t-1) \Big] \\ &= \lim_{t\to-\infty} \Big[-1 + e^t(1-t) \Big] \\ &= -1 + \lim_{t\to-\infty} \frac{1-t}{e^{-t}} \qquad \text{“}\frac{\infty}{\infty}\text{”} \\ &\overset{\text{l'H}}{=} -1 + \lim_{t\to-\infty} \frac{1}{e^{-t}} \\ &= -1 + 0 \end{aligned}$$

since $e^{-t} \to \infty$ as $t \to -\infty$ gives $\dfrac{1}{e^{-t}} \to 0$.

Hence $\displaystyle\int_{-\infty}^{0} xe^x\,dx$ converges and $\displaystyle\int_{-\infty}^{0} xe^x\,dx = -1$.

4.

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx.$$

Solution. Write

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx = \int_{-\infty}^{0} \frac{e^x}{1+e^{2x}}\,dx + \int_{0}^{\infty} \frac{e^x}{1+e^{2x}}\,dx.$$

Evaluate each of the right improper integrals separately:

$$\begin{aligned}
\int_{-\infty}^{0} \frac{e^x}{1+e^{2x}}\,dx &= \lim_{t\to-\infty} \int_{t}^{0} \frac{e^x}{1+e^{2x}}\,dx \\
&= \lim_{t\to-\infty} \arctan e^x \Big]_t^0 \\
&= \lim_{t\to-\infty} \left(\arctan e^0 - \arctan e^t\right) \\
&= \lim_{t\to-\infty} \left(\arctan 1 - \arctan e^t\right) \\
&= \lim_{t\to-\infty} \left(\frac{\pi}{4} - \arctan e^t\right) \\
&= \frac{\pi}{4} - 0 \qquad\qquad (\because t \to -\infty \Rightarrow e^t \to 0 \Rightarrow \arctan e^t \to \arctan(0) = 0) \\
&= \frac{\pi}{4}
\end{aligned}$$

and

$$\begin{aligned}
\int_{0}^{\infty} \frac{e^x}{1+e^{2x}}\,dx &= \lim_{t\to\infty} \int_{0}^{t} \frac{e^x}{1+e^{2x}}\,dx \\
&= \lim_{t\to\infty} \arctan e^x \Big]_0^t \\
&= \lim_{t\to\infty} \left(\arctan e^t - \arctan e^0\right) \\
&= \lim_{t\to\infty} \left(\arctan e^t - \arctan 1\right) \\
&= \lim_{t\to\infty} \left(\arctan e^t - \frac{\pi}{4}\right) \\
&= \frac{\pi}{2} - \frac{\pi}{4} \qquad\qquad \left(\because \lim_{x\to\infty} \arctan x = \frac{\pi}{2}\right) \\
&= \frac{\pi}{4}
\end{aligned}$$

Hence both $\displaystyle\int_{-\infty}^{0} \frac{e^x}{1+e^{2x}}\,dx$ and $\displaystyle\int_{0}^{\infty} \frac{e^x}{1+e^{2x}}\,dx$ converge so that $\displaystyle\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx$ converges, and

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}.$$

Theorem 10.1.

$$\int_{1}^{\infty} \frac{1}{x^p}\,dx = \begin{cases} \dfrac{1}{p-1} & \text{if } p > 1 \\ \text{diverges} & \text{if } p \le 1. \end{cases}$$

Proof: If $p = 1$, then

$$\begin{aligned}\int_1^\infty \frac{1}{x^p}\,dx &= \lim_{t\to\infty}\int_1^t \frac{1}{x}\,dx\\ &= \lim_{t\to\infty} \ln x\Big]_1^t\\ &= \lim_{t\to\infty}[\ln t - \ln 1]\\ &= \lim_{t\to\infty} \ln t\\ &= \infty.\end{aligned}$$

Thus for $p = 1$, $\displaystyle\int_1^\infty \frac{1}{x^p}\,dx$ diverges.

Now suppose that $p \neq 1$. Then

$$\begin{aligned}\int_1^\infty \frac{1}{x^p}\,dx &= \lim_{t\to\infty}\int_1^t \frac{1}{x^p}\,dx\\ &= \lim_{t\to\infty}\left[\frac{x^{-p+1}}{-p+1}\right]_1^t\\ &= \lim_{t\to\infty}\left[\frac{t^{1-p}}{1-p} - \frac{1}{1-p}\right]\\ &= \lim_{t\to\infty}\left(\frac{1}{1-p}\right)\left[\frac{1}{t^{p-1}} - 1\right].\end{aligned}$$

If $p < 1$, then $p - 1 < 0$ so that as $t \to \infty$, $\dfrac{1}{t^{p-1}} \to \infty$. Thus for $p < 1$,

$$\int_1^\infty \frac{1}{x^p}\,dx = \infty.$$

Hence,

$$\int_1^\infty \frac{1}{x^p}\,dx \text{ diverges for } p < 1.$$

If $p > 1$, then $p - 1 > 0$ so that as $t \to \infty$, $t^{p-1} \to \infty$ and hence $\dfrac{1}{t^{p-1}} \to 0$. Thus for $p > 1$,

$$\begin{aligned}\int_1^\infty \frac{1}{x^p}\,dx &= \lim_{t\to\infty}\left(\frac{1}{1-p}\right)\left[\frac{1}{t^{p-1}} - 1\right]\\ &= \frac{1}{1-p}\left(\lim_{t\to\infty}\frac{1}{t^{p-1}} - 1\right)\\ &= \frac{1}{1-p}\,(0-1)\\ &= \frac{1}{p-1}\,.\end{aligned}$$

Hence,

$$\int_1^\infty \frac{1}{x^p}\,dx = \begin{cases}\dfrac{1}{p-1} & \text{if } p > 1\\ \text{diverges} & \text{if } p \le 1.\end{cases}$$

10.2 Type II: Infinite Discontinuities

Definition of Improper Integrals with Infinite Discontinuities: Type II

1. If f is continuous on $[a, b)$ and f has an infinite discontinuity at b, then $\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$.

2. If f is continuous on $(a, b]$ and f has an infinite discontinuity at a, then $\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$.

3. If f is continuous on $[a, b]$ except for some $c \in (a, b)$ at which f has an infinite discontinuity, then $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$.

In the first two cases, the improper integral **converges** if the limit exists (and is finite). In this case $\lim_{t\to b^-} \int_a^t f(x)\,dx = L$ or $\lim_{t\to a^+} \int_t^b f(x)\,dx = M$. Then:

$\int_a^b f(x)\,dx$ *converges* to L, and we write $\int_a^b f(x)\,dx = L$;

$\int_a^b f(x)\,dx$ *converges* to M, and we write $\int_a^b f(x)\,dx = M$.

If the limits do not exist, then the improper integrals **diverge**.

In the third case, the improper integral $\int_a^b f(x)\,dx$ diverges if either $\int_a^c f(x)\,dx$ or $\int_c^b f(x)\,dx$ diverges.

If both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ converge, then $\int_a^b f(x)\,dx$ is convergent.

Worked Example 10.2.1. Determine whether the following improper integrals converge or diverge.

1. **Left endpoint infinite discontinuity**

$$\int_0^1 \ln x\,dx.$$

Solution. Since the domain of $\ln x$ is $(0, \infty)$ and $\ln x$ has an infinite discontinuity at $x = 0$, the given integral is improper of Type II. We use the fact that

$$\int \ln x\,dx = x\ln x - x + C = x(\ln x - 1) + C.$$

$$\begin{aligned}
\int_0^1 \ln x\,dx &= \lim_{t\to 0^+}\int_t^1 \ln x\,dx \\
&= \lim_{t\to 0^+}\Big[x(\ln x - 1)\Big]_t^1 \\
&= \lim_{t\to 0^+}\Big[1(\ln 1 - 1) - t(\ln t - 1)\Big] \\
&= \lim_{t\to 0^+}\Big[-1 - t\ln t + t\Big] \\
&= -1 - \lim_{t\to 0^+} t\ln t + \lim_{t\to 0^+} t \\
&= -1 - \lim_{t\to 0^+} \frac{\ln t}{\frac{1}{t}} \qquad \text{“}\frac{\infty}{\infty}\text{”} \\
&\overset{\text{l'H}}{=} -1 - \lim_{t\to 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}} \\
&= -1 + \lim_{t\to 0^+} t \\
&= -1 + 0 \\
&= -1.
\end{aligned}$$

Hence, $\int_0^1 \ln x\,dx$ converges and $\int_0^1 \ln x\,dx = -1$.

2. **Right endpoint infinite discontinuity**

(a)

$$\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx.$$

Solution. Since the function $\frac{1}{\sqrt{1-x^2}}$ has an infinite discontinuity at the right endpoint $x = 1$, the integral is improper of Type II.

Then

$$\begin{aligned}
\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx &= \lim_{t\to 1^-}\int_0^t \frac{1}{\sqrt{1-x^2}}\,dx \\
&= \lim_{t\to 1^-} \arcsin x\Big]_0^t \\
&= \lim_{t\to 1^-}[\arcsin t - \arcsin 0] \\
&= \lim_{t\to 1^-} \arcsin t \\
&= \arcsin 1 \qquad (\because \arcsin x \text{ is continuous at } x = 1) \\
&= \frac{\pi}{2}.
\end{aligned}$$

Hence, $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$ is convergent and $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx = \frac{\pi}{2}$.

(b)

$$\int_0^{\frac{\pi}{2}} \sec x\,dx.$$

Solution. Since $\sec x$ has an infinite discontinuity at $x = \dfrac{\pi}{2}$, the integral is improper of Type II. Then

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \sec x\,dx &= \lim_{t\to\left(\frac{\pi}{2}\right)^-} \int_0^t \sec x\,dx \\
&= \lim_{t\to\left(\frac{\pi}{2}\right)^-} \ln|\sec x + \tan x|\Big|_0^t \\
&= \lim_{t\to\left(\frac{\pi}{2}\right)^-} \Big[\ln|\sec t + \tan t| - \ln|\sec 0 + \tan 0|\Big] \\
&= \lim_{t\to\left(\frac{\pi}{2}\right)^-} \Big[\ln|\sec t + \tan t| - \ln 1\Big] \\
&= \lim_{t\to\left(\frac{\pi}{2}\right)^-} \Big[\ln|\sec t + \tan t|\Big] \\
&= \infty \qquad (\because \sec t \to \infty \text{ and } \tan t \to \infty \text{ as } t \to \left(\frac{\pi}{2}\right)^-.
\end{aligned}$$

3. **Interior infinite discontinuity**
(a)

$$\int_0^4 \frac{1}{(x-3)^2}\,dx.$$

Solution. The integrand $\dfrac{1}{(x-3)^2}$ is not defined at $x = 3 \in (0, 4)$ and has an infinite discontinuity there. Thus, the given integral has an interior infinite discontinuity, hence is improper of Type II. Then

$$\int_0^4 \frac{1}{(x-3)^2}\,dx = \int_0^3 \frac{1}{(x-3)^2}\,dx + \int_3^4 \frac{1}{(x-3)^2}\,dx.$$

The given integral converges if both improper integrals on the right converge. If one of the improper integrals on the right diverges, then the given improper integral diverges. Now

$$\begin{aligned}
\int_0^3 \frac{1}{(x-3)^2}\,dx &= \lim_{t\to 3^-} \int_0^t \frac{1}{(x-3)^2}\,dx \\
&= \lim_{t\to 3^-} \left[-\frac{1}{x-3}\right]_0^t \\
&= \lim_{t\to 3^-} \left[-\frac{1}{t-3} - \left(-\frac{1}{0-3}\right)\right] \\
&= \lim_{t\to 3^-} \left[-\frac{1}{t-3} - \frac{1}{3}\right] \\
&= \infty \qquad \left(\because \frac{1}{t-3} \to -\infty \text{ as } t \to 3^-\right).
\end{aligned}$$

Hence, $\displaystyle\int_0^3 \frac{1}{(x-3)^2}\,dx$ diverges and consequently, $\displaystyle\int_0^4 \frac{1}{(x-3)^2}\,dx$ diverges.

Note: There is no need to evaluate $\displaystyle\int_3^4 \frac{1}{(x-3)^2}\,dx$. Also, the FTC2 cannot apply here. If we did not identify that the integral is improper we would have concluded the incorrect result

$$\int_0^4 \frac{1}{(x-3)^2}\,dx = -\frac{1}{x-3}\Big]_0^4 = -1 - \frac{1}{3} = -\frac{4}{3} \qquad \textbf{WRONG!}$$

(b)

$$\int_{-2}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx.$$

Solution. The integral has an infinite discontinuity at $x = -1 \in (-2, 7)$, hence is improper of Type II. Then

$$\int_{-2}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx = \int_{-2}^{-1} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx + \int_{-1}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx.$$

Now

$$\begin{aligned}
\int_{-2}^{-1} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx &= \lim_{t\to -1^-} \int_{-2}^{t} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx \\
&= \lim_{t\to -1^-} 3(x+1)^{\frac{1}{3}}\Big]_{-2}^{t} \\
&= \lim_{t\to -1^-} \left[3(t+1)^{\frac{1}{3}} - 3(-1)^{\frac{1}{3}}\right] \\
&= \lim_{t\to -1^-} \left[3(t+1)^{\frac{1}{3}} + 3\right] \\
&= 0 + 3 \\
&= 3
\end{aligned}$$

and

$$\begin{aligned}
\int_{-1}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx &= \lim_{t\to -1^+} \int_{t}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx \\
&= \lim_{t\to -1^+} 3(x+1)^{\frac{1}{3}}\Big]_{t}^{7} \\
&= \lim_{t\to -1^+} \left[3(7+1)^{\frac{1}{3}} - 3(1+t)^{\frac{1}{3}}\right] \\
&= \lim_{t\to -1^+} \left[6 - 3(1+t)^{\frac{1}{3}}\right] \\
&= 6 - 0 \\
&= 6.
\end{aligned}$$

Thus both $\int_{-2}^{-1} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx$ and $\int_{-1}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx$ converge so that $\int_{-2}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx$ converges and

$$\int_{-2}^{7} \frac{1}{(x+1)^{\frac{2}{3}}}\,dx = 3 + 6 = 9.$$

An integral can be doubly improper, i.e., it is improper of Type I and improper of Type II.

Worked Example 10.2.2. (Doubly Improper) Determine whether the following integrals are convergent or divergent.

1.

$$\int_{0}^{\infty} \frac{dx}{\sqrt{x}(x+1)}.$$

Solution. We note that the integral is doubly improper since the interval is infinite and the integrand has an infinite discontinuity at the left endpoint $x = 0$. We split the interval at a convenient point, say $x = 1$. Then

$$\int_0^\infty \frac{dx}{\sqrt{x}(x+1)} = \int_0^1 \frac{dx}{\sqrt{x}(x+1)} + \int_1^\infty \frac{dx}{\sqrt{x}(x+1)}.$$

We first calculate the indefinite integral $\displaystyle\int \frac{dx}{\sqrt{x}(x+1)}$. Put $u^2 = x$. Then $2u\,du = dx$ and thus

$$\int \frac{dx}{\sqrt{x}(x+1)} = \int \frac{2u}{u(u^2+1)}\,du = 2\int \frac{1}{1+u^2}\,du = 2\arctan u + C = 2\arctan\sqrt{x} + C.$$

Then

$$\begin{aligned}
\int_0^1 \frac{dx}{\sqrt{x}(x+1)} &= \lim_{t\to 0^+} \int_t^1 \frac{dx}{\sqrt{x}(x+1)} \\
&= \lim_{t\to 0^+} 2\arctan\sqrt{x}\Big]_t^1 \\
&= \lim_{t\to 0^+} 2\left[\arctan\sqrt{1} - \arctan\sqrt{t}\right] \\
&= \lim_{t\to 0^+} 2\left[\frac{\pi}{4} - \arctan\sqrt{t}\right] \\
&= 2\left[\frac{\pi}{4} - \arctan 0\right] \\
&= \frac{\pi}{2}.
\end{aligned}$$

and

$$\begin{aligned}
\int_1^\infty \frac{dx}{\sqrt{x}(x+1)} &= \lim_{t\to\infty} \int_1^t \frac{dx}{\sqrt{x}(x+1)} \\
&= \lim_{t\to\infty} 2\arctan\sqrt{x}\Big]_1^t \\
&= \lim_{t\to\infty} 2\left[\arctan\sqrt{t} - \arctan\sqrt{1}\right] \\
&= \lim_{t\to\infty} 2\left[\arctan\sqrt{t} - \frac{\pi}{4}\right] \\
&= 2\left[\frac{\pi}{2} - \frac{\pi}{4}\right] \\
&= \frac{\pi}{2}.
\end{aligned}$$

Hence $\displaystyle\int_0^\infty \frac{dx}{\sqrt{x}(x+1)}$ converges since both $\displaystyle\int_0^1 \frac{dx}{\sqrt{x}(x+1)}$ and $\displaystyle\int_1^\infty \frac{dx}{\sqrt{x}(x+1)}$ converge and

$$\int_0^\infty \frac{dx}{\sqrt{x}(x+1)} = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

2.

$$\int_0^\infty \frac{1}{x+x^2}\,dx$$

Solution. The integral is doubly improper. Choose $x = 1$. Then

$$\int_0^\infty \frac{1}{x+x^2}\,dx = \int_0^1 \frac{1}{x+x^2}\,dx + \int_1^\infty \frac{1}{x+x^2}\,dx.$$

Now

$$\int \frac{1}{x+x^2}\,dx = \int \frac{1}{x(1+x)}\,dx.$$

We decompose the integrand into partial fractions.

$$\frac{1}{x(1+x)} = \frac{A}{x} + \frac{B}{1+x}$$

gives

$$1 = A(1+x) + Bx.$$

Then

$$x = 0 \Rightarrow A = 1 \text{ and } x = -1 \Rightarrow B = -1.$$

Thus

$$\frac{1}{x(1+x)} = \frac{1}{x} - \frac{1}{1+x}.$$

Then

$$\begin{aligned}
\int_0^1 \frac{1}{x+x^2}\,dx &= \lim_{t\to 0^+} \int_t^1 \frac{1}{x+x^2}\,dx \\
&= \lim_{t\to 0^+} \int_t^1 \left[\frac{1}{x} - \frac{1}{1+x}\right] dx \\
&= \lim_{t\to 0^+} \Big[\ln|x| - \ln|1+x|\Big]_t^1\,dx \\
&= \lim_{t\to 0^+} \Big[\ln|x| - \ln|1+x|\Big]_t^1\,dx \\
&= \lim_{t\to 0^+} \left[\ln\left|\frac{x}{1+x}\right|\right]_t^1\,dx \\
&= \lim_{t\to 0^+} \left[\ln\left(\frac{1}{2}\right) - \ln\left(\frac{t}{1+t}\right)\right]_t^1\,dx \\
&= \infty \qquad\qquad \because \frac{t}{1+t} \to 0 \text{ as } t \to 0^+ \Rightarrow \ln\left(\frac{t}{1+t}\right) \to -\infty \text{ as } t \to 0^+.
\end{aligned}$$

Thus $\int_0^1 \frac{1}{x+x^2}\,dx$ diverges and hence $\int_0^\infty \frac{1}{x+x^2}\,dx$ diverges.

Tutorial 10.2.1. 1. Thomas Exercises 11th ed. 8.8 p. 615 **or** 12th ed. 8.7 p. 505: 1, 4, 10,11, 13, 16, 18, 22, 25, 31, 35, 42, 45, 51, 60, 65.

Chapter 11

Infinite Sequences and Series

11.1 Sequences

LEARNING OUTCOMES:

On completion of this section you should

1. be able to distinguish between convergent and divergent sequences,
2. know the limit laws for sequences,
3. know the Squeeze Theorem for sequences,
4. know how to calculate the limit of convergent sequences,
5. know the definitions of increasing, decreasing, monotonic and bounded sequences.

A **sequence** is an ordered list of infinitely many numbers

$$a_1, a_2, a_3, a_4, \ldots, a_n, \ldots$$

The number a_n is called the n-th term of the sequence. The subscript n is called the index. Often, one omits the general n-th term and simply writes

$$a_1, a_2, a_3, a_4, \ldots$$

In case of particularly given sequences, the latter notation may be ambiguous, whereas the first one is quite clumsy. Therefore, one also uses the notations

$$\{a_n\} \qquad \text{and} \qquad \{a_n\}_{n=1}^{\infty}$$

Note. The index of the sequence does not have to start at 1, e. g.,

$$\{a_0, a_1, a_2, \ldots\}, \quad \{a_2, a_3, a_4, \ldots\}, \quad \{a_2, a_4, a_6, \ldots\}$$

are sequences, say with $a_n = n$, although different sequences.

Example 11.1.1. In this example, each line gives several different notations for a particular sequence:

(a) $\left\{\dfrac{n}{n+1}\right\}$; $\left\{\dfrac{n}{n+1}\right\}_{n=1}^{\infty}$; $\left\{\dfrac{1}{2}, \dfrac{2}{3}, \dfrac{3}{4}, \ldots\right\}$; $\left\{\dfrac{1}{2}, \dfrac{2}{3}, \dfrac{3}{4}, \ldots, \dfrac{n}{n+1}, \ldots\right\}$

(b) $\left\{\dfrac{(-1)^n(n+1)}{5^n}\right\}$; $a_n = \dfrac{(-1)^n(n+1)}{5^n}$; $\left\{-\dfrac{2}{5}, \dfrac{3}{25}, -\dfrac{4}{125}, \ldots\right\}$

(c) $\left\{\cos\dfrac{n\pi}{6}\right\}_{n=0}^{\infty}$; $a_n = \cos\dfrac{n\pi}{6}, n \geq 0$; $a_n = \cos\dfrac{n\pi}{6}, n = 0, 1, 2, \ldots$; $\left\{1, \dfrac{\sqrt{3}}{2}, \dfrac{1}{2}, 0. \ldots, \cos\dfrac{n\pi}{6}, \ldots\right\}$

Note that in (c) above we have written $n \geq 0$ to indicate which indices n we take to build the sequence. Here it is understood that indices are integers [unless otherwise stated].

Worked Example 11.1.2. Consider the sequences
(i) $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$, (ii) $\left\{\frac{n}{n+1}\right\}_{n=0}^{\infty}$, (iii) $\left\{\frac{n+1}{n+2}\right\}_{n=0}^{\infty}$.
Which of these sequences are identical?

Solution. (i) and (ii) are different; the first term of (ii) is not present in (i).
The sequences (i) and (iii) are identical: Put $m = n + 1$ in (iii) and then rename m to n:

$$\begin{aligned}\left\{\frac{n+1}{n+2}\right\}_{n=0}^{\infty} &= \left\{\frac{m}{m+1}\right\}_{m=1}^{\infty} \\ &= \left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}\end{aligned}$$

Worked Example 11.1.3. Find the general term of the sequence

$$\left\{\frac{3}{5}, -\frac{4}{25}, \frac{1}{25}, -\frac{6}{625}, \frac{7}{3125}, \dots\right\}$$

Solution. Denominators are powers of 5: $5, 5^2, 5^2, 5^4, 5^5$, which is 5^n if we start with $n = 1$, once in the third term we multiply numerator and denominator by 5.
Then the numerators start at 3 and increase by 1; the numerators are therefore $n + 2$. Finally, the signs are alternatively positive and negative, starting with positive, which can be written as multiplication by $(-1)^{n+1}$. Hence the general term is

$$a_n = (-1)^{n+1}\frac{n+2}{5^n}$$

Worked Example 11.1.4. Many sequences cannot easily be guessed from their first terms; indeed, only the simplest sequences with an "obvious" rule should be written in "dots" notation. Try to guess a rule for the following sequences:

(i) $\{7, 1, 8, 2, 8, 1, 8, 2, 8, 4, 5, \dots\}$, (ii) $\{1, 1, 2, 3, 5, 8, 13, 21, \dots\}$.

Solution. (i) Here is no apparent rule. But these numbers happen to be the first decimal places of the number e.
(ii) From the third term onwards, the term is the sum of its two predecessors. That is, the sequence $\{f_n\}_{n=1}^{\infty}$ is defined by

$$f_1 = 1, \quad f_2 = 1, \quad f_n = f_{n-2} + f_{n-1} \text{ for } n \geq 3.$$

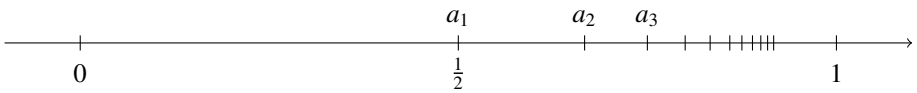
This sequence is called the Fibonacci sequence.

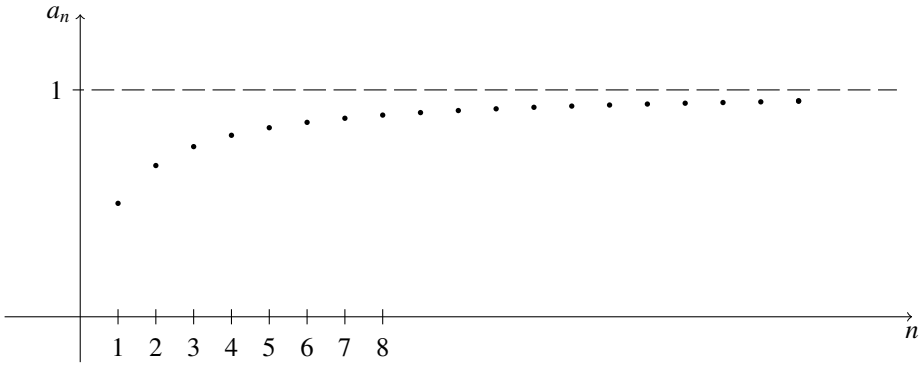
Note. (i) A sequence can be identified with a function on the (positive) integers:

$$a_n = f(n).$$

(ii) A sequence can be plotted as points of the real axis or as the graph of the function, see (i).

Below is a plot for the sequence from Example 11.1.1 (a): $a_n = \frac{n}{n+1}$.





From the above graph it appears to be obvious that the line $y = 1$ (denoting the co-ordinate direction of the a_n with y) is a horizontal asymptote. That is, for sufficiently large n, a_n comes as close to 1 as we want, and we say the sequence $\{a_n\}$ converges to 1.

This is formalized in the following definition:

Definition. A sequence $\{a_n\}$ is said to **converge** if there is a number L such that a_n is as close to L as we like for all sufficiently large n. We write

$$\boxed{\lim_{n\to\infty} a_n = L \quad \text{or} \quad a_n \to L \text{ as } n \to \infty}$$

A sequence which does not converge is said to **diverge**.

Compare this definition with the definition of horizontal asymptotes, see Calculus First Semester Study Guide [CFSSG], page 8.

Theorem 11.1. *If* $\lim_{x\to\infty} f(x) = L$ *and* $a_n = f(n)$ *for integers n, then* $\lim_{n\to\infty} a_n = L$.

In particular, CFFSG, Theorem 2.1 immediately gives

Theorem 11.2 (Limit Laws)**.** *Let* $c \in \mathbb{R}$ *and suppose that* $\lim_{n\to\infty} a_n = L$ *and* $\lim_{n\to\infty} b_n = M$ *both exist. Then*

1. $\lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = L + M.$
2. $\lim_{n\to\infty}(a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n = L - M.$
3. $\lim_{n\to\infty}(ca_n) = c \lim_{n\to\infty} a_n = cL.$
4. $\lim_{n\to\infty}(a_n b_n) = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right) = LM.$
5. *if* $M \neq 0$, $\lim_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} = \dfrac{L}{M}.$
6. *if* $L \neq 0$ *and* $M = 0$, *then* $\lim_{n\to\infty} \dfrac{a_n}{b_n}$ *does not exist.*
7. *if* $k \in \mathbb{Z}^+$, $\lim_{n\to\infty} a_n^k = \left(\lim_{n\to\infty} a_n\right)^k = L^k.$
8. $\lim_{n\to\infty} c = c.$
9. *if* $k \in \mathbb{Z}^+$, $\lim_{n\to\infty} \sqrt[k]{a_n} = \sqrt[k]{\lim_{n\to\infty} a_n} = \sqrt[k]{L}$. *If k is even, we assume* $L > 0$.
10. *if* $\lim_{n\to\infty} |a_n| = 0$, *then* $\lim_{n\to\infty} a_n = 0.$

The Sandwich theorem, CFSSG, Theorem 2.7 has a variant for sequences

Theorem 11.3 (**Sandwich Theorem**)**.** *If* $a_n \leq b_n \leq c_n$ *and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, *then* $\lim_{n\to\infty} b_n = L$.

Theorem 2.3 in CFSSG leads to

Theorem 11.4. *If $r > 0$, then*

$$\lim_{n\to\infty} \frac{1}{n^r} = 0.$$

Definition. If a_n becomes larger than any positive number for sufficiently large n, i. e., if for every $M > 0$ there is an integer N such that

$$a_n > M \quad \text{whenever} \quad n > N,$$

then the sequence $\{a_n\}$ is said to diverge to ∞, and we write

$$\lim_{n\to\infty} a_n = \infty.$$

Theorem 11.5. *If $\lim_{n\to\infty} |a_n| = \infty$, then $\{a_n\}$ diverges.*

For a proof see bounded sequences below.

Worked Example 11.1.5. For each of the following sequences, determine if it converges or diverges, and find the limit if it exists.

1. $a_n = \dfrac{n}{n+1}$

   Solution.

$$\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = \frac{\lim_{n\to\infty} 1}{1 + \lim_{n\to\infty} \frac{1}{n}}$$
$$= \frac{1}{1+0} = 1$$

   by Theorem 11.4. So the limit exists and equals 1.

2. $a_n = \dfrac{\ln n}{n}$

   Solution. Both numerator and denominator approach ∞ as $n \to \infty$. In order to be able to apply l'Hôpital's rule, we replace the integer n by the real number x:

$$\lim_{x\to\infty} \frac{\ln x}{x} \qquad \text{“}\frac{\infty}{\infty}\text{”}$$
$$\overset{\text{l'H}}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{1}$$
$$= \lim_{x\to\infty} \frac{1}{x} = 0.$$

   By Theorem 11.1 it follows that $\{a_n\}$ converges, and

$$\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{x\to\infty} \frac{\ln x}{x} = 0.$$

3. $\{-1, 1, -1, 1, -1, 1, \dots\}$

   Solution. Infinitely many terms are 1 and infinitely many terms are -1. Hence the sequence does not approach a single number: the sequence $\{(-1)^n\}$ diverges.

4. $a_n = \dfrac{(-1)^n}{n}$

Solution.

$$\lim_{n\to\infty}\left|\frac{(-1)^n}{n}\right| = \lim_{n\to\infty}\frac{1}{n} = 0$$

exists. Hence, by Theorem 11.2.10,

$$\lim_{n\to\infty}\frac{(-1)^n}{n} = 0.$$

5. $a_n = \dfrac{n!}{n^n}$

Solution. The n-th term is

$$a_n = \frac{n!}{n^n} = \frac{1\cdot 2\cdot 3\cdot\cdots\cdot n}{n\cdot n\cdot n\cdot\cdots\cdot n} = \frac{1}{n}\left(\frac{2\cdot 3\cdot\cdots\cdot n}{n\cdot n\cdot\cdots\cdot n}\right)$$

The term in brackets is at most 1. Hence

$$0 < a_n \le \frac{1}{n}.$$

Since $\frac{1}{n}\to 0$ as $n\to\infty$ by Theorem 11.4,

$$\lim_{n\to\infty}\frac{n!}{n^n} = 0$$

by the squeeze theorem.

Worked Example 11.1.6. For which values of r does the sequence $\{r^n\}$ converge?

Solution. By CFSSG, Theorem 2.5, and Theorem 11.1,

$$\lim_{n\to\infty} a^n = \begin{cases} 0 & \text{if } 0 \le a < 1, \\ \infty & \text{if } a > 1. \end{cases}$$

Now put $a = |r|$. If $-1 < r < 1$ then $0 \le a < 1$, and by Theorem 11.2.10, $\lim\limits_{n\to\infty} r^n = 0$. By Theorem 11.5, the sequence $\{r^n\}$ diverges if $|r| > 1$. Finally, $\lim\limits_{n\to\infty} 1^n = 1$ by Theorem 11.2.8, and $\{(-1)^n\}$ does not converge by Example 11.1.5.3, above. Hence $\{r^n\}$ converges for $-1 < r \le 1$ and diverges for all other real r.

Definition. A sequence $\{a_n\}_{n=1}^{\infty}$ is called

increasing if $a_n < a_{n+1}$ for all $n \ge 1$,
decreasing if $a_n > a_{n+1}$ for all $n \ge 1$,
monotonic if it is either increasing or decreasing.

Worked Example 11.1.7. Determine if the following sequences are decreasing or increasing:

1. $\left\{\dfrac{3}{n+5}\right\}$

Solution. This sequence is decreasing because

$$\frac{3}{n+5} > \frac{3}{n+6} = \frac{3}{(n+1)+5}$$

2. $\left\{\dfrac{n}{2n^2-1}\right\}$

Solution 1.
This sequence is decreasing if

$$\frac{n}{2n^2-1} > \frac{n+1}{2(n+1)^2-1}.$$

By cross-multiplication, this is equivalent to

$$\begin{aligned} n((2(n+1)^2-1) > (n+1)(2n^2-1) &\Leftrightarrow 2n^3+4n^2+n > 2n^3+2n^2-n-1 \\ &\Leftrightarrow 2n^2+2n > -1, \end{aligned}$$

which is true for all $n \geq 1$. Hence the sequence is decreasing.

Solution 2.
Consider the function $f(x) = \dfrac{x}{2x^2-1}$:

$$f'(x) = \frac{2x^2-1-x(4x)}{(2x^2-1)^2} = \frac{-2x^2-1}{(2x^2-1)^2} < 0 \quad \text{for all } x > 1.$$

Thus f is decreasing on $[1,\infty)$, and therefore $f(n) > f(n+1)$ for all $n \geq 1$. Therefore, $\left\{\dfrac{n}{2n^2-1}\right\}$ is decreasing.

Definition. A sequence $\{a_n\}_{n=1}^{\infty}$ is called

bounded above if there is a number M such that $a_n \leq M$ for all $n \geq 1$,
bounded below if there is a number m such that $a_n \geq m$ for all $n \geq 1$,
bounded if it is bounded above and below.

We will see applications of the following important theorem in later sections.

Theorem 11.6 (**Monotonic Sequence Theorem**). *Every bounded, monotonic sequence is convergent.*

Tutorial 11.1.1. Thomas Exercises 11th ed. 11.1 pp. 741–742: 1, 3, 4, 6, 17, 21, 25, 29, 33, 35, 42, 62, 73, 81–84 **or** 12th ed. 10.1 pp. 559–560: 1, 3, 4, 6, 17, 21, 29, 33, 37, 39, 46, 66, 73, 87–90.

11.2 Series

LEARNING OUTCOMES:

On completion of this section you should be able to

1. know how to use the n-th partial sum s_n of a series to determine if a series is convergent or divergent,
2. be able to calculate the sum of a convergent series,
3. be able to prove the theorem that if the series $\sum_{n=1}^{\infty} a_n$ is convergent, then limit of the sequence $\{a_n\}$ is 0,
4. know how to apply the test for divergence,
5. know the properties of convergent series.

Given a sequence $\{a_n\}_{n=1}^{\infty}$, we can formally consider the expression

$$a_1 + a_2 + a_3 + \cdots + a_n + \ldots$$

and called it a **series**, written symbolically as

$$\sum_{n=1}^{\infty} a_n \qquad \text{or} \qquad \sum a_n.$$

But how do we add up infinitely many numbers? Does it make sense at all?

Example 11.2.1. Consider the sequence $\{1, -1, 1, -1, 1, \dots\}$. Adding up the first n terms, we arrive at

$$1, 0, 1, 0, 1, \dots \quad \text{for } n = 1, 2, 3, 4, 5, \dots$$

and it does not make sense to assign a unique value to the infinite sum.

However, what we obtain in this way is a sequence of numbers

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{i=1}^{n} a_i, \qquad n = 1, 2, 3, \dots$$

of well-defined sums, called **partial sums** of the series $\sum_{n=1}^{\infty} a_n$. If by adding more and more terms of the series a certain number is approached, we will assign this number as the value of the series. Formally, we have the following definition:

Definition. Given a series $\sum_{n=1}^{\infty} a_n$, and denoting by $s_n = \sum_{i=1}^{n} a_i$ its n-th partial sum, the series $\sum_{n=1}^{\infty} a_n$ is said to **converge** if the sequence $\{s_n\}$ converges. In this case, the number $s = \lim_{n\to\infty} s_n$ is called the **sum of the series** and we write

$$\sum_{n=1}^{\infty} a_n = s$$

A series which does not converge is said to **diverge** or to be **divergent**.

Note. 1. Observe that $\sum_{n=1}^{\infty} a_n$ denotes a series (convergent or divergent) as well as its sum if it converges.
2. A series does not have to start at $n = 1$. The change of notation is obvious for other starting indices.

Example 11.2.2. Consider the **geometric series**

$$\sum_{n=0}^{\infty} ar^n, \quad a \neq 0,\ r \in \mathbb{R}.$$

Recall the partial sums for $r \neq 1$:

$$\begin{aligned} s_n &= a + ar + ar^2 + \cdots + ar^n, \\ rs_n &= \quad ar + ar^2 + \cdots + ar^n + ar^{n+1}. \end{aligned}$$

Subtracting these equations gives

$$s_n - rs_n = a - ar^{n+1},$$

and so

$$s_n = \frac{a(1 - r^{n+1})}{1 - r}.$$

By Example 11.1.6, $\{s_n\}$ and thus the geometric series converges if $|r| < 1$, with

$$\sum_{n=0}^{\infty} ar^n = \lim_{n\to\infty} s_n = \frac{a}{1 - r},$$

and diverges if $|r| > 1$ or $r = -1$. Finally, for $r = 1$, $s_n = a(n + 1)$, and thus $|s_n| \to \infty$ as $n \to \infty$.

Thus we have shown

Theorem 11.7. *The geometric series*

$$\sum_{n=0}^{\infty} ar^n, \quad a \neq 0,$$

is convergent if $|r| < 1$*, with sum*

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}\,.$$

If $|r| \geq 1$*, the geometric series diverges.*

Worked Example 11.2.3. Find the sum of the geometric series

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \dots$$

Solution. The first term is $a = 5$ and the common ratio is $r = -\frac{2}{3}$. Then $|r| = \frac{2}{3} < 1$. Hence, by Theorem 11.7,

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots = \frac{5}{1-\left(-\frac{2}{3}\right)} = \frac{5}{\frac{5}{3}} = 3.$$

Worked Example 11.2.4. Is the series $\displaystyle\sum_{n=1}^{\infty} 2^{2n}3^{1-n}$ convergent or divergent?

Solution. The n-th term of the series is obviously of the form ar^n:

$$a_n = 2^{2n}3^{1-n} = \frac{3\cdot 4^n}{3^n} = 3\left(\frac{4}{3}\right)^n.$$

So $r = \dfrac{4}{3} > 1$, and the series diverges.

Worked Example 11.2.5. Write the number $2.3\overline{17} = 2.3171717\dots$ as a ratio of integers.

Solution. Write

$$2.3\overline{17} = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \dots.$$

Starting with the second term we have a geometric series with $a = \frac{17}{10^3}$ and $r = 10^{-2}$. Hence the series converges, and

$$\begin{aligned} 2.3\overline{17} &= 2.3 + \frac{\frac{17}{10^3}}{1-10^{-2}} = 2.3 + \frac{17}{10^3 - 10} \\ &= \frac{23}{10} + \frac{17}{990} = \frac{1147}{495}\,. \end{aligned}$$

Worked Example 11.2.6. Show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges and find its sum.

Solution. Using partial fractions

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$$

the partial sums become

$$\begin{aligned} s_n &= \sum_{i=1}^{n} \frac{1}{i(i+1)} = \sum_{i=1}^{n}\left(\frac{1}{i} - \frac{1}{i+1}\right) \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \end{aligned}$$

since adjacent terms from different brackets cancel (telescoping sum). Alternatively,

$$\begin{aligned} s_n &= \sum_{i=1}^{n} \frac{1}{i(i+1)} = \sum_{i=1}^{n} \left(\frac{1}{i} - \frac{1}{i+1} \right) \\ &= \sum_{i=1}^{n} \frac{1}{i} - \sum_{i=2}^{n+1} \frac{1}{i} \\ &= 1 + \sum_{i=2}^{n} \frac{1}{i} - \sum_{i=2}^{n} \frac{1}{i} - \frac{1}{n+1} \\ &= 1 - \frac{1}{n+1} \,. \end{aligned}$$

Thus

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1} \right) = 1 - 0 = 1$$

Theorem 11.8. *If the series* $\sum_{n=1}^{\infty} a_n$ *converges, then* $\lim_{n\to\infty} a_n = 0$.

Proof. Let

$$s_n = a_1 + a_2 + \cdots + a_n \,.$$

Then

$$a_n = s_n - s_{n-1} \,.$$

Since $\sum_{n=1}^{\infty} a_n$ converges,

$$\lim_{n\to\infty} s_n = s$$

exists. Since also $n - 1 \to \infty$ as $n \to \infty$,

$$\lim_{n\to\infty} s_{n-1} = s.$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (s_n - s_{n-1}) = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = s - s = 0.$$

Note. The converse of Theorem 11.8 is not true, see the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ done later.

The contrapositive statement to Theorem 11.8 is very useful:

Theorem 11.9 (**Test for Divergence**). *If* $\lim_{n\to\infty} a_n$ *does not exist or if* $\lim_{n\to\infty} a_n \neq 0$*, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*

Note. From $\lim_{n\to\infty} a_n = 0$ nothing can be concluded about the convergence of the series $\sum_{n=1}^{\infty} a_n$.

Worked Example 11.2.7. Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$ diverges.

Solution.

$$\lim_{n\to\infty} \frac{n^2}{5n^2+4} = \lim_{n\to\infty} \frac{1}{5 + \frac{4}{n^2}} = \frac{1}{5} \neq 0.$$

Hence the series diverges by the Test for Divergence.

From Theorem 11.2 we immediately infer

Theorem 11.10 (Sum Laws). *Let $c \in \mathbb{R}$ and suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge. Then also $\sum_{n=1}^{\infty}(ca_n)$, $\sum_{n=1}^{\infty}(a_n + b_n)$ and $\sum_{n=1}^{\infty}(a_n - b_n)$ converge, and*

1. $\sum_{n=1}^{\infty}(ca_n) = c\sum_{n=1}^{\infty} a_n$,
2. $\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$,
3. $\sum_{n=1}^{\infty}(a_n - b_n) = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$,

Worked Example 11.2.8. Find the sum of the series $\sum_{n=1}^{\infty}\left(\frac{3}{n(n+1)} + \frac{1}{2^n}\right)$.

Solution. We already know that

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

And $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a geometric series with $a = \frac{1}{2}$ and $r = \frac{1}{2}$. Hence

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2}\frac{1}{1-\frac{1}{2}} = 1.$$

Therefore

$$\sum_{n=1}^{\infty}\left(\frac{3}{n(n+1)} + \frac{1}{2^n}\right) = 3\sum_{n=1}^{\infty} \frac{1}{n(n+1)} + \sum_{n=1}^{\infty} \frac{1}{2^n} = 3 \cdot 1 + 1 = 4.$$

Note. For convergence it does not matter at which index the series starts. But it matters for the sum. More precisely:

Theorem 11.11. *Let $k < m$ be integers. Then the series $\sum_{n=k}^{\infty} a_n$ converges if and only if the series $\sum_{n=m}^{\infty} a_n$ converges, and*

$$\sum_{n=k}^{\infty} a_n = \sum_{n=k}^{m-1} a_n + \sum_{n=m}^{\infty} a_n\,.$$

Tutorial 11.2.1. 1. Thomas Exercises 11th ed. 11.2 pp. 753–754: 1, 3, 5, 7, 9, 13, 19, 23, 25, 51, 58 **or** 12th ed. 10.2 pp. 569–570: 1, 3, 5, 7, 9, 13, 15, 16, 19, 26, 27 ,29, 31, 33, 37, 41, 49, 51, 63, 64.

11.3 Convergence Tests for Series

LEARNING OUTCOMES:

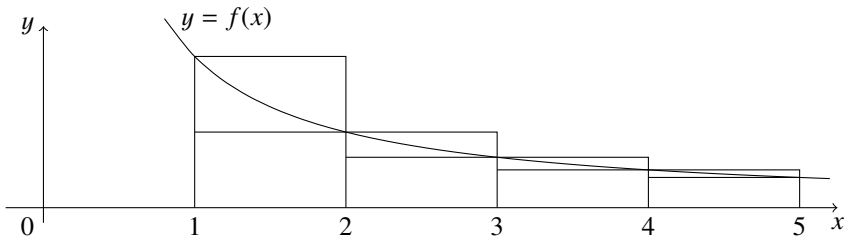
On completion of this section you should

1. know how to apply the Integral Test to test if a series $\sum a_n$ is convergent or divergent,

2. know how to compare a given series with a series that is known to be convergent or divergent,
3. know how to deal with series whose terms are alternately positive and negative,
4. know how to test if an alternating series converges,
5. be able to distinguish between series which are absolutely convergent and those which are conditionally convergent,
6. know the statement of the Absolute Convergence Test which shows that absolute convergence implies convergence,
7. know how to apply the Ratio Test and the Root Test.

11.3.1 The Integral Test

Let f be a continuous, positive, decreasing function on $[1, \infty)$. Considering lower and upper sums, L and U, for the integral $\int_1^n f(x)\,dx$



it follows that

$$\sum_{i=2}^{n} f(i) = L \leq \int_1^n f(x)\,dx \leq U = \sum_{i=1}^{n-1} f(i).$$

Using Theorem 11.11 and the Monotonic Sequence Theorem, one can show that

Theorem 11.12 (Integral Test). *Let f be a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the integral $\int_1^{\infty} f(x)\,dx$ converges.*

Note. We can start the series or the integral at any suitable integer. In particular, f only has to be positive and decreasing for sufficiently large x.

Worked Example 11.3.1. Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ for convergence.

Solution. The function $x \mapsto x^2 + 1$ is continuous, positive, increasing on $[1, \infty)$.
Hence $f(x) = \dfrac{1}{x^2+1}$ is continuous, positive, decreasing on $[1, \infty)$.
The Integral Test is applicable to f, and from

$$\int_1^{\infty} \frac{dx}{x^2+1} = \lim_{t\to\infty} \int_1^t \frac{dx}{x^2+1} = \lim_{t\to\infty} \arctan x \Big|_1^t$$
$$= \lim_{t\to\infty} \left(\arctan t - \frac{\pi}{4}\right) = \frac{\pi}{2} - \frac{\pi}{4} < \infty$$

it follows that the sum converges.

Theorem 11.13 (**p-series Test**). *Let $p \in \mathbb{R}$. The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.*

Proof. If $p \leq 0$ then $\frac{1}{n^p} \not\to 0$ as $n \to \infty$, so the p-series does not converge.

If $p > 0$, then $f(x) = \frac{1}{x^p}$ is continuous, positive, and decreasing. So the Integral Test is applicable. By Theorem 5.1, $\int_1^{\infty} \frac{dx}{x^p}$ converges if and only if $p > 1$, and the result follows.

Worked Example 11.3.2. Determine whether the following series converge:
(i) $\sum_{n=1}^{\infty} \frac{1}{n^3}$, (ii) $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}}$, (iii) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

Solution.(i) and (ii) are p-series with $p = 3$ and $p = \frac{1}{3}$, respectively, so the series (i) converges, whereas the series (ii) diverges.
(iii): $f(x) = \frac{1}{x \ln x}$ is continuous, positive, decreasing on $[2, \infty]$; either use that both factors in the denominator are increasing or calculate

$$f'(x) = -\frac{\ln x + 1}{(x \ln x)^2} < 0 \text{ for } x > 1.$$

Now

$$\lim_{t\to\infty} \int_2^t f(x)\,dx = \lim_{t\to\infty} \int_2^t \frac{1}{x \ln x}\,dx = \ln(\ln x)\Big|_2^t = \infty.$$

Hence the sum (iii) does not converge.
Note. For $0 < p \leq 1$, $\lim_{n\to\infty} \frac{1}{n^p} = 0$, but $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges.
In particular, the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$ diverges.

Tutorial 11.3.1. Thomas Exercises 12th ed. only: 10.3 p. 575: 3, 5, 6, 7, 9.

11.3.2 The Comparison Test

Theorem 11.14 (**Comparison Test**). *Let $\sum a_n$ and $\sum b_n$ be series with positive terms.*
(i) *If $\sum b_n$ converges and $a_n \leq b_n$ for all n, then also $\sum a_n$ converges.*
(ii) *If $\sum b_n$ diverges and $a_n \geq b_n$ for all n, then also $\sum a_n$ diverges.*

Note. The most common series with known convergence behaviour to compare with are p-series and geometric series.

Worked Example 11.3.3. Determine whether the following series converge:
(i) $\sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}$, (ii) $\sum_{n=1}^{\infty} \frac{\ln n}{n}$.

Solution. (i) The dominant term in the denominator is $2n^2$, and

$$\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

is a multiple of a p-series with $p = 2$ and hence convergent. Because of

$$\frac{5}{2n^2 + 4n + 3} < \frac{5}{2n^2}$$

the Comparison Test shows that $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ converges.
(ii) We try to compare with the p-series for $p = 1$, which diverges. Therefore we estimate

$$\frac{\ln n}{n} \geq \frac{1}{n} \quad \text{for } n \geq 3 (> e)$$

and obtain that $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges by the Comparison Test.

Tutorial 11.3.2. Thomas Exercises 12 th ed. only: 10.4 p. 580: 1, 3, 5, 6, 7. 8.

11.3.3 Alternating Series

An alternating series is a series of the form

$$\sum_{n=1}^{\infty}(-1)^n b_n \qquad \text{or} \qquad \sum_{n=1}^{\infty}(-1)^{n-1} b_n \qquad \text{with } b_n > 0,$$

that is, the terms of the series are alternately positive or negative. For example,

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Theorem 11.15 (Alternating series test). *If the alternating series* $\sum_{n=1}^{\infty}(-1)^n b_n$ *or* $\sum_{n=1}^{\infty}(-1)^{n-1} b_n$ *satisfies*
(i) $b_n \geq b_{n+1}$ *for all* n,
(ii) $\lim_{n\to\infty} b_n = 0$,
then the series converges.

Note. $\lim_{n\to\infty} b_n = 0$ is necessary by Theorem 11.9 since $\lim_{n\to\infty}(-1)^n b_n = 0 \Leftrightarrow \lim_{n\to\infty} b_n = 0 \Leftrightarrow \lim_{n\to\infty}(-1)^{n-1} b_n = 0$.

Example 11.3.4. The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

satisfies (i) $\frac{1}{n} > \frac{1}{n+1}$ and (ii) $\lim_{n\to\infty} \frac{1}{n} = 0$. Hence the series converges by the Alternating Series Test.

Example 11.3.5. The alternating series $\sum_{n=1}^{\infty}(-1)^{n-1} b_n$ with

$$b_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is odd,} \\ \frac{1}{n^2} & \text{if } n \text{ is even.} \end{cases}$$

satisfies (ii) but not (i). Note that the series of the terms with even indices is the negative of the p-series for $p = 2$ and thus converges, whereas the series of the odd terms is the divergent harmonic series. Note that filling up a series with zeros does not change its convergence behaviour. Since the sum of a convergent and a divergent series is divergent (this follows by proof by contradiction from Theorem 11.10), $\sum_{n=1}^{\infty}(-1)^{n-1} b_n$ diverges.

Example 11.3.6. The alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{3n}{4n-1}$ does not converge since

$$\lim_{n\to\infty}\frac{3n}{4n-1}=\frac{3}{4}\neq 0$$

and thus (ii) is not satisfied, which is necessary for convergence. See the note following the statement of the Alternating Series Test.

Worked Example 11.3.7. Find whether the series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{n^2}{n^3+1}$ is convergent.

Solution. Clearly, we have an alternating series with

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{n^2}{n^3+1}=0.$$

To show that $b_n \geq b_{n+1}$ we have at least the following 2 choices:
1. Show the inequality directly by cross-multiplication and simplification (see Example 11.1.7.2).
2. Show that $f(x)=\frac{x^2}{x^3+1}$ has a negative derivative for sufficiently large x.
Alternatively, write

$$\frac{n^2}{n^3+1}=\frac{n^2+\frac{1}{n}}{n^3+1}-\frac{\frac{1}{n}}{n^3+1}=\frac{1}{n}-\frac{1}{n(n^3+1)}.$$

Then

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{n^2}{n^3+1}=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}+\sum_{n=1}^{\infty}\frac{(-1)^n}{n(n^3+1)}$$

is the sum of two alternating series which converge by the alternating series test; hence the series converges.

Tutorial 11.3.3. Thomas Exercises 12th ed. only: 10.6 p. 591: 1, 3, 5, 7, 9, 11.

11.3.4 Absolute Convergence and the Ratio and Root Tests

Definition. A series $\sum a_n$ is called **absolutely convergent** if the series of its absolute values $\sum |a_n|$ converges.

Example 11.3.8. The series

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}$$

is absolutely convergent since

$$\sum_{n=1}^{\infty}\left|\frac{(-1)^n}{n^2}\right|=\sum_{n=1}^{\infty}\frac{1}{n^2}$$

is a convergent p-series ($p=2$).

Definition. A series $\sum a_n$ is called **conditionally convergent** if it is convergent but not absolutely convergent.

Example 11.3.9. The alternating harmonic series $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$ is conditionally convergent.

Theorem 11.16. *If a series $\sum a_n$ is absolutely convergent, then it is convergent.*

Worked Example 11.3.10. Determine whether the series $\sum_{n=1}^{\infty}\frac{\sin n}{n^2}$ is convergent.

Solution. The series has infinitely many positive and negative terms, but it is not alternating. However,

$$\left|\frac{\sin n}{n^2}\right| = \frac{|\sin n|}{n^2} \leq \frac{1}{n^2}\,.$$

Since $\sum_{n=1}^{n} \frac{1}{n^2}$ is a convergent p-series, the Comparison Test shows that $\sum_{n=1}^{n} \left|\frac{\sin n}{n^2}\right|$ converges. Hence $\sum_{n=1}^{n} \frac{\sin n}{n^2}$ is absolutely convergent.

Theorem 11.17 (Ratio Test). (i) *If* $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *converges absolutely,*

(ii) *If* $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *diverges,*

(iii) *If* $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L = 1$, *then no conclusion about convergence of the series* $\sum_{n=1}^{\infty} a_n$ *can be drawn fron the Ratio Test.*

Note. The Ratio Test is useful if the n-th term contains many factors, including factors of the form r^n and $n!$. If it contains only factors of the form n^p, then it is normally not successful.

Worked Example 11.3.11. Determine if the series $\sum_{n=1}^{\infty}(-1)^n \frac{n^3}{3^n}$ converges.

Solution. Use the ratio test with $a_n = (-1)^n \frac{n^3}{3^n}$:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{(-1)^{n+1}(n+1)^3}{3^{n+1}}}{(-1)^n \frac{n^3}{3^n}}\right| = \frac{(n+1)^3}{3^{n+1}} \cdot \frac{3^n}{n^3}$$
$$= \frac{1}{3}\left(\frac{n+1}{n}\right)^3 = \frac{1}{3}\left(1+\frac{1}{n}\right)^3 \to \frac{1}{3} < 1 \text{ as } n \to \infty,$$

and hence the given series is (absolutely) convergent by the Ratio Test.

Worked Example 11.3.12. Determine if the series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$ converges.

Solution. Since the terms a_n are positive, we can drop the absolute value sign:

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(n+1)^{n+1}}{(n+1)!}}{\frac{n^n}{n!}} = \frac{(n+1)(n+1)^n}{(n+1)n!} \cdot \frac{n!}{n^n}$$
$$= \left(\frac{n+1}{n}\right)^n = \left(1+\frac{1}{n}\right)^n \to e > 1 \text{ as } n \to \infty,$$

and hence the given series is divergent by the ratio test. For a shorter proof, see Worked Example 11.1.5.5.

Theorem 11.18 (Root Test). (i) *If* $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *converges absolutely,*

(ii) *If* $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *diverges,*

(iii) *If* $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L = 1$, *then no conclusion about convergence of the series* $\sum_{n=1}^{\infty} a_n$ *can be drawn from the Root Test.*

The root test is most useful if a_n contains (only) n-th powers.

Worked Example 11.3.13. Determine if the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$ converges.

Solution. Here $a_n = \left(\frac{2n+3}{3n+2}\right)^n$. Hence

$$\sqrt[n]{|a_n|} = \frac{2n+3}{3n+2} = \frac{2+\frac{3}{n}}{3+\frac{2}{n}} \to \frac{2}{3} < 1 \text{ as } n \to \infty,$$

and hence the given series is (absolutely) convergent by the root test.

Tutorial 11.3.4. 1. Thomas Exercises 12th ed. only: 10.5 p. 585: 1, 3, 5, 7, 9, 11, 13, 15.
2. Thomas Exercises 11th ed. 11.6 p. 776: 13, 15, 19, 33, 39, 42 **or** 12th ed. 10.6 p. 591: 17, 19, 23, 35, 41, 44.

11.3.5 Strategy for Testing Series

There are no straightforward rules as to which test to apply in order to determine convergence or divergence of a series. However, the shape of the n-th term may give an educated guess which test may work. Note that more than one convergence test may work, or none of those we have learnt. Here is a list of patterns and suggested tests.

- If the series is of the form $\sum \frac{1}{n^p}$, use the p-series test.
- If the series is of the form $\sum ar^n$, it is a geometric series, which converges if and only if $|r| < 1$.
- For rational functions and roots of rational functions, consider "highest powers", which should give a p-series, and then apply the Comparison Test.
- Look out for $\lim_{n\to\infty} a_n \neq 0$. Then apply the Divergence Test.
- If the series is of the from $\sum(-1)^n b_n$ or $\sum(-1)^{n-1} b_n$ with positive b_n, then try the Alternating Series Test.
- If a_n contains factorials or n-th powers, the Ratio Test may be successful.
- If $a_n = (b_n)^n$, the Root Test may give a result.
- Cases for the p-series Test are normally inconclusive with the Ratio and Root Tests; in particular if a_n is a rational function of n, the Root and Ratio Test are inconclusive.
- If $a_n = f(n)$ with a function f for which the convergence behaviour of $\int_1^{\infty} f(x)\,dx$ can be found, the Integral Test may apply.

Worked Example 11.3.14. Complete the following table.

Series	Suggested Test	Reason
$\sum_{n=1}^{\infty} \frac{n-1}{2n-1}$	Divergence Test	$a_n \to \frac{1}{2} \neq 0$ as $n \to \infty$
$\sum_{n=1}^{\infty} \frac{\sqrt{n^3+1}}{3n^3+4n^2+2}$	p-series, Comparison Test	rational function and root (algebraic function)
$\sum_{n=1}^{\infty} ne^{-n^2}$	Integral Test	$\int_1^{\infty} xe^{-x^2}\,dx$ can be evaluated
$\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n^4+1}$	Alternating Series Test	
$\sum_{n=1}^{\infty} \frac{2^k}{k!}$	Ratio Test	terms of the series contain $k!$
$\sum_{n=1}^{\infty} \frac{1}{2+3^n}$	Geometric series, Comparison Test	compare with $\sum_{n=1}^{\infty} \frac{1}{3^n}$

Tutorial 11.3.5. 1. Thomas Exercises 11th ed. 11.3 pp. 759–760: 5, 9, 10, 16, 25 **or** 12th ed. 10.3 p. 575: 15, 99, 20, 26, 35.
2. Thomas Exercises 11th ed. 11.4 p. 765: 1, 3, 7, 11, 22, 31 **or** 12th ed. 10.4 p. 580: 17, 19, 25, 28, 41, 49.
3. Thomas Exercises 11th ed. 11.5 p. 770: 1, 3, 5, 13, 17, 21 **or** 12th ed. 10.5 p. 585: 17, 19, 21, 29, 33, 37, 43.

11.4 Power Series

LEARNING OUTCOMES:

On completion of this section you should

1. be able to recognize the form of a power series: $\sum_{n=0}^{\infty} c_n x^n$ or $\sum_{n=0}^{\infty} c_n (x-a)^n$,
2. know how to apply the tests for convergence to calculate the values of x for which a given power series converges,
3. know how to calculate the radius and interval of convergence.

A series

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

is called a **power series in** $(x-a)$ or a **power series centered at** a or a **power series about** a.

Note that $(x-a)^0 = 1$. If $a = 0$, the power series simplifies to

$$\sum_{n=0}^{\infty} c_n x^n .$$

For $x = a$, all terms from the second onwards are 0, so the series converges to c_0 for $x = a$.

Note that if a power series converges (for more than one x), then it represents a function of x. Two natural questions arise:

- For which x does the series converge?
- Has the series a "closed form" representation?

Example 11.4.1. The geometric series

$$\sum_{n=0}^{\infty} x^n$$

converges if $-1 < x < 1$ and diverges otherwise, see Theorem 11.7.

Worked Example 11.4.2. For which values of x does the series $\displaystyle\sum_{n=0}^{\infty} n!x^n$ converge?

Solution. We use the Ratio Test with $a_n = n!x^n$. From

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| = \lim_{n\to\infty}(n+1)|x| = \infty \qquad \text{if } x \neq 0$$

it thus follows that the series diverges if $x \neq 0$. It therefore converges only for $x = 0$.

Worked Example 11.4.3. For which values of x does the Bessel function of order 0 defined by

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$$

converge?

Solution. Let $a_n = \dfrac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$. From

$$\begin{aligned}\left|\frac{a_{n+1}}{a_n}\right| &= \left|\frac{(-1)^{n+1}x^{2(n+1)}}{2^{2(n+1)}[(n+1)!]^2}\cdot\frac{2^{2n}(n!)^2}{(-1)^n x^{2n}}\right| \\ &= \frac{x^{2n+2}}{2^{2n+2}(n+1)^2(n!)^2}\cdot\frac{2^{2n}(n!)^2}{x^{2n}} = \frac{x^2}{4(n+1)^2} \to 0 < 1 \text{ as } n\to\infty\end{aligned}$$

it follows that the series converges for all x. Hence the domain of the Bessel function J_0 is $(-\infty, \infty)$.

Worked Example 11.4.4. For which values of x does the series $\displaystyle\sum_{n=0}^{\infty} \frac{(x-3)^n}{n}$ converge?

Solution. Let $a_n = \dfrac{(x-3)^n}{n}$. From

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(x-3)^{n+1}}{n+1}\cdot\frac{n}{(x-3)^n}\right| = \frac{1}{1+\dfrac{1}{n}}|x-3| \to |x-3| \qquad \text{as } n\to\infty$$

it follows that the series converges if $|x-3| < 1$ and diverges if $|x-3| > 1$. If $|x-3| = 1$, we have the two case $x - 3 = 1$ and $x - 3 = -1$. For $x - 3 = 1$ we have $(x-3)^n = 1$ for all n, and the series becomes the (divergent) harmonic series. For $x - 3 = -1$ we have $(x-3)^n = (-1)^n$ for all n, and the series becomes the (convergent) alternating harmonic series. Since

$$|x-3| < 1 \Leftrightarrow -1 < x-3 < 1 \Leftrightarrow 2 < x < 4$$

it follows that the series converges exactly for $x \in [2, 4)$.

Theorem 11.19. *There are three alternatives for the domain of a power series* $\sum_{n=0} c_n(x-a)^n$*:*

(i) *The series converges only for* $x = a$.
(ii) *The series converges for all* $x \in \mathbb{R}$.
(iii) *There is* $R > 0$ *such that the series converges if* $|x-a| < R$ *and diverges if* $|x-a| > R$.

The number R in case (iii) is called the **radius of convergence**. It is convenient to put the radius of convergence as $R = 0$ in (i) and $R = \infty$ in (ii).
Note that (iii) says that the series converges for x in the interval $(a-R, a+R)$ and diverges outside $[a-R, a+R]$. For $x = a-R$ and $x = a+R$ anything can happen. In any case, the domain of the series is an interval, called the **interval of convergence**, and for $0 < R < \infty$, there are four possibilities for the interval of convergence:

$$(a-R, a+R), \quad (a-R, a+R], \quad [a-R, a+R), \quad [a-R, a+R].$$

Note that the Ratio and Root Tests fail at the endpoints of the interval of convergence, and other tests have to be used.

Worked Example 11.4.5. Find the radius of convergence and the interval of convergence for the series

1. $\displaystyle\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n^3+1}}$

Solution. Let $a_n = \dfrac{(-3)^n x^n}{\sqrt{n^3+1}}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-3)^{n+1}x^{n+1}}{\sqrt{(n+1)^3+1}} \cdot \frac{\sqrt{n^3+1}}{(-3)^n x^n}\right| = \left|-3x\sqrt{\frac{n^3+1}{(n+1)^3+1}}\right|$$

$$= 3|x|\sqrt{\frac{1+\dfrac{1}{n^3}}{\left(1+\dfrac{1}{n}\right)^3+\dfrac{1}{n^3}}} \to 3|x| \qquad \text{as } n \to \infty$$

and the Ratio Test gives that the series converges for $|x| < \dfrac{1}{3}$ and diverges for $|x| > \dfrac{1}{3}$. Hence the radius of convergence is $R = \dfrac{1}{3}$.

For $|x| = \dfrac{1}{3}$,

$$|a_n| = \frac{3^n|x|^n}{\sqrt{n^3+1}} = \frac{1}{\sqrt{n^3+1}} \le \frac{1}{n^{\frac{3}{2}}}$$

and therefore the series is a convergent by the Comparison Test and the p-series Test ($p = \dfrac{3}{2}$). Hence the interval of convergence is $\left[-\dfrac{1}{3}, \dfrac{1}{3}\right]$.

2. $\displaystyle\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}$

Solution. Let $a_n = \dfrac{n(x+2)^n}{3^{n+1}}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)(x+2)^{n+1}}{3^{n+2}} \cdot \frac{3^{n+1}}{n(x+2)^n}\right|$$

$$= \left(1+\frac{1}{n}\right)\frac{|x+2|}{3} \to \frac{|x+2|}{3} \text{ as } n \to \infty$$

and the Ratio Test gives that the series converges for $|x+2| < 3$ and diverges for $|x+2| > 3$. Hence the radius of convergence is $R = 3$.

For $|x+2| = 3$, the sequence

$$|a_n| = \frac{n|x+2|^n}{3^{n+1}} = \frac{n}{3}$$

does not converge to 0. Hence the series does not converge for these x by the Divergence Test. Thus the interval of convergence is $(-5, 1)$.

Below is a summary of the examples of this section.

	Series	Radius of convergence	Interval of convergence
Geometric series	$\sum_{n=0}^{\infty} x^n$	$R = 1$	$(-1, 1)$
Example 11.4.2	$\sum_{n=0}^{\infty} n!x^n$	$R = 0$	$\{0\}$
Bessel function J_0	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$	$R = \infty$	$(-\infty, \infty)$
Example 11.4.4	$\sum_{n=0}^{\infty} \frac{(x-3)^n}{n}$	$R = 1$	$[2, 4)$
Example 11.4.5.1	$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n^3+1}}$	$R = \frac{1}{3}$	$\left[-\frac{1}{3}, \frac{1}{3}\right]$
Example 11.4.5.2	$\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}$	$R = 3$	$(-5, 1)$

Tutorial 11.4.1. Thomas Exercises 11th ed. 11.7 p. 788: 1, 5, 9, 13 **or** 12th ed. 10.7 pp. 600–601: 1, 5, 9, 13, 37, 39.

11.5 Representations of Functions as Power Series

LEARNING OUTCOMES:

On completion of this section you should

1. know how to represent certain types of functions as sums of power series,
2. know how to differentiate and integrate power series term-by-term.

Recall that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad |x| < 1$$

and that the geometric series converges exactly for $|x| < 1$.

Worked Example 11.5.1. Express $\frac{1}{1+x^2}$ as a sum of a power series about 0.

Solution. Replace x by $-x^2$ in the geometric series:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \ldots$$

This series converges for $|-x^2| < 1$, i. e., $|x| < 1$. Hence the interval of convergence is $(-1, 1)$.

Worked Example 11.5.2. Find a power series representation for $1/(x+2)$ about 0.

Solution. Make the constant in the denominator 1 in order to apply the geometric series:

$$\frac{1}{2+x} = \frac{1}{2\left(1+\frac{x}{2}\right)} = \frac{1}{2}\,\frac{1}{1-\left(-\frac{x}{2}\right)}$$
$$= \frac{1}{2}\sum_{n=0}^{\infty}\left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}x^n.$$

The series converges if and only if $\left|-\frac{x}{2}\right| < 1$, i. e., $|x| < 2$. Hence the interval of convergence is $(-2, 2)$.

Worked Example 11.5.3. Find a power series representation for $x^3/(x+2)$ about 0.

Solution. This is x^3 times the function in Example 11.5.2. Hence its interval of convergence is $(-2, 2)$ as there and

$$\frac{x^3}{x+2} = x^3\cdot\frac{1}{x+2} = x^3\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}x^n = \sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}x^{n+3}$$
$$= \frac{1}{2}x^3 - \frac{1}{4}x^4 + \frac{1}{8}x^5 - \frac{1}{16}x^6 + \dots$$

Writing powers as x^n, not x^{n+3}, we get

$$\frac{x^3}{x+2} = \sum_{n=3}^{\infty}\frac{(-1)^{n-1}}{2^{n-2}}x^n.$$

Theorem 11.20. *If the power series* $\sum_{n=1}^{\infty} c_n(x-a)^n$ *has radius of convergence* $R > 0$*, then the function* f *defined by*

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

is differentiable (and thus continuous) on the interval $(a-R, a+R)$*, and*

(i) $f'(x) = \sum_{n=0}^{\infty} nc_n(x-a)^{n-1} = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots,$

(ii) $\int f(x)\,dx = C + \sum_{n=0}^{\infty} c_n\frac{(x-a)^{n+1}}{n+1} = C + c_0(x-a) + c_1\frac{(x-a)^2}{2} + c_2\frac{(x-a)^3}{3} + \dots$

The radii of convergence of the power series in (i) *and* (ii) *are* R*.*

Note. Theorem 11.20 says that a power series can be differentiated and integrated term by term. This is seen by rewriting (i) and (ii) there in the form

(iii) $\frac{d}{dx}\left[\sum_{n=0}^{\infty} c_n(x-a)^n\right] = \sum_{n=1}^{\infty}\frac{d}{dx}[c_n(x-a)^n],$

(iv) $\int\left[\sum_{n=0}^{\infty} c_n(x-a)^n\right]dx = \sum_{n=1}^{\infty}\int c_n(x-a)^n\,dx.$

Note. Although the radius of convergence does not change, the interval of convergence may change if we differentiate or integrate a power series.

Example 11.5.4. The derivative of the Bessel function

$$J_0(x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$$

is represented by the power series

$$J_0'(x) = \sum_{n=0}^{\infty} \frac{d}{dx} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2} = \sum_{n=1}^{\infty} \frac{(-1)^n 2n x^{2n-1}}{2^{2n}(n!)^2}$$

Worked Example 11.5.5. Find a power series expansion of $\dfrac{1}{(1-x)^2}$ about 0 by differentiating the power series of $\dfrac{1}{1-x}$. Find its radius of convergence.

Solution.

$$\begin{aligned}\frac{1}{(1-x)^2} &= \frac{d}{dx}\frac{1}{1-x} = \frac{d}{dx}\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty}\frac{d}{dx}x^n \\ &= \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty}(n+1)x^n.\end{aligned}$$

The radius of convergence is 1, as with the geometric series.

Worked Example 11.5.6. Find a power series expansion of $\ln(1-x)$ about 0 and its radius of convergence.

Solution. Observe

$$\frac{d}{dx}\ln(1-x) = -\frac{1}{1-x}\,.$$

Then

$$\begin{aligned}-\ln(1-x) &= \int \frac{dx}{1-x} = \int \sum_{n=0}^{\infty} x^n\,dx \\ &= C + \sum_{n=0}^{\infty}\frac{x^{n+1}}{n+1} = C + \sum_{n=1}^{\infty}\frac{x^n}{n}\,.\end{aligned}$$

The radius of convergence is 1, as with the geometric series.
To find C, put $x = 0$ in this equation. Then $-\ln(1) = C$, i. e., $C = 0$. Hence

$$\ln(1-x) = -\sum_{n=1}^{\infty}\frac{x^n}{n}\,.$$

Note. Putting $x = \dfrac{1}{2}$ above we obtain

$$\ln 2 = -\ln\frac{1}{2} = \sum_{n=1}^{\infty}\frac{1}{n2^n}.$$

Worked Example 11.5.7. Find a power series representation for $f(x) = \arctan x$.

Solution. Since

$$f'(x) = \frac{1}{1+x^2}\,,$$

we conclude from Example 11.5.1 that

$$\begin{aligned}\arctan x &= \int \frac{dx}{1+x^2} = \int \sum_{n=0}^{\infty}(-1)^n x^{2n}\,dx \\ &= C + \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1} = C + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots.\end{aligned}$$

Putting $x = 0$, we get $C = 0$. Hence

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots.$$

Note that this series has radius of convergence 1.

Worked Example 11.5.8. Find $\displaystyle\int \frac{dx}{1+x^7}$ as a power series

Solution. Since

$$\frac{1}{1+x^7} = \frac{1}{1-(-x^7)} = \sum_{n=0}^{\infty} (-1)^n x^{7n},$$

we conclude that

$$\int \frac{dx}{1+x^7} = \int \sum_{n=0}^{\infty} (-1)^n x^{7n} dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{7n+1}}{7n+1}.$$

Note that this series has radius of convergence 1.

Tutorial 11.5.1. 1. Find power series representations for the following functions and indefinite integrals:

(a) $\dfrac{1}{5-x}$ (b) $\dfrac{1}{(1+x)^2}$ (c) $\dfrac{1}{(1+x)^3}$ (d) $\dfrac{x^2}{(1+x)^3}$

(e) $\ln\left(\dfrac{1+x}{1-x}\right)$ (f) $\displaystyle\int \frac{t}{1-t^8}$ (g) $\dfrac{x - \arctan x}{x^3}$

Also find the intervals of convergence.

2. Use the power series of arctan to prove the following series representation of π:

$$\pi = 2\sqrt{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)3^n}.$$

11.6 Taylor and Maclaurin Series

LEARNING OUTCOMES:

On completion of this section you should

1. be able to identify Taylor series of functions centred at a,
2. be able to identify Maclaurin series (Taylor series centred at 0),
3. be able to identify n-th degree Taylor polynomials of f at a, and Maclaurin polynomials,
4. know the Maclaurin series of the elementary functions and their intervals of convergence.

Recursively differentiating in Theorem 11.20 and substituting $x = a$, we find

Theorem 11.21. *If f has a power series representation at a, written as*

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \quad |x-a| < R,$$

then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

That is, every function f which has a power series expansion can be written as

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \\ &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots, \end{aligned}$$

and this power series is called the **Taylor series of the function** f **at** a (or **about** a or **centred at** a).
The polynomial

$$\begin{aligned} T_n(x) &= \sum_{i=0}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i \\ &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n \end{aligned}$$

is called the n**-th degree Taylor polynomial at** a.
In the special case $a = 0$, the expansion reads

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \\ &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots, \end{aligned}$$

and in this case the power series is also called the **Maclaurin series of the function** f.

We are now going to find the Maclaurin series for some important functions. We assume, without proof, that these functions coincide with their Maclaurin series expansions. The existence of a series representation will be shown in second year. We will only show that the Maclaurin series converge.

Worked Example 11.6.1. Find the Maclaurin series of the function $f(x) = e^x$ and its radius of convergence.

Solution. We know

$$f^{(n)}(x) = e^x,$$

and so the Maclaurin series expansion of e^x is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

With $a_n = \dfrac{x^n}{n!}$ the Ratio Test gives

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \frac{|x|}{n} \to 0 < 1.$$

Hence the Maclaurin series of e^x converges for all x.
The Maclaurin series of e^x is

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for all } x.}$$

In particular,

$$\boxed{e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots}$$

Worked Example 11.6.2. Find the Taylor series for $f(x) = e^x$ at 2.

Solution. We know $f^{(n)}(2) = e^2$. Hence

$$\boxed{e^x = \sum_{n=0}^{\infty} \frac{e^2}{n!}(x-2)^n \quad \text{for all } x.}$$

Worked Example 11.6.3. Find the Maclaurin series for $f(x) = \sin x$.

Solution. We know

$$f(x) = \sin x, \quad f'(x) = \cos x, \quad f''(x) = -\sin x,$$

and therefore, since $f^{(n+2)} = (f'')^{(n)} = -f^{(n)}$,

$$f^{(2n)}(x) = (-1)^n f(x) = (-1)^n \sin x, \quad f^{(2n+1)}(x) = (-1)^n f'(x) = (-1)^n \cos x.$$

It follows that

$$f^{(2n)}(0) = 0, \quad f^{(2n+1)}(x) = (-1)^n.$$

Hence, only terms with odd powers occur in the Maclaurin expansion of $\sin x$:

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n &= \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\end{aligned}$$

Thus

$$\boxed{\begin{aligned}\sin x &= \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\end{aligned}}$$

Worked Example 11.6.4. Find the Maclaurin series for $f(x) = \cos x$.

Solution. Since we can differentiate power series term by term, we have

$$\begin{aligned}\cos x = \frac{d}{dx}\sin x &= \frac{d}{dx}\left[\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}\right] \\ &= \sum_{n=0}^{\infty}(-1)^n \frac{(2n+1)x^{2n}}{(2n+1)!}.\end{aligned}$$

Hence

$$\boxed{\begin{aligned}\cos x &= \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!} \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\end{aligned}}$$

Worked Example 11.6.5. Find the Maclaurin series for $x\cos x$.

Solution. From above,

$$x\cos x = x\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!} = \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n)!}.$$

Note. By Theorem 11.21, if a function has a power series representation about a, then the coefficients of the power series are uniquely determined.

Below is a table of important Maclaurin series we have found so far, together with their intervals of convergence.

$\dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \ldots$	$(-1, 1)$
$e^x = \displaystyle\sum_{n=0}^{\infty}\frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$	$(-\infty, \infty)$
$\sin x = \displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$	$(-\infty, \infty)$
$\cos x = \displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$	$(-\infty, \infty)$
$\arctan x = \displaystyle\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$	$(-1, 1)$

Worked Example 11.6.6. Evaluate $\displaystyle\int e^{-x^2}\,dx$ as an infinite series.

Solution. The Maclaurin series expansion of e^{-x^2} is

$$e^{-x^2} = \sum_{n=0}^{\infty}\frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}.$$

Hence

$$\int e^{-x^2}\,dx = C + \sum_{n=0}^{\infty}\int\frac{(-1)^n x^{2n}}{n!}\,dx = C + \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)n!}.$$

Worked Example 11.6.7. Evaluate $\displaystyle\lim_{x\to 0}\frac{e^x - 1 - x}{x^2}$.

Solution. Repeated application of l'Hôpital's rule would work, but we use Maclaurin series here:

$$\begin{aligned}
\lim_{x\to 0}\frac{e^x - 1 - x}{x^2} &= \lim_{x\to 0}\frac{\left(1 + \dfrac{x}{1!} + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \ldots\right) - 1 - x}{x^2} \\
&= \lim_{x\to 0}\frac{\dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \ldots}{x^2} \\
&= \lim_{x\to 0}\left(\frac{1}{2!} + \frac{x}{3!} + \ldots\right) \\
&= \frac{1}{2}
\end{aligned}$$

because power series are continuous functions.

Power series can be multiplied and divided to obtain new power series.

Worked Example 11.6.8. Find the first 3 terms of the power series for

1. $e^x \sin x$

   Solution.

$$\begin{aligned} e^x \sin x &= \left(1 + x + \frac{x^2}{2} + \dots\right)\left(x - \frac{x^3}{6} + \dots\right) \\ &= x + x^2 + \frac{x^3}{2} - \frac{x^3}{6} + \text{terms in } x^4 \text{ and higher powers of } x \\ &= x + x^2 + \frac{x^3}{3} + \dots \end{aligned}$$

2. $\sec x$

   Solution. Here we use a "geometric series" approach:

$$\begin{aligned} \sec x = \frac{1}{\cos x} &= \frac{1}{1 - \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \dots\right)} \\ &= 1 + \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \dots\right) + \left(\frac{x^2}{2!} - \frac{x^4}{4!} + \dots\right)^2 + \dots \\ &= 1 + \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^4}{4} + \dots \\ &= 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots \end{aligned}$$

3. $\tan x$

   Solution. Using the result from 2. and the technique from 1., we obtain

$$\begin{aligned} \tan x = \frac{\sin x}{\cos x} &= \left(x - \frac{x^3}{6} + \frac{x^5}{120} + \dots\right)\left(1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots\right) \\ &= x - \frac{x^3}{6} + \frac{x^5}{120} + \frac{1}{2}x^3 - \frac{1}{12}x^5 + \frac{5}{24}x^5 + \dots \\ &= x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \dots \end{aligned}$$

Tutorial 11.6.1. Thomas Exercises 11th ed. 11.8 pp. 794–795: 1, 7, 9, 13, 17, 21, 25 **or** 12th ed. 10.8 p. 606: 1, 3, 9, 11, 15, 19, 23, 27, 33, 35.

11.7 Binomial Series

LEARNING OUTCOMES:

On completion of this section you should

1. know how to write the Binomial Series in expanded form as an infinite series and in abbreviated form using sigma notation,
2. know under which conditions the Binomial Series converges or diverges,
3. know the definition of the binomial coefficient $\binom{k}{n}$ if k is no longer a positive integer and $n \geq 1$.

Consider the function $f(x) = (1+x)^k$ for any real k. Then

$$\begin{aligned} f(x) &= (1+x)^k \\ f'(x) &= k(1+x)^{k-1} \\ f''(x) &= k(k-1)(1+x)^{k-2} \\ &\vdots \\ f^{(n)}(x) &= k(k-1)\cdots(k-n+1)(1+x)^{k-n} \end{aligned}$$

and therefore

$$f^{(n)}(0) = k(k-1)\cdots(k-n+1).$$

The Maclaurin series of $(1+x)^k$ is called the **Binomial Series** and we have

Theorem 11.22 (**Binomial Series**). *For $k \in \mathbb{R}$ and $|x| < 1$,*

$$\begin{aligned} (1+x)^k &= 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \ldots \\ &= \sum_{n=0}^{\infty} \binom{k}{n} x^n, \end{aligned}$$

where $\binom{k}{n} = \dfrac{k(k-1)\cdots(k-n+1)}{n!}$ $(n \geq 1)$ *and* $\binom{k}{0} = 1.$

Note. The binomial coefficent $\binom{k}{n}$ defined in Theorem 11.22 coincides with the usual binomial coefficient when $k \geq n$ is a nonnegative integer.

Worked Example 11.7.1. Find the Maclaurin series and radii of convergence for the following functions:

1. $\dfrac{1}{(1+x)^2}$

   Solution. We find the binomial coefficient for $k = -2$:

   $$\begin{aligned} \binom{-2}{n} &= \frac{(-2)(-3)(-4)\cdots(-2-n+1)}{n!} \\ &= \frac{(-1)^n 2\cdot 3\cdot 4\cdots n(n+1)}{n!} = (-1)^n(n+1). \end{aligned}$$

   The Maclaurin series converges for $|x| < 1$, and

   $$\begin{aligned} \frac{1}{(1+x)^2} &= (1+x)^{-2} = \sum_{n=0}^{\infty} \binom{-2}{n} x^n \\ &= \sum_{n=0}^{\infty} (-1)^n(n+1)x^n = 1 - 2x + 3x^2 - 4x^3 + \ldots \end{aligned}$$

2. $\dfrac{1}{\sqrt{4-x}}$

Solution. First rewrite

$$\frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{4\left(1-\frac{x}{4}\right)}} = \frac{1}{2\sqrt{1-\frac{x}{4}}} = \frac{1}{2}\left(1-\frac{x}{4}\right)^{-\frac{1}{2}}.$$

The Binomial Theorem with $k = -\dfrac{1}{2}$ and x replaced with $-\frac{x}{4}$ gives

$$\frac{1}{\sqrt{4-x}} = \frac{1}{2}\left(1-\frac{x}{4}\right)^{-\frac{1}{2}} = \frac{1}{2}\sum_{n=0}^{\infty}\binom{-\frac{1}{2}}{n}\left(-\frac{x}{4}\right)^n$$

Calculating

$$\begin{aligned}\binom{-\frac{1}{2}}{n} &= \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\cdots\left(-\frac{1}{2}-n+1\right)}{n!}\\ &= \frac{(-1)^n\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\cdots\left(\frac{2n-1}{2}\right)}{n!}\\ &= \frac{(-1)^n 1\cdot 2\cdot\ 3\cdots(2n-1)\cdot(2n)\cdot 2^{-2n}}{(n!)^2}\\ &= (-1)^n\frac{(2n)!}{2^{2n}(n!)^2}.\end{aligned}$$

it follows that

$$\frac{1}{\sqrt{4-x}} = \sum_{n=0}^{\infty}\frac{(2n)!}{(n!)^2 2^{4n+1}}x^n.$$

The series converges when $\left|-\dfrac{x}{4}\right| < 1$, i. e., when $|x| < 4$. So the radius of convergence is $R = 4$.

Tutorial 11.7.1. Thomas Exercises 11th ed. 11.10 pp. 815–817: 3, 7, 9, 63 **or** 12th ed. 10.10 pp. 620–622: 3, 7, 9, 10, 59.

Chapter 12

Differential Equations

LEARNING OUTCOMES:

On completion of this section you should be able to

1. identify the type of differential equation under consideration,
2. know the technique of solving such a differential equation,
3. find the general solution to the given differential equation,
4. find the particular solution to a given initial value problem,
5. know how to solve the examinable worked out problems from this section.

A *differential equation* is an equation that involves derivatives or differentials of unknown functions. An equation that involves $x, y, y', y'', \ldots, y^{(n)}$, where y is a function of x is called an *ordinary differential equation of order n* (n^{th} order ODE).

Worked Example 12.0.1. 1. $\dfrac{dy}{dx} = -\cos x$ or $y' = -\cos x$ is a 1st order ODE.
2. $\dfrac{d^2y}{dx^2} + kx - \sin x = 0$ or $y'' + kx - \sin x = 0$ is a 2nd order ODE.
3. $(y''')^4 - x^2(y'')^5 + 4xy = xe^x$ is a 3rd order ODE.

The order of an ODE is thus the highest order derivative occuring in the ODE. In this course we will primarily deal with solving 1st order ODE's.
If a function $f(x)$ has the property that substituting f and its derivatives into a differential equation satisfies the ODE, then f is called a solution to the ODE.

Worked Example 12.0.2. $f(x) = \cos 2x$ and $g(x) = A\sin 2x + B\cos 2x$, $A, B \in \mathbb{R}$ are solutions to the ODE

$$y'' + 4y = 0.$$

Solution.

$$f(x) = \cos 2x \Rightarrow f'(x) = -2\sin 2x \Rightarrow f''(x) = -4\cos 2x.$$

Then $f''(x) + 4f(x) = -4\cos 2x + 4(\cos 2x) = 0$ so that f satisfies $y'' + 4y = 0$. Hence, f is a solution to the given ODE.
Also,

$$\begin{aligned} g(x) = A\sin 2x + B\cos 2x &\Rightarrow g'(x) = 2A\cos 2x - 2B\sin 2x \\ &\Rightarrow g''(x) = -4A\sin 2x - 4B\cos 2x. \end{aligned}$$

Then $g''(x) + 4g(x) = (-4A\sin 2x - 4B\cos 2x) + 4(A\sin 2x + B\cos 2x) = 0$ so that $g(x)$ also satisfies $y'' + 4y = 0$. Hence, g is also a solution to the given ODE.

Tutorial 12.0.1. 1. Verify that $f(x) = \sin 2x$ and $g(x) = R\cos(2x - \alpha)$, $\alpha, R \in \mathbb{R}$ are solutions to $\dfrac{d^2y}{dx^2} + 4y = 0$.

2. Prove that $y = Cx^{-\frac{2}{3}}$ with $C \in \mathbb{R}$ is a solution to $2xy^3 + 3x^2y^2\dfrac{dy}{dx} = 0$.

3. If $C \in \mathbb{R}$ show that $y = Cx^3$ is a solution to $x^3y''' + x^2y'' - 3xy' - 3y = 0$.

4. If A and B are any real constants, show that $y = \sin x - 3\cos x + Ae^{-x} + Be^{-2x}$ is a solution to $y'' + 3y' + 2y = 10\sin x$.

5. Identify the order of each ODE in the above tutorial examples.

6. (a) Show that any function of the form

$$y = Ae^{kx} + Be^{-kx}$$

or

$$y = C\sinh mx + D\cosh mx$$

satisfies the differential equation $y'' = k^2y$.

(b) Find $y = y(x)$ such that $y'' = 9y$, $y(0) = -4$, $y'(0) = 6$.

As can be noted, solutions to ODE's are not unique. We next focus on finding a solution to various types of ODE's.

12.1 Separable Differential Equations

The *separable* ODE's are the simplest type. As the name suggests if the ODE can be arranged such that all x's are on the left-hand side of the equation and all y's are on the right-hand side of the equation (or vice versa), then the ODE is called separable. The general form of a separable ODE is

$$M(x) + N(y)\frac{dy}{dx} = 0 \quad \text{or} \quad M(x)\,dx + N(y)\,dy = 0.$$

The technique to find a solution is

$$M(x)\,dx + N(y)\,dy = 0 \Rightarrow \int M(x)\,dx + \int N(y)\,dy = C.$$

Worked Example 12.1.1. Solve

$$\frac{dy}{dx} = \frac{\sin x}{\cos y}.$$

Solution. This ODE is separable. Thus we solve as follows:

$$\begin{aligned}\frac{dy}{dx} = \frac{\sin x}{\cos y} &\Rightarrow \cos y\,dy = \sin x\,dx\\ &\Rightarrow \cos y\,dy - sinx\,dx = 0\\ &\Rightarrow \int \cos y\,dy - \int \sin x\,dx = C\\ &\sin y + \cos x = C \text{ is the general solution.}\end{aligned}$$

Note that:

1. The solution given above, with the arbitrary constant C, is the general solution. It gives a general family of solutions.
2. The constant arising from $\int \cos y\,dy$ and the constant from $\int \sin x\,dx$ are combined into the one constant C.
3. In general, an nth order ODE will have n arbitrary constants in its solution.
4. Often the general solution is given in implicit form; it may be left that way.

Worked Example 12.1.2. Solve

$$y^4 e^{2x} + y' = 0.$$

Solution:

$$\begin{aligned} y^4 e^{2x} + y' = 0 &\Rightarrow y^4 e^{2x} + \frac{dy}{dx} = 0 \\ &\Rightarrow \frac{dy}{dx} = -y^4 e^{2x} \\ &\Rightarrow -y^{-4}\, dy = e^{2x}\, dx \\ &\Rightarrow -\int y^{-4}\, dy = \int e^{2x}\, dx \\ &\Rightarrow \frac{1}{3}y^{-3} = \frac{1}{2}e^{2x} + C \text{ is the general solution in implicit form.} \end{aligned}$$

We may simplify the implicit form to give the general solution explicitly as follows:

$$\begin{aligned} \frac{1}{3}y^{-3} = \frac{1}{2}e^{2x} + C &\Rightarrow 2y^{-3} = 3e^{2x} + 6C \\ &\Rightarrow \frac{y^3}{2} = \frac{1}{3e^{2x} + K} \quad (6C = K) \\ &\Rightarrow y = \left(\frac{2}{3e^{2x} - K}\right)^{\frac{1}{3}} \text{ is the explicit general solution.} \end{aligned}$$

An *initial value problem* arises when we wish to solve an ODE with an *initial condition*. This allows one to calculate a particular value for the constant C in the general solution. The resulting solution is then a **particular** solution to the ODE.

Worked Example 12.1.3. Find the particular solution to the ODE

$$2x(y+1)\, dx - y\, dy = 0$$

that satisfies the initial condition $y(0) = -2$, i. e., $y = -2$ when $x = 0$.

Solution.

$$\begin{aligned} 2x(y+1)\, dx - y\, dy = 0 &\Rightarrow 2x\, dx - \frac{y}{y+1}\, dy = 0 \quad (y \neq -1) \\ &\Rightarrow \int 2x\, dx - \int \frac{y}{y+1}\, dy = C \\ &\Rightarrow \int 2x\, dx - \int \left(1 - \frac{1}{y+1}\right) dy = C \quad \text{(long division – partial fractions)} \\ &\Rightarrow x^2 - (y - \ln|y+1|) = C \\ &\Rightarrow x^2 - y + \ln|y+1| = C \text{ is the general implicit solution.} \end{aligned}$$

Since $y = -2$ when $x = 0$, we then substitute these values into the general solution above to find the value of C.

$$\begin{aligned} x = 0,\ y = -2 \Rightarrow C &= x^2 - y + \ln|y+1| \\ &= 0 - (-2) + \ln|-2+1| \\ &= 2 + \ln 1 \\ &= 2. \end{aligned}$$

Hence the particular implicit solution is

$$x^2 - y + \ln|y+1| = 2.$$

Tutorial 12.1.1. Solve the following separable ODE's.
1. $y' = 2xy$.
2. $xy\dfrac{dy}{dx} = 1 - x^2$.
3. $(1 + 3x^2)\,dy = (1 - 3y^2)\,dx$. Find the particular solution that satisfies $y = \sqrt{3}$ when $x = 1$.
4. $xy + \dfrac{dy}{dx}e^{x^2}\ln y = 0$.
5. $\dfrac{dy}{dx} = \dfrac{1}{(3x+2)^2}$.
6. $y(x + 1)y' = x^2 + 2$, $y(0) = 1$.

12.2 Linear Differential Equations

Equations of the form

$$\frac{dy}{dx} + p(x)y = q(x)$$

are called **first-order linear** differential equations. If the functions p and q are continuous, we can solve this type of ODE by multiplying the equation by $e^{\int p(x)\,dx}$, called an **integrating factor**. Then we get

$$\frac{dy}{dx}e^{\int p(x)\,dx} + p(x)ye^{\int p(x)\,dx} = q(x)e^{\int p(x)\,dx}$$

However, the LHS of the above equation is

$$\frac{dy}{dx}e^{\int p(x)\,dx} + p(x)ye^{\int p(x)\,dx} = \frac{d}{dx}\left(ye^{\int p(x)\,dx}\right) \quad \text{(verify this identity)}$$

so that the linear ODE is now reduced to

$$\frac{d}{dx}\left(ye^{\int p(x)\,dx}\right) = q(x)e^{\int p(x)\,dx}$$

which we then solve by direct integration.
If

$$\mu(x) = e^{\int p(x)\,dx}$$

is the integrating factor, then the general solution is

$$y(x) = \frac{1}{\mu(x)}\left(\int q(x)\mu(x)\,dx + C\right).$$

Worked Example 12.2.1. Solve the differential equation

$$y' + \frac{1}{x}y = \frac{1}{x^2}, \quad x > 0.$$

Solution. The ODE is linear:

$$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = \frac{1}{x^2}$$

with $p(x) = \dfrac{1}{x}$ and $q(x) = \dfrac{1}{x^2}$ continuous on $(0, \infty)$. We multiply across by the integrating factor

$$e^{\int p(x)\,dx} = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x.$$

and then solve accordingly:

$$\begin{aligned}
\frac{dy}{dx} + \left(\frac{1}{x}\right)y = \frac{1}{x^2} &\Rightarrow x\frac{dy}{dx} + x\left(\frac{1}{x}\right)y = x\frac{1}{x^2} \\
&\Rightarrow \frac{dy}{dx}(xy) = \frac{1}{x} \\
&\Rightarrow xy = \ln x + C \\
&\Rightarrow y = \frac{1}{x}\ln x + \frac{C}{x} \text{ is the general explicit solution.}
\end{aligned}$$

Worked Example 12.2.2. Find the particular solution to

$$y' - y\tan x = \sec x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

which satisfies $y\left(\frac{\pi}{4}\right) = 0$.
Solution. The ODE is linear:

$$\frac{dy}{dx} - y\tan x = \sec x$$

with $p(x) = -\tan x$ and $q(x) = \sec x$ continuous for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. We then multiply by the integrating factor

$$e^{\int p(x)\,dx} = e^{\int(-\tan x)\,dx} = e^{\ln|\cos x|} = \cos x.$$

Note that $-\frac{\pi}{2} < x < \frac{\pi}{2} \Rightarrow \cos x > 0 \Rightarrow |\cos x| = \cos x$. The ODE then reduces as follows:

$$\begin{aligned}\frac{dy}{dx} - y\tan x = \sec x &\Rightarrow \cos x\frac{dy}{dx} - y\cos x\tan x = \cos x\sec x\\ &\Rightarrow \frac{d}{dx}(y\cos x) = 1\\ &\Rightarrow y\cos x = \int dx\\ &\Rightarrow y\cos x = x + C \text{ is the general solution.}\end{aligned}$$

Since $y\left(\frac{\pi}{4}\right) = 0$,

$$\begin{aligned}x = \frac{\pi}{4},\ y = 0 &\Rightarrow (0)\cos\frac{\pi}{4} = \frac{\pi}{4} + C\\ &\Rightarrow C = -\frac{\pi}{4}.\end{aligned}$$

Hence the particular solution which satisfies $y\left(\frac{\pi}{4}\right) = 0$ is

$$y = \left(x - \frac{\pi}{4}\right)\sec x.$$

Tutorial 12.2.1. Thomas Exercises 9.2 11th ed. pp. 641–643: 1, 3, 5, 9, 15, 19, 35 **or** 12th ed. pp. 526–528: 1, 3, 5, 9, 15, 19, 31.

12.3 Homogeneous Differential Equations

A function of two variables $f(x, y)$ is said to be homogeneous of degree n if

$$f(tx, ty) = t^n f(x, y)$$

for each $t > 0$.

Worked Example 12.3.1. Show that $f(x, y) = 2x^5 - x^2y^3 + 5xy^4$ is homogeneous of degree 5.
Solution.

$$\begin{aligned}f(tx, ty) &= 2(tx)^5 - (tx)^2(ty)^3 + 5(tx)(ty)^4\\ &= 2t^5x^5 - t^5x^2y^3 + 5t^5xy^4\\ &= t^5(2x^5 - x^2y^3 + 5xy^4)\\ &= t^5 f(x, y).\end{aligned}$$

It is easier to note that if f is a polynomial in x and y such that the sum of the powers of x and y in each term is the same value n, then f is homogeneous of degree n.

Worked Example 12.3.2. $g(x,y) = \dfrac{1}{x^2+y^2}e^{\frac{x}{y}}$ is homogeneous of degree -2.

Solution.

$$\begin{aligned} g(tx,ty) &= \frac{1}{(tx)^2+(ty)^2}e^{\frac{tx}{ty}} \\ &= \frac{1}{t^2(x^2+y^2)}e^{\frac{x}{y}} \\ &= t^{-2}g(x,y). \end{aligned}$$

Thus g is homogeneous of degree -2.

Worked Example 12.3.3. $h(x,y) = y - x\cot\left(\dfrac{x}{y}\right)$ is homogeneous of degree 1.

Solution.

$$h(tx,ty) = ty - (tx)\cot\left(\frac{tx}{ty}\right) = t\left(y - x\cot\left(\frac{x}{y}\right)\right) = th(x,y).$$

A **homogeneous differential equation** is an equation of the form

$$f(x,y)\,dx + g(x,y)\,dy = 0$$

where f and g are homogeneous functions of the *same degree*. To solve such type of equations we make the appropriate substitution $y = vx$ (or $x = vy$). Then $dy = v\,dx + x\,dv$ (or $dx = v\,dy + y\,dv$). We then substitute back y and dy (or x and dx) into the original ODE which reduces the homogeneous ODE to a separable ODE.

Worked Example 12.3.4. Solve the differential equation

$$(x^2+y^2)\,dx - x^2dy = 0.$$

Solution. Here $f(x,y) = x^2+y^2$ is homogeneous of degree 2 and $g(x,y) = -x^2$ is also homogeneous of degree 2. Hence this is a homogeneous ODE of degree 2. Let $y = vx$. Then $dy = v\,dx + x\,dv$. Substituting back into the ODE gives

$$\begin{aligned} (x^2+y^2)\,dx - x^2\,dy = 0 &\Rightarrow (x^2+(xv)^2)\,dx - x^2(v\,dx + x\,dv) = 0 \\ &\Rightarrow x^2(1+v^2)\,dx - x^2v\,dx - x^3\,dv = 0 \\ &\Rightarrow x^2(1+v^2-v)\,dx - x^3\,dv = 0 \\ &\Rightarrow (v^2-v+1)\,dx - x\,dv = 0 \\ &\Rightarrow \frac{1}{x}\,dx = \frac{1}{v^2-v+1}\,dv \qquad \text{(separable ODE)} \\ &\Rightarrow \frac{1}{x}\,dx = \frac{1}{\left(v-\frac{1}{2}\right)^2+\frac{3}{4}}\,dv \qquad \text{(complete square)} \\ &\Rightarrow \int \frac{1}{x}\,dx = \int \frac{1}{\left(v-\frac{1}{2}\right)^2+\frac{3}{4}}\,dv \\ &\Rightarrow \ln|x| = \frac{2}{\sqrt{3}}\arctan\left(\frac{2(v-\frac{1}{2})}{\sqrt{3}}\right) + C \end{aligned}$$

Since $y = vx$, $v = \dfrac{y}{x}$. Substituting back gives

$$\ln|x| = \frac{2}{\sqrt{3}}\arctan\left(\frac{2v-1}{\sqrt{3}}\right) + C \Rightarrow \ln|x| = \frac{2}{\sqrt{3}}\arctan\left(\frac{2\left(\frac{y}{x}\right)-1}{\sqrt{3}}\right) + C$$
$$\Rightarrow \ln|x| = \frac{2}{\sqrt{3}}\arctan\left(\frac{2y-x}{\sqrt{3}x}\right) + C \qquad \text{is the general solution.}$$

Worked Example 12.3.5. Solve the differential equation

$$3y\,dx + (x+2y)\,dy = 0$$

by using the substitution $x = vy$.
Solution. This is a homogeneous ODE of degree 1. Let $x = vy$. Then $dx = v\,dy + y\,dv$. Substituting back into the ODE reduces it as follows:

$$\begin{aligned}
3y\,dx + (x+2y)\,dy = 0 &\Rightarrow 3y(v\,dy + y\,dv) + (vy+2y)\,dy = 0\\
&\Rightarrow 3vy\,dy + 3y^2\,dv + y(v+2)\,dy = 0\\
&\Rightarrow 3y^2\,dv + y(3v+v+2)\,dy = 0\\
&\Rightarrow 3y\,dv + (4v+2)\,dy = 0 \qquad \text{(separable ODE)}\\
&\Rightarrow \frac{3}{2v+1}\,dv + \frac{2}{y}\,dy = 0\\
&\Rightarrow 3\int\frac{1}{2v+1}\,dv + 2\int\frac{1}{y}\,dy = 0\\
&\Rightarrow \frac{3}{2}\ln|2v+1| + 2\ln|y| = C\\
&\Rightarrow 3\ln|2v+1| + 4\ln|y| = 2C\\
&\Rightarrow 3\ln\left|2\left(\frac{x}{y}\right)+1\right| + 4\ln|y| = 2C \quad (\because x = vy)\\
&\Rightarrow 3\ln\left|\frac{2x+y}{y}\right| + 4\ln|y| = 2C\\
&\Rightarrow 3\ln|2x+y| - 3\ln|y| + 4\ln|y| = 2C\\
&\Rightarrow 3\ln|2x+y| + \ln|y| = K \quad \text{is the general solution.}
\end{aligned}$$

Tutorial 12.3.1. State the degree of the following homogeneous ODE's and then solve them.
1. $x(x+y)\,dy - y^2\,dx = 0$.
2. $x\,dy + \left(xe^{-\frac{y}{x}} - y\right)\,dx = 0$.
3. $\left(y + \sqrt{x^2-y^2}\right)\,dx = x\,dy$.
4. $\left[y\sin\left(\dfrac{y}{x}\right) + x\cos\left(\dfrac{y}{x}\right)\right]\,dx - x\sin\left(\dfrac{y}{x}\right)\,dy = 0$.
5. Use the substitution $x = vy$ to solve $2xy\,dx + (y^2 - x^2)\,dy = 0$.

12.4 Exact Differential Equations

A prerequisite for this section is the notion of a *partial derivative* of a function of more than one variable. A more comprehensive reading into partial derivatives, Section 14.3 pp. 782–790 of Thomas, will facilitate your understanding of this section.
We will be concerned mainly with functions of two variables $f(x,y)$ as seen in the previous section on homogeneous ODE's. The graph of a function $f(x,y)$ of two variables x and y is the set of all points (x,y,z) in three-dimensional space such that $z = f(x,y)$. Generally, $f(x,y)$ represents a surface in three-dimensional space $\mathbb{R}^3$.

Worked Example 12.4.1. Let $f(x,y) = \dfrac{\sin(x+y)}{x}$. Find $f(2,-2)$ and $f(\pi, \frac{\pi}{2})$.

Solution.

$$f(2,-2) = \frac{\sin(2+(-2))}{2} = \frac{\sin(0)}{2} = \frac{0}{2} = 0$$

and

$$f\left(\pi, \frac{\pi}{2}\right) = \frac{\sin(\pi + \frac{\pi}{2})}{\pi} = \frac{\sin(\frac{3\pi}{2})}{\pi} = -\frac{1}{\pi}.$$

You may plot $f(x,y) = \dfrac{\sin(x+y)}{x}$ using Mathematica. The calculation $f(2,-2) = 0$ means that $z = 0$ when $x = 2$ and $y = -2$ on the given surface.

If f is a function of two variables, then the **first partial derivative of** $f(x,y)$ **with respect to** x is

$$f_x(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h} \quad (y \text{ is fixed})$$

and the **first partial derivative of** $f(x,y)$ **with respect to** y is

$$f_y(x,y) = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h} \quad (x \text{ is fixed})$$

provided that these limits exist.
By the above limit definitions,

to find $f_x(x,y)$ **we may regard** y **as a constant and differentiate** $f(x,y)$ **with respect to** x **in the usual way**

and

to find $f_y(x,y)$ **we may regard** x **as a constant and differentiate** $f(x,y)$ **with respect to** y **in the usual way**.

Other common notations for the two first partial derivatives of $f(x,y)$ are

$$f_x = \frac{\partial f}{\partial x} \qquad \text{and} \qquad f_y = \frac{\partial f}{\partial y}.$$

The geometric interpretation of the first partial derivatives are given at the bottom of page 783–784 in Thomas.

Worked Example 12.4.2. If $f(x,y) = x^2y^3$, evaluate $f_x(3,5)$ and $f_y(3,5)$.

Solution. To find f_x we use the rule:
treat y as a constant and differentiate f in the usual way with respect to x:

$$\begin{aligned} f(x,y) = x^2y^3 \Rightarrow f_x(x,y) &= 2xy^3 \qquad (y \text{ is a constant}) \\ \Rightarrow f_x(3,5) &= 2(3)(5^3) = 750. \end{aligned}$$

To find f_y we use the rule:
treat x as a constant and differentiate f in the usual way with respect to y:

$$\begin{aligned} f(x,y) = x^2y^3 \Rightarrow f_y(x,y) &= 3x^2y^2 \qquad (x \text{ is a constant}) \\ \Rightarrow f_y(3,5) &= 3(3^2)(5^2) = 675. \end{aligned}$$

Note that $f_x = 2xy^3$ and $f_y = 3x^2y^2$ so that $f_x \neq f_y$ in general!

Worked Example 12.4.3. Let $f(x,y) = x^2y + \sin(xy^2)$. Find f_x and f_y.

Solution.

$$f(x,y) = x^2y + \sin(xy^2) \Rightarrow f_x = 2xy + y^2\cos(xy^2) \qquad (y \text{ constant, note chain rule})$$

and

$$f(x,y) = x^2y + \sin(xy^2) \Rightarrow f_y = x^2 + 2xy\cos(xy^2) \qquad (x \text{ constant, note chain rule}).$$

Worked Example 12.4.4. Let $z = xe^{xy}$. Find z_x and z_y.

Solution.

$$z_x = x(e^{xy}y) + (1)e^{xy} = e^{xy}(xy + 1) \qquad \text{and} \qquad z_y = x(e^{xy}x) = x^2e^{xy}.$$

Note the use of the product and the chain rules.

If f is a function of two variables x and y, then f_x and f_y are also functions of two variables and we may consider their first partial derivatives. These are called the second partial derivatives of f and are denoted as follows:

1. $f_{xx} = (f_x)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial x^2}$. To find f_{xx} we differentiate f partially with respect to x twice whilst keeping y constant.

2. $f_{xy} = (f_x)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right) = \dfrac{\partial^2 f}{\partial y \partial x}$. To find f_{xy} we differentiate f partially with respect to x whilst keeping y constant, then differentiate again partially with respect to y keeping x constant.

3. $f_{yx} = (f_y)_x = \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial x \partial y}$. To find f_{yx} we differentiate f partially with respect to y whilst keeping x constant, then differentiate again partially with respect to x keeping y constant.

4. $f_{yy} = (f_y)_y = \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial y}\right) = \dfrac{\partial^2 f}{\partial y^2}$. To find f_{yy} we differentiate f partially with respect to y twice whilst keeping x constant.

Worked Example 12.4.5. Find the second partial derivatives of f if $f(x, y) = x^3y^2 - 2x^2y + 3x$.

Solution. We first find the first partial deriviatives f_x and f_y.

$$f_x = 3x^2y^2 - 4xy + 3 \quad \text{and} \quad f_y = 2x^3y - 2x^2.$$

Then

$$\begin{aligned}
f_{xx} &= \frac{\partial}{\partial x}f_x = \frac{\partial}{\partial x}(3x^2y^2 - 4xy + 3) = 6xy^2 - 4y,\\
f_{xy} &= \frac{\partial}{\partial y}f_x = \frac{\partial}{\partial y}(3x^2y^2 - 4xy + 3) = 6x^2y - 4x,\\
f_{yx} &= \frac{\partial}{\partial x}f_y = \frac{\partial}{\partial x}(2x^3y - 2x^2) = 6x^2y - 4x,\\
f_{yy} &= \frac{\partial}{\partial y}f_y = \frac{\partial}{\partial y}(2x^3y - 2x^2) = 2x^3.
\end{aligned}$$

Third and higher partial derivatives are defined in a similar fashion, e. g.,

$$f_{xxx} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x^2}\right) = \frac{\partial^3 f}{\partial x^3}$$

and

$$f_{xyx} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial^3 f}{\partial x \partial y \partial x}.$$

For functions of three or more variables, the technique for calculating the partial derivatives works in the same way as we have done above, with all variables being kept constant except the one with respect to which we want to differentiate.

Tutorial 12.4.1. 1. Thomas Exercises 14.3 11th ed. pp. 978–980: 1, 7, 11, 17, 19, 23, 25, 29, 41, 47, 63, 66, 68, 69, 74 **or** 12th ed. pp. 790-792: 1, 7, 11, 17, 19, 23, 25, 29, 41, 47, 73, 76, 79, 80, 81, 86.

Partial derivatives are used in solving a particular type of first order ODE's called *exact differential equations*. Let M and N be functions of two variables x and y which have continuous first partial derivatives. The differential equation

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0 \quad \text{or} \quad M(x,y)\,dx + N(x,y)\,dy = 0$$

is an **exact differential equation** provided that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{or} \quad M_y = N_x.$$

We may thus test whether a given ODE

$$M\,dx + N\,dy = 0$$

is exact by calculating M_y and N_x. Accordingly,

$$M_y = N_x \Rightarrow \textbf{ODE IS EXACT},$$
$$M_y \neq N_x \Rightarrow \textbf{ODE IS NOT EXACT}.$$

Worked Example 12.4.6. Show that the differential equation

$$(2x-2)\,dx + (8y-16)\,dy = 0$$

is exact.

Solution. Let $M = 2x-2$ and $N = 8y-16$. Then $M_y = 0 = N_x$. Thus $M_y = N_x$ and hence the given ODE is exact.

Theorem 12.1. *If $M\,dx + N\,dy = 0$ is an exact differential equation, then there exists a function $f(x,y)$ such that $f_x = M$ and $f_y = N$ and the differential equation has a solution of the form $f(x,y) = C$ for a constant C.*

The above theorem provides a technique for finding a solution to the exact ODE

$$M\,dx + N\,dy = 0.$$

We illustrate by the following examples.

Worked Example 12.4.7. Solve the differential equation

$$(2x-2)\,dx + (8y-16)\,dy = 0.$$

Solution. The ODE was shown in Example 12.4.6 to be exact with $M = 2x-2$ and $N = 8y-16$. Thus by Theorem 12.1 there exists a function $f(x,y)$ such that

$$f_x = M = 2x-2$$

and

$$f_y = N = 8y-16.$$

Using the above two formulae $f(x,y)$ may be found as follows. Since $f_x(x,y) = 2x-2$, integrating f_x with respect to x (keeping y constant) gives

$$f_x(x,y) = 2x-2 \Rightarrow \int f_x(x,y)\,dx = \int (2x-2)\,dx$$
$$\Rightarrow f(x,y) = x^2 - 2x + g(y),$$

where $g(y)$ is a function of y only. Since $f_y = N$, from the calculation of $f(x,y)$ above we may now find f_y.

$$f(x,y) = x^2 - 2x + g(y) \Rightarrow f_y = \frac{\partial}{\partial y}(x^2 - 2x + g(y)) = g'(y).$$

Then

$$\begin{aligned} f_y = N &\Rightarrow g'(y) = 8y - 16 \\ &\Rightarrow g(y) = \int (8y - 16)\, dy \\ &\Rightarrow g(y) = 4y^2 - 16y + K. \end{aligned}$$

Hence

$$f(x, y) = x^2 - 2x + g(y) = x^2 - 2x + 4y^2 - 16y + K.$$

By Theorem 12.1, the solution to the exact ODE is thus

$$f(x, y) = C,$$

i. e.,

$$x^2 - 2x + 4y^2 - 16y = C.$$

The constant K is absorbed in the constant C.

Worked Example 12.4.8. Solve the differential equation

$$(y \cos x + 2xe^y)\, dx + (\sin x + x^2 e^y + 2)\, dy = 0.$$

Solution. Let $M = y \cos x + 2xe^y$ and $N = \sin x + x^2 e^y + 2$. Then

$$M_y = \cos x + 2xe^y = N_x$$

so that the ODE is exact. Thus there is $f(x, y)$ such that $f_x = M$ and $f_y = N$. Then

$$\begin{aligned} f_x = M &\Rightarrow f_x = y \cos x + 2xe^y \\ &\Rightarrow f(x, y) = \int (y \cos x + 2xe^y)\, dx \\ &\Rightarrow f(x, y) = y \sin x + x^2 e^y + g(y) \qquad \text{(keeping } y \text{ constant)} \\ &\Rightarrow f_y = \sin x + x^2 e^y + g'(y) \\ &\Rightarrow N = \sin x + x^2 e^y + g'(y) \qquad (\because f_y = N) \\ &\Rightarrow \sin x + x^2 e^y + 2 = \sin x + x^2 e^y + g'(y) \\ &\Rightarrow g'(y) = 2 \\ &\Rightarrow g(y) = \int 2\, dy \\ &\Rightarrow g(y) = 2y + K. \end{aligned}$$

Hence $f(x, y) = y \sin x + x^2 e^y + g(y) = y \sin x + x^2 e^y + 2y + K$ so that a solution to the ODE is $f(x, y) = C$, i. e.,

$$y \sin x + x^2 e^y + 2y = C.$$

In the above example we could have started with $f_y = N$ instead as the following example illustrates.

Worked Example 12.4.9. Solve the differential equation

$$(2x + y^3)\, dx + (3xy^2 - e^{2y})\, dy = 0$$

subject to the initial condition $y(-1) = 0$.
Solution. Let $M = 2x + y^3$ and $N = 3xy^2 - e^{2y}$. Then

$$M_y = 3y^2 = N_x$$

so that the ODE is exact. Thus there is $f(x,y)$ such that $f_x = M$ and $f_y = N$. Then

$$\begin{aligned}
f_y = N &\Rightarrow f_y = 3xy^2 - e^{2y} \\
&\Rightarrow f(x,y) = \int (3xy^2 - e^{2y})\,dy \\
&\Rightarrow f(x,y) = xy^3 - \frac{1}{2}e^{2y} + g(x) \qquad \text{(keeping } x \text{ constant)} \\
&\Rightarrow f_x = y^3 + g'(x) \\
&\Rightarrow M = y^3 + g'(x) \qquad (\because f_x = M) \\
&\Rightarrow 2x + y^3 = y^3 + g'(x) \\
&\Rightarrow 2x = g'(x) \\
&\Rightarrow g(x) = \int 2x\,dx = x^2 + K.
\end{aligned}$$

Thus $f(x,y) = xy^3 - \frac{1}{2}e^{2y} + g(x)$ becomes $f(x,y) = xy^3 - \frac{1}{2}e^{2y} + x^2 + K$. Then the general solution to the ODE is

$$xy^3 - \frac{1}{2}e^{2y} + x^2 = C.$$

Applying the initial conditions gives

$$\begin{aligned}
y(-1) = 0 &\Rightarrow (-1)(0)^3 - \frac{1}{2}e^{2(0)} + (-1)^2 = C \\
&\Rightarrow C = \frac{1}{2}
\end{aligned}$$

Hence the particular solution to the ODE is

$$xy^3 - \frac{1}{2}e^{2y} + x^2 = \frac{1}{2}.$$

Tutorial 12.4.2. Show that the following differential equations are exact and then solve them.
1. $(2x + y^3)\,dx + (3xy^2 + ye^{2y})\,dy = 0$.
2. $(y\cos x + 2xe^y + x\sin x)\,dx + (\sin x + x^2e^y + 2y)\,dy = 0$.
3. $(2x + 3y + 1)\,dx + (3x + 4y - 1)\,dy = 0$.
4. $(x + 3y)\,dx + (3x - 2y)\,dy = 0$. Show that this is also a homogeneous ODE and also solve it using the technique learned for homogeneous ODE's.